# JESS Thermodynamic Database v8.9

## Samarium

## 24-Jan-23

| Reaction No. 2678 | | | | | | | |
|---|---|---|---|---|---|---|---|
| F-1 + Sm+3 = Sm+3_F-1 | | | | | | | |
| **No** | **t** | **I-Str** | **Medium** | **Value (Dev)** | **W** | **Tech.** | **Ref. #** |
| 1 | 25 | 0 | Inf. Dilution | lgK:4.18(3SF) | 4 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.17(3SF) | 4 | EOD | 7141 |
| 3 | 25 | 0 | Inf. Dilution | lgK:3.58(3SF) | 3 | CRV | 8781 |
| 4 | 25 | 0 | Inf. Dilution | lgK:4.06(3SF) | 4 | EDH | 8917 |
| 5 | 25 | 0 | Inf. Dilution | lgK:3.80(0.30SD) | 4 | CRV | 32222 |
| 6 | 25 | 0.015 | NaClO4 | lgK:3.91(3SF) | 5 | MCE | 15996 |
| 7 | 25 | 0.025 | HNO3 | lgK:3.59(0.05SD) | 6 | MCE | 8221 |
| 8 | 25 | 0.1 | NaClO4 | lgK:3.500(0.002SD) | 6 | MCE | 15996 |
| 9 | 25 | 0.1 | Unknown | lgK:3.54(3SF) | 6 | CRV | 552 |
| 10 | 25 | 0.4 | NaClO4 | lgK:3.290(0.005SD) | 6 | MCE | 15996 |
| 11 | 25 | 0.5 | NH4NO3 | lgK:3.15(0.02SD) | 4 | MIS | 5387 |
| 12 | 25 | 0.68 | NaClO4 | lgK:3.17(3SF) | 5 | EFE | 8917 |
| 13 | 25 | 0.7 | NaClO4 | lgK:3.225(0.008SD) | 6 | MCE | 15996 |
| 14 | 25 | 1 | NaCl | lgK:3.092(0.02SD) | 5 | MPT | 5385 |
| 15 | 25 | 1 | NaClO4 | lgK:3.1(0.05SD) | 6 | MRX | 437 |
| 16 | 25 | 1 | Unknown | lgK:3.12(3SF) | 5 | CRV | 6761 |
| 17 | 25 | 1.5 | NaClO4 | lgK:3.243(0.006SD) | 6 | MCE | 15996 |
| 18 | 25 | 3 | NaClO4 | lgK:3.309(0.005SD) | 6 | MCE | 15996 |
| 19 | 25 | 5 | NaClO4 | lgK:3.576(0.007SD) | 6 | MCE | 15996 |
| 20 | 25 | 6 | NaClO4 | lgK:3.710(4SF) | 5 | MCE | 15996 |
| 21 | 25 | 0 | Inf. Dilution | dH:26.520(6.290SD)kJ | 4 | CRV | 32222 |
| 22 | 25 | 1 | NaClO4 | dH:9.(1.SD) | 0 | MCL | 437 |
| 23 | 25 | 1 | NaClO4 | dH:10.8(0.49SD)kJ | 5 | MCL | 5146 |
| 24 | 25 | 1 | Unknown | dH:2.6(2SF) | 6 | CRV | 552 |
| 25 | 25 | 1 | Unknown | dH:9.39(3SF) | 0 | CRV | 6761 |

| Reaction No. 3632 |
|---|
| [18]O6 + Sm+3 = Sm+3_[18]O6 |

doi: 10.5281/zenodo.7700024

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:8.10(3SF) | 5 | MIS | 3696 |

| Reaction No. 5482 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 = Sm+3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 18:20 | 0.04 | Unknown | lgK:16.34(4SF) | 0 | MSP | 931 |
| 2 | 20 | 0.01 | Unknown | lgK:16.97(4SF) | 0 | MGL | 140 |
| 3 | 20 | 0.1 | K+1 salt | lgK:17.(0.3SD) | 7 | EIU | 903 |
| 4 | 20 | 0.1 | KCl | lgK:16.55(4SF) | 0 | MGL | 140 |
| 5 | 20 | 0.1 | KCl | lgK:17.2(3SF) | 4 | MGL | 140 |
| 6 | 20 | 0.1 | KCl | lgK:16.9(3SF) | 5 | MGL | 197 |
| 7 | 20 | 0.1 | KCl | lgK:17.0(0.08SD) | 5 | MGL | 903 |
| 8 | 20 | 0.1 | KNO3 | lgK:17.14(4SF) | 3 | MPL | 136 |
| 9 | 20 | 0.1 | KNO3 | lgK:16.3(3SF) | 0 | MGL | 140 |
| 10 | 20 | 0.1 | KNO3 | lgK:16.7(0.2SD) | 0 | MGL | 903 |
| 11 | 20 | 0.1 | KNO3 | lgK:17.3(0.05SD) | 3 | MPL | 903 |
| 12 | 20 | 0.1 | KNO3 | lgK:17.3(0.05SD) | 3 | MGL | 1995 |
| 13 | 20 | 0.1 | KNO3 | lgK:17.14(4SF) | 6 | MPL | 6950 |
| 14 | 23 | | | lgK:17.(0.5SD) | 4 | MEP | 906 |
| 15 | 25 | 0.1 | KCl | lgK:16.53(4SF) | 0 | MIX | 140 |
| 16 | 25 | 0.1 | KCl | lgK:16.94(4SF) | 5 | MIX | 903 |
| 17 | 25 | 0.5 | NaClO4 | lgK:16.2(0.06SD) | 5 | MGL | 2650 |
| 18 | 25 | 0.5 | NaNO3 | lgK:16.92(4SF) | 0 | MGL | 11516 |
| Note (18): Kopyrin A et al., Radiokhim., 1984, 26, 303 | | | | | | | |
| 19 | 25 | | | lgK:16.43(4SF) | 0 | MCN | 931 |
| 20 | 25 | 0.1 | KNO3 | dH:-3.35(3SF) | 4 | MCL | 903 |
| 21 | 25 | 0.44 | NaNO3 | dH:-4.6(2SF) | 4 | MCL | 903 |
| 22 | 25 | 0.5 | NaClO4 | dH:-26.4(0.5SD)kJ | 4 | MCL | 2651 |
| 23 | 25 | 1 | Unknown | dH:-5.6(2SF) | 6 | CRV | 552 |

| Reaction No. 5483 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + H+1 = Sm+3_H+1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.01 | KCl | lgK:1.69(3SF) | 6 | MGL | 2333 |
| 2 | 25 | 0.1 | KCl | lgK:1.7(0.05SD) | 3 | EIU | 903 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0.1 | KCl | lgK:1.7(0.05SD) | 7 | EIU | 903 |
| 4 | 25 | | | lgK:2.22(3SF) | 0 | MCN | 931 |

**Reaction No. 5522**

His-1 + Sm+3 = Sm+3_His-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 3 | NaClO4 | lgK:4.5(0.07SD) | 4 | EIU | 904 |
| 2 | 35 | 0.1 | KNO3 | lgK:3.85(3SF) | 4 | MGL | 1245 |
| 3 | 37 | 3 | NaClO4 | lgK:4.4(0.08SD) | 4 | EIU | 904 |
| 4 | 25 | 3 | NaClO4 | dH:0.55(0.02SD) | 5 | MCL | 904 |
| 5 | 37 | 3 | NaClO4 | dH:0.05(0.02SD) | 5 | MCL | 904 |

**Reaction No. 5523**

2<His-1> + Sm+3 = Sm+3_His-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 3 | NaClO4 | lgK:8.7(0.08SD) | 4 | EIU | 904 |
| 2 | 37 | 3 | NaClO4 | lgK:8.8(0.1SD) | 4 | EIU | 904 |
| 3 | 25 | 3 | NaClO4 | dH:-3.61(0.02SD) | 5 | MCL | 904 |
| 4 | 37 | 3 | NaClO4 | dH:-4.8(0.03SD) | 5 | MCL | 904 |

**Reaction No. 5543**

H+1 + His-1 + Sm+3 = Sm+3_H+1_His-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:11.7(3SF) | 1 | EDH | |
| 2 | 25 | 3 | NaClO4 | lgK:11.8(0.05SD) | 0 | EIU | 904 |
| 3 | 37 | 3 | NaClO4 | lgK:11.2(0.2SD) | 4 | EIU | 904 |
| 4 | 25 | 3 | NaClO4 | dH:-2.50(0.02SD) | 3 | MCL | 904 |
| 5 | 37 | 3 | NaClO4 | dH:-1.19(0.02SD) | 3 | MCL | 904 |

**Reaction No. 5575**

Thr-1 + Sm+3 = Sm+3_Thr-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KCl | lgK:3.1(2SF) | 4 | MGL | 905 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.72(3SF) | 2 | EAE | |

**Reaction No. 5705**

Met-1 + Sm+3 = Sm+3_Met-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KCl | lgK:4.6(2SF) | 4 | MGL | 905 |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.22(3SF) | 2 | EAE | |

**Reaction No. 5743**

Ser-1 + Sm+3 = Sm+3_Ser-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KCl | lgK:3.4(2SF) | 4 | MGL | 905 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.02(3SF) | 2 | EAE | |

**Reaction No. 6722**

Acetic-1 + Sm+3 = Sm+3_Acetic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:1.95(3SF) | 5 | MSP | 4440 |
| 2 | 20 | 0.1 | NaClO4 | lgK:2.30(3SF) | 5 | MGL | 999 |
| 3 | 20 | 0.1 | Unknown | lgK:2.30(3SF) | 5 | ENS | 1539 |
| 4 | 20 | 0.2 | NaClO4 | lgK:2.01(0.01SD) | 4 | MQH | 999 |
| 5 | 25 | 0 | Inf. Dilution | lgK:2.839(4SF) | 4 | MGL | 381 |
| 6 | 25 | 2 | NaClO4 | lgK:2.03(3SF) | 5 | ENS | 5483 |
| 7 | 25 | 2 | NaClO4 | dH:1.453(4SF) | 5 | MCL | 5483 |

**Reaction No. 6723**

2<Acetic-1> + Sm+3 = Sm+3_Acetic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:3.74(3SF) | 5 | MSP | 4440 |
| 2 | 20 | 0.1 | Unknown | lgK:3.88(3SF) | 6 | ENS | 1539 |
| 3 | 20 | 0.2 | NaClO4 | lgK:3.3(0.05SD) | 4 | MQH | 999 |
| 4 | 20 | 2 | NaClO4 | lgK:3.26(3SF) | 5 | MQH | 11516 |
| Note (4): Sonesson A, Acta Chem. Scand., 1958, 12, 1937 | | | | | | | |
| 5 | 25 | 0 | Unknown | lgK:1.96(3SF) | 0 | MGL | 381 |
| 6 | 25 | 2 | NaClO4 | lgK:3.29(3SF) | 5 | ENS | 5483 |
| 7 | 25 | 2 | NaClO4 | dH:2.876(4SF) | 5 | MCL | 5483 |

**Reaction No. 6724**

3<Acetic-1> + Sm+3 = Sm+3_Acetic-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:5.28(3SF) | 5 | MSP | 4440 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 20 | 0.1 | Unknown | lgK:5.30(3SF) | 5 | ENS | 1539 |
| 3 | 20 | 0.2 | NaClO4 | lgK:3.9(0.08SD) | 3 | MQH | 999 |
| 4 | 25 | 2 | NaClO4 | lgK:3.90(3SF) | 5 | ENS | 5483 |
| 5 | 25 | 2 | NaClO4 | dH:3.773(4SF) | 5 | MCL | 5483 |

**Reaction No. 6725**

4<Acetic-1> + Sm+3 = Sm+3_Acetic-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.2 | NaClO4 | lgK:3.8(2SF) | 4 | MQH | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.3(3SF) | 2 | EAE | |

**Reaction No. 6726**

Sm+3_Acetic-1 + Acetic-1 = Sm+3_Acetic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:1.79(3SF) | 5 | MSP | 4440 |
| 2 | 20 | 0.1 | NaClO4 | lgK:1.58(3SF) | 5 | MGL | 999 |
| 3 | 20 | 2 | NaClO4 | lgK:1.25(3SF) | 5 | MQH | 11516 |
| Note (3): Sonesson A, Acta Chem. Scand., 1958, 12, 1937 | | | | | | | |
| 4 | 25 | 0 | Inf. Dilution | lgK:1.96(3SF) | 4 | MGL | 381 |

**Reaction No. 6784**

Glycolic-1 + Sm+3 = Sm+3_Glycolic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:2.914(4SF) | 3 | MGL | 999 |
| 2 | 20 | 2 | NaClO4 | lgK:2.56(3SF) | 5 | MQH | 999 |
| 3 | 25 | 0.2 | NaClO4 | lgK:3.9(0.06SD) | 0 | MPT | 3677 |
| 4 | 25 | 0.5 | NaClO4 | lgK:3.(0.4SD) | 2 | MGL | 1731 |
| 5 | 25 | 2 | NaClO4 | lgK:2.46(3SF) | 2 | MGL | 2130 |
| 6 | 25 | 2 | Unknown | dG:-3.47(3SF) | 3 | EOD | 31164 |
| 7 | 25 | 2 | NaClO4 | dH:-3.89(3SF)kJ | 5 | MCL | 7703 |
| 8 | 25 | 2 | NaClO4 | dH:-4.35(3SF)kJ | 5 | MCL | 7703 |

**Reaction No. 6785**

2<Glycolic-1> + Sm+3 = Sm+3_Glycolic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:5.01(3SF) | 6 | MGL | 999 |
| 2 | 20 | 2 | NaClO4 | lgK:4.53(3SF) | 5 | MQH | 999 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0.5 | NaClO4 | lgK:4.6(0.07SD) | 5 | MGL | 1731 |
| 4 | 25 | 2 | Unknown | dG:-6.14(3SF) | 3 | EOD | 31164 |
| 5 | 25 | 2 | NaClO4 | dH:-7.24(3SF)kJ | 5 | MCL | 7703 |
| 6 | 25 | 2 | NaClO4 | dH:-10.06(4SF)kJ | 5 | MCL | 7703 |

**Reaction No. 6786**

3<Glycolic-1> + Sm+3 = Sm+3_Glycolic-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:6.57(3SF) | 3 | MGL | 999 |
| 2 | 20 | 2 | NaClO4 | lgK:5.9(2SF) | 4 | MQH | 999 |
| 3 | 25 | 0.5 | NaClO4 | lgK:5.7(0.08SD) | 2 | MGL | 1731 |
| 4 | 25 | 2 | Unknown | dG:-7.93(3SF) | 3 | EOD | 31164 |
| 5 | 25 | 2 | NaClO4 | dH:-15.4(3SF)kJ | 5 | MCL | 7703 |

**Reaction No. 6787**

4<Glycolic-1> + Sm+3 = Sm+3_Glycolic-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 2 | NaClO4 | lgK:6.4(2SF) | 4 | MQH | 999 |
| 2 | 25 | 0.5 | NaClO4 | lgK:7.2(0.1SD) | 5 | MGL | 1731 |
| 3 | 25 | 2 | Unknown | dG:-8.71(3SF) | 3 | EOD | 31164 |
| 4 | 25 | 2 | NaClO4 | dH:-20.67(4SF)kJ | 5 | MCL | 7703 |

**Reaction No. 7091**

Malonic-2 + Sm+3 = Sm+3_Malonic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.193(4SF) | 2 | MGL | 4388 |
| 2 | 25 | 0.1 | NaClO4 | lgK:4.63(3SF) | 3 | MGL | 906 |
| 3 | 25 | 0.1 | NaClO4 | lgK:4.5(0.03SD) | 6 | MGL | 1499 |
| 4 | 25 | 0.1 | NaClO4 | lgK:4.73(3SF) | 2 | MKN | 2619 |
| 5 | 25 | 0.1 | Unknown | lgK:4.19(3SF) | 3 | CRV | 7271 |
| 6 | 25 | 0.2 | NaClO4 | lgK:4.87(0.01SD) | 0 | MPT | 3677 |
| 7 | 25 | 1 | Na+1 salt | lgK:3.67(3SF) | 3 | CRV | 7271 |
| 8 | 25 | 1 | NaClO4 | lgK:3.67(3SF) | 6 | MGL | 999 |
| 9 | 25 | 0.1 | NaClO4 | dH:13.77(0.3SD)kJ | 5 | MCL | 1499 |
| 10 | 25 | 1 | Na+1 salt | dH:3.0(2SF) | 3 | CRV | 7271 |

**Reaction No. 7092**

| 2<Malonic-2> + Sm+3 = Sm+3_Malonic-2(2) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:6.845(4SF) | 6 | MGL | 4388 |
| 2 | 25 | 1 | NaClO4 | lgK:6.07(3SF) | 6 | MGL | 999 |
| 3 | 25 | 1 | Na+1 salt | dH:5.1(2SF) | 3 | CRV | 7271 |

| Reaction No. 7274 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Propanoic-1 + Sm+3 = Sm+3_Propanoic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.21(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:2.0(0.03SD) | 4 | MGL | 999 |

| Reaction No. 7275 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Propanoic-1 + Propanoic-1 = Sm+3_Propanoic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:1.49(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:1.2(0.05SD) | 3 | MGL | 999 |

| Reaction No. 7307 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3OHPropan-1 + Sm+3 = Sm+3_3OHPropan-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:1.75(3SF) | 6 | MGL | 999 |

| Reaction No. 7357 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_2SHAcet-2 = Sm+3_H+1_2SHAcet-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.11(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:1.8(0.05SD) | 4 | MGL | 3749 |

| Reaction No. 7358 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_2SHAcet-2 + H+1_2SHAcet-2 = Sm+3_H+1(2)_2SHAcet-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:1.36(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:0.9(0.1SD) | 4 | MGL | 3749 |

| Reaction No. 7448 |
|---|
| Lactic-1 + Sm+3 = Sm+3_Lactic-1 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:2.88(3SF) | 5 | MGL | 11516 |
| Note (1): Powell J et al., Rare Earth Research II, 1964 | | | | | | | |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.88(3SF) | 6 | MGL | 999 |
| 3 | 25 | 0.2 | NaClO4 | lgK:2.5(2SF) | 4 | MGL | 999 |
| 4 | 25 | 0.2 | NaClO4 | lgK:4.1(0.03SD) | 0 | MPT | 3677 |
| 5 | 25 | 0.5 | NaClO4 | lgK:2.63(3SF)w | 5 | MPH | 7703 |
| 6 | 25 | 2 | NaClO4 | lgK:2.56(3SF) | 5 | MGL | 2130 |
| 7 | 25 | 2 | NaClO4 | dH:-9.75(3SF)kJ | 5 | MCL | 7703 |

| Reaction No. 7449 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Lactic-1> + Sm+3 = Sm+3_Lactic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:5.09(3SF) | 5 | MGL | 11516 |
| Note (1): Powell J et al., Rare Earth Research II, 1964 | | | | | | | |
| 2 | 25 | 0.1 | NaClO4 | lgK:5.09(3SF) | 6 | MGL | 999 |
| 3 | 25 | 0.2 | NaClO4 | lgK:4.54(3SF) | 5 | MGL | 11516 |
| Note (3): Deelstra H et al., Anal. Chim. Acta, 1964, 31, 251 | | | | | | | |
| 4 | 25 | 0.5 | NaClO4 | lgK:4.66(3SF)w | 5 | MPH | 7703 |
| 5 | 25 | 2 | NaClO4 | lgK:4.58(3SF) | 5 | MQH | 7703 |
| 6 | 25 | 2 | NaClO4 | dH:-7.9(2SF)kJ | 5 | MCL | 7703 |

| Reaction No. 7450 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Lactic-1 + Lactic-1 = Sm+3_Lactic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.21(3SF) | 5 | MGL | 11516 |
| Note (1): Powell J et al., Rare Earth Research II, 1964 | | | | | | | |
| 2 | 25 | 0.2 | NaClO4 | lgK:2.04(3SF) | 6 | MGL | 999 |
| 3 | 25 | 2 | NaClO4 | lgK:2.02(3SF) | 6 | MGL | 2130 |

| Reaction No. 7451 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Lactic-1(2) + Lactic-1 = Sm+3_Lactic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:0.90(2SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:1.32(3SF) | 6 | MGL | 2130 |

| Reaction No. 7452 |
|---|

| Sm+3_Lactic-1(3) + Lactic-1 = Sm+3_Lactic-1(4) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:0.41(2SF) | 6 | MGL | 999 |

| Reaction No. 7453 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<Lactic-1> + Sm+3 = Sm+3_Lactic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:6.35(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.5 | NaClO4 | lgK:5.82(3SF)w | 5 | MPH | 7703 |
| 3 | 25 | 2 | NaClO4 | lgK:5.90(3SF) | 5 | MQH | 7703 |
| 4 | 25 | 2 | NaClO4 | dH:-38.00(4SF)kJ | 5 | MCL | 7703 |

| Reaction No. 7703 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DTTAP-5 + Sm+3 = Sm+3_DTTAP-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:20.(0.3SD) | 3 | MGL | 1067 |

| Reaction No. 7704 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_DTTAP-5 = Sm+3_H+1_DTTAP-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:13.(0.3SD) | 3 | MGL | 1067 |

| Reaction No. 9161 | | | | | | | |
|---|---|---|---|---|---|---|---|
| NTA-3 + Sm+3 = Sm+3_NTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 15 | 0.1 | KNO3 | lgK:11.53(4SF) | 5 | MGL | 1118 |
| 2 | 18:20 | 0.02 | Unknown | lgK:10.78(4SF) | 0 | MSP | 1118 |
| 3 | 20 | 0.1 | KCl | lgK:11.33(4SF) | 5 | MGL | 6950 |
| 4 | 20 | 0.1 | KNO3 | lgK:11.35(4SF) | 6 | CMN | 1118 |
| 5 | 20 | 0.1 | KNO3 | lgK:11.51(4SF) | 5 | MGL | 1118 |
| 6 | 20 | 0.1 | KNO3 | lgK:11.25(4SF) | 5 | MPL | 1118 |
| 7 | 20 | 0.1 | KNO3 | lgK:11.33(4SF) | 5 | MPL | 1118 |
| 8 | 25 | 0 | Inf. Dilution | lgK:13.21(4SF) | 4 | EDH | 8698 |
| 9 | 25 | 0.1 | K+1 salt | lgK:11.3(0.1SD) | 5 | CRV | 552 |
| 10 | 25 | 0.1 | KNO3 | lgK:11.53(4SF) | 5 | MGL | 1118 |
| 11 | 25 | 0.1 | NaClO4 | lgK:11.40(4SF) | 5 | MMC | 8698 |
| 12 | 25 | 0.5 | NaClO4 | lgK:11.2(0.09SD) | 0 | MGL | 2650 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Note (12): Suspect other species | | | | | | | |
| 13 | 25 | 0.7 | NaClO4 | lgK:10.21(4SF) | 5 | MMC | 8698 |
| 14 | 25 | 1 | NaClO4 | lgK:10.06(4SF) | 5 | MMC | 8698 |
| 15 | 25 | 2 | NaClO4 | lgK:10.04(4SF) | 5 | MMC | 8698 |
| 16 | 25 | 3 | NaClO4 | lgK:10.09(4SF) | 5 | MMC | 8698 |
| 17 | 25 | 5 | NaClO4 | lgK:10.69(4SF) | 5 | MMC | 8698 |
| 18 | 30 | 0.1 | KNO3 | lgK:11.55(4SF) | 5 | MGL | 1118 |
| 19 | 35 | 0.1 | KNO3 | lgK:11.52(4SF) | 5 | MGL | 1118 |
| 20 | 40 | 0.1 | KNO3 | lgK:11.54(4SF) | 5 | MGL | 1118 |
| 21 | 20 | 0.1 | KNO3 | dH:-1.047(4SF) | 5 | MCL | 1118 |
| 22 | 25 | 0.1 | Unknown | dH:-1.1(2SF) | 5 | CRV | 552 |
| 23 | 25 | 0.5 | NaClO4 | dH:-7.36(0.33SD)kJ | 5 | MCL | 2651 |

| Reaction No. 9162 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_NTA-3 + NTA-3 = Sm+3_NTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 15 | 0.1 | KNO3 | lgK:9.17(3SF) | 5 | MGL | 1118 |
| 2 | 19 | 0.02 | Unknown | lgK:9.76(3SF) | 5 | MSP | 11516 |
| Note (2): Astakhov K et al., Zhur. Neorg. Khim., 1961, 6, 1057 | | | | | | | |
| 3 | 20 | 0.1 | KCl | lgK:9.15(3SF) | 5 | MDG | 6950 |
| 4 | 20 | 0.1 | KNO3 | lgK:9.10(3SF) | 6 | CMN | 1118 |
| 5 | 20 | 0.1 | KNO3 | lgK:9.05(3SF) | 5 | MGL | 1118 |
| 6 | 20 | 1 | KNO3 | lgK:8.09(3SF) | 5 | MGL | 11516 |
| Note (6): Gfeller Y et al., Inorg. Chim. Acta, 1978, 29, 217 | | | | | | | |
| 7 | 25 | 0.1 | KNO3 | lgK:9.00(3SF) | 5 | MGL | 1118 |
| 8 | 30 | 0.1 | KNO3 | lgK:8.97(3SF) | 5 | MGL | 1118 |
| 9 | 35 | 0.1 | KNO3 | lgK:8.91(3SF) | 5 | MGL | 1118 |
| 10 | 40 | 0.1 | KNO3 | lgK:8.87(3SF) | 5 | MGL | 1118 |

| Reaction No. 9187 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_NTA-3 + OH-1 = Sm+3_OH-1_NTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | K+1 salt | lgK:6.16(3SF) | 4 | CRV | 552 |
| 2 | 20 | 0.2 | KNO3 | lgK:6.17(0.01SD) | 4 | MGL | 1118 |
| 3 | 25 | 0.1 | K+1 salt | lgK:6.59(3SF) | 4 | CRV | 552 |

| Reaction No. 9245 |
|---|

Sm+3_NTA-3 + Citric-3 = Sm+3_Citric-3_NTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:3.(1SF)w | 3 | MPH | 1118 |

**Reaction No. 9369**

Sm+3_EDTA-4 + NTA-3 = Sm+3_NTA-3_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:5.00(3SF) | 5 | MGL | 1118 |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.2(2SF) | 1 | ESC | |
| 3 | 20 | 0.1 | KNO3 | dH:-6.23(3SF) | 5 | MCL | 1118 |

**Reaction No. 10271**

Sm+3 + H+1_3SHPropan-2 = Sm+3_H+1_3SHPropan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 2 | NaClO4 | lgK:1.74(3SF) | 6 | MGL | 3234 |
| 2 | 31 | 2 | NaClO4 | lgK:2.2(0.05SD) | 4 | MGL | 999 |
| 3 | 25 | 2 | NaClO4 | dH:2.(0.3SD) | 5 | MCL | 3234 |

**Reaction No. 10422**

Maleic-2 + Sm+3 = Sm+3_Maleic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.82(0.01SD) | 5 | MGL | 906 |
| 2 | 25 | 0.1 | NaClO4 | lgK:3.82(0.01SD) | 5 | MGL | 999 |
| 3 | 25 | 0.2 | NaClO4 | lgK:5.2(0.03SD) | 0 | MPT | 3677 |
| Note (3): Suspect other value | | | | | | | |
| 4 | 25 | 1 | NaClO4 | lgK:3.00(3SF) | 4 | MGL | 999 |

**Reaction No. 10423**

Sm+3_Maleic-2 + Maleic-2 = Sm+3_Maleic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.19(0.02SD) | 4 | MGL | 999 |

**Reaction No. 10424**

2<Maleic-2> + Sm+3 = Sm+3_Maleic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:6.01(3SF) | 3 | MGL | 11516 |
| Note (1): Roulet R et al., Helv. Chim. Acta, 1970, 53, 1876 | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 25 | 1 | NaClO4 | lgK:4.90(3SF) | 6 | MGL | 999 |

| Reaction No. 10437 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Fumaric-2 + Sm+3 = Sm+3_Fumaric-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.83(0.01SD) | 6 | MGL | 999 |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.82(0.02SD) | 6 | MGL | 3848 |

| Reaction No. 10491 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Succinic-2 = Sm+3_H+1_Succinic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaClO4 | lgK:2.00(0.10SD) | 3 | MIX | 5698 |

| Reaction No. 10545 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Diglycol-2 + Sm+3 = Sm+3_Diglycol-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 5 | 1 | NaClO4 | lgK:5.62(3SF) | 6 | MQH | 999 |
| 2 | 20 | 1 | NaClO4 | lgK:5.62(3SF) | 6 | MQH | 999 |
| 3 | 20 | 1 | NaClO4 | lgK:5.55(3SF) | 6 | MQH | 23149 |
| 4 | 25 | 0 | Inf. Dilution | lgK:5.7(2SF) | 1 | ESC | 6422 |
| 5 | 35 | 1 | NaClO4 | lgK:5.61(3SF) | 6 | MQH | 999 |
| 6 | 50 | 1 | NaClO4 | lgK:5.52(3SF) | 6 | MQH | 999 |

| Reaction No. 10546 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Diglycol-2> + Sm+3 = Sm+3_Diglycol-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 5 | 1 | NaClO4 | lgK:10.06(4SF) | 6 | MQH | 999 |
| 2 | 20 | 1 | NaClO4 | lgK:10.01(4SF) | 6 | MQH | 999 |
| 3 | 20 | 1 | NaClO4 | lgK:9.89(3SF) | 6 | MQH | 23149 |
| 4 | 25 | 0 | Inf. Dilution | lgK:10.2(3SF) | 1 | ESC | 6422 |
| 5 | 35 | 1 | NaClO4 | lgK:9.92(3SF) | 6 | MQH | 999 |
| 6 | 50 | 1 | NaClO4 | lgK:9.79(3SF) | 6 | MQH | 999 |

| Reaction No. 10547 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<Diglycol-2> + Sm+3 = Sm+3_Diglycol-2(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 5 | 1 | NaClO4 | lgK:13.09(4SF) | 4 | MQH | 999 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 20 | 1 | NaClO4 | lgK:13.03(4SF) | 4 | MQH | 999 |
| 3 | 20 | 1 | NaClO4 | lgK:12.79(4SF) | 4 | MQH | 23149 |
| 4 | 25 | 0 | Inf. Dilution | lgK:13.2(3SF) | 1 | ESC | 6422 |
| 5 | 35 | 1 | NaClO4 | lgK:12.87(4SF) | 4 | MQH | 999 |
| 6 | 50 | 1 | NaClO4 | lgK:12.63(4SF) | 4 | MQH | 999 |

Reaction No. 10601

Malic-2 + Sm+3 = Sm+3_Malic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:4.76(0.01SD) | 5 | MGL | 6670 |
| 2 | 22 | 2 | NaClO4 | lgK:3.90(3SF) | 5 | MQH | 999 |
| 3 | 25 | 0.1 | NaClO4 | lgK:4.9(0.03SD) | 4 | MGL | 999 |
| 4 | 25 | 0.2 | KCl | lgK:4.4(0.03SD)w | 4 | MPH | 999 |
| 5 | 25 | 0.2 | NaClO4 | lgK:4.5(0.03SD) | 5 | MPT | 3677 |

Reaction No. 10602

Sm+3_Malic-2 + Malic-2 = Sm+3_Malic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:3.06(3SF) | 5 | MGL | 11516 |
| Note (1): Samir A et al., Zhur. Neorg. Khim., 1980, 25, 2977 | | | | | | | |
| 2 | 22 | 2 | NaClO4 | lgK:2.58(3SF) | 6 | MQH | 999 |
| 3 | 25 | 0.1 | NaClO4 | lgK:3.3(0.06SD) | 4 | MGL | 999 |

Reaction No. 10728

IButanoic-1 + Sm+3 = Sm+3_IButanoic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:2.05(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:2.0(0.03SD) | 4 | MGL | 999 |
| 3 | 25 | 2 | NaClO4 | dH:2.7(0.08SD) | 5 | MCL | 999 |

Reaction No. 10729

2<IButanoic-1> + Sm+3 = Sm+3_IButanoic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:3.30(3SF) | 6 | MGL | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:3.25(3SF) | 5 | MGL | 11516 |
| Note (2): Choppin G et al., Inorg. Chem., 1965, 4, 1254 | | | | | | | |
| 3 | 25 | 2 | Unknown | dH:4.7(2SF) | 3 | CRV | 7271 |

| Reaction No. 10730 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_IButanoic-1 + IButanoic-1 = Sm+3_IButanoic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.5 | NaClO4 | lgK:1.25(3SF) | 5 | MGL | 11516 |
| Note (1): Stagg R et al., Inorg. Chem., 1964, 3, 242 | | | | | | | |
| 2 | 25 | 2 | NaClO4 | lgK:1.3(0.05SD) | 3 | MGL | 999 |
| 3 | 25 | 2 | NaClO4 | dH:2.1(0.2SD) | 5 | MCL | 999 |

| Reaction No. 10881 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2OH2MePropan-1 + Sm+3 = Sm+3_2OH2MePropan-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.993(4SF) | 6 | MGL | 999 |
| 2 | 25 | 0.2 | NaClO4 | lgK:2.75(0.02SD) | 4 | MGL | 999 |
| 3 | 25 | 0.5 | NaClO4 | lgK:2.63(3SF) | 6 | MGL | 999 |
| 4 | 25 | 1 | NaClO4 | lgK:2.63(3SF) | 6 | MQH | 999 |
| 5 | 25 | 2 | NaClO4 | lgK:2.75(3SF) | 6 | MGL | 2130 |

| Reaction No. 10882 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2OH2MePropan-1 + 2OH2MePropan-1 = Sm+3_2OH2MePropan-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:2.0(0.03SD) | 4 | MGL | 999 |
| 2 | 25 | 1 | NaClO4 | lgK:2.13(3SF) | 6 | MQH | 999 |
| 3 | 25 | 2 | NaClO4 | lgK:2.15(3SF) | 6 | MGL | 2130 |

| Reaction No. 10883 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2OH2MePropan-1(2) + 2OH2MePropan-1 = Sm+3_2OH2MePropan-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:1.4(0.07SD) | 3 | MGL | 999 |
| 2 | 25 | 1 | NaClO4 | lgK:1.33(3SF) | 6 | MQH | 999 |
| 3 | 25 | 2 | NaClO4 | lgK:1.58(3SF) | 6 | MGL | 2130 |

| Reaction No. 10884 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2OH2MePropan-1(3) + 2OH2MePropan-1 = Sm+3_2OH2MePropan-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:1.2(0.1SD) | 3 | MGL | 999 |
| 2 | 25 | 1 | NaClO4 | lgK:1.15(3SF) | 6 | MQH | 999 |

| Reaction No. 10885 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2OH2MePropan-1> + Sm+3 = Sm+3_2OH2MePropan-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:5.39(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.2 | NaClO4 | lgK:4.77(3SF) | 5 | MGL | 7703 |
| 3 | 25 | 0.5 | NaClO4 | lgK:4.60(3SF) | 6 | MGL | 999 |
| 4 | 25 | 1 | NaClO4 | lgK:4.76(3SF) | 5 | MQH | 7703 |
| 5 | 25 | 2 | NaClO4 | lgK:4.90(3SF) | 5 | MQH | 7703 |

| Reaction No. 10886 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<2OH2MePropan-1> + Sm+3 = Sm+3_2OH2MePropan-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:6.77(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.2 | NaClO4 | lgK:6.17(3SF) | 5 | MGL | 7703 |
| 3 | 25 | 1 | NaClO4 | lgK:6.09(3SF) | 5 | MQH | 7703 |
| 4 | 25 | 2 | NaClO4 | lgK:6.48(3SF) | 5 | MQH | 7703 |

| Reaction No. 10961 | | | | | | | |
|---|---|---|---|---|---|---|---|
| ThioDiAcet-2 + Sm+3 = Sm+3_ThioDiAcet-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:2.90(3SF) | 6 | MGL | 999 |

| Reaction No. 10962 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<ThioDiAcet-2> + Sm+3 = Sm+3_ThioDiAcet-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:4.68(3SF) | 6 | MGL | 999 |

| Reaction No. 10963 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1 + ThioDiAcet-2 = Sm+3_H+1_ThioDiAcet-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:5.59(3SF) | 6 | MGL | 999 |

| Reaction No. 10964 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1 + 2<ThioDiAcet-2> = Sm+3_H+1_ThioDiAcet-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:7.82(3SF) | 6 | MGL | 999 |

| Reaction No. 11036 | | | | | | | |
|---|---|---|---|---|---|---|---|
| NAcGly-1 + Sm+3 = Sm+3_NAcGly-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.9(0.04SD)w | 4 | MPH | 999 |

| Reaction No. 11056 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Asp-2 + Sm+3 = Sm+3_Asp-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:5.6(0.04SD) | 6 | MGL | 906 |
| 2 | 25 | 0.1 | Unknown | lgK:5.55(3SF) | 4 | CRV | 7271 |
| 3 | 25 | 0.2 | NaClO4 | lgK:6.2(0.03SD) | 0 | MPT | 3677 |
| 4 | 30 | 0.1 | NaClO4 | lgK:7.83(3SF) | 0 | MGL | 906 |
| 5 | 30 | 0.1 | NaClO4 | lgK:5.13(3SF) | 0 | MGL | 2261 |
| 6 | 40 | 0.1 | NaClO4 | lgK:9.45(3SF) | 0 | MGL | 906 |
| 7 | 50 | 0.1 | NaClO4 | lgK:10.6(3SF) | 0 | MGL | 906 |

| Reaction No. 11057 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Asp-2> + Sm+3 = Sm+3_Asp-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:9.7(0.2SD) | 3 | MGL | 999 |
| 2 | 25 | 0.1 | Unknown | lgK:9.7(2SF) | 4 | CRV | 7271 |
| 3 | 30 | 0.1 | NaClO4 | lgK:9.64(3SF) | 5 | MGL | 2261 |
| 4 | 40 | 0.1 | NaClO4 | lgK:15.85(4SF) | 0 | MGL | 11516 |
| Note (4): Trivedi C et al., J. Indian Chem. Soc., 1971, 48, 803 | | | | | | | |

| Reaction No. 11215 | | | | | | | |
|---|---|---|---|---|---|---|---|
| IDA-2 + Sm+3 = Sm+3_IDA-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:6.64(0.02SD) | 4 | MGL | 999 |
| 2 | 25 | 0.1 | KNO3 | lgK:6.6(0.1SD) | 5 | CRV | 13242 |

| Reaction No. 11216 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_IDA-2 + IDA-2 = Sm+3_IDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:5.24(0.02SD) | 4 | MGL | 999 |

| Reaction No. 11250 |
|---|

| EtThioAcet-1 + Sm+3 = Sm+3_EtThioAcet-1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 31 | 2 | NaClO4 | lgK:1.9(0.05SD) | 4 | MGL | 999 |
| 2 | 30 | 0.1 | Acetone:Water<br>75:25Wgt<br>Unknown | lgK:7.20(3SF) | 5 | MGL | 906 |

| Reaction No. 11251 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EtThioAcet-1 + EtThioAcet-1 = Sm+3_EtThioAcet-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 31 | 2 | NaClO4 | lgK:0.9(0.1SD) | 4 | MGL | 999 |
| 2 | 30 | 0.1 | Acetone:Water<br>75:25Wgt<br>Unknown | lgK:5.96(3SF) | 5 | MGL | 906 |

| Reaction No. 11339 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2Furoic-1 + Sm+3 = Sm+3_2Furoic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.9(0.2SD) | 4 | MGL | 999 |

| Reaction No. 11340 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2Furoic-1> + Sm+3 = Sm+3_2Furoic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.1(0.04SD) | 4 | MGL | 999 |

| Reaction No. 11397 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Acac-1 + Sm+3 = Sm+3_Acac-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:5.60(3SF) | 6 | ENS | 1539 |
| 2 | 25 | 2 | NaClO4 | lgK:5.3(0.05SD) | 4 | MGL | 999 |
| 3 | 30 | 0 | Inf. Dilution | lgK:5.9(2SF) | 4 | MGL | 999 |
| 4 | 30 | 0.1 | Unknown | lgK:5.59(3SF) | 6 | MGL | 999 |
| 5 | 23 | 0.005 | Methanol:Water<br>5:95Wgt<br>Unknown | lgK:5.8(0.07SD) | 4 | MGL | 906 |
| 6 | 23 | 0.005 | Methanol:Water<br>50:50Wgt<br>Unknown | lgK:6.89(0.02SD) | 4 | MGL | 906 |
| 7 | 23 | 0.005 | Methanol:Water<br>80:20Wgt<br>Unknown | lgK:8.1(0.06SD) | 4 | MGL | 906 |

| Reaction No. 11398 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Acac-1 + Acac-1 = Sm+3_Acac-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:4.40(3SF) | 6 | MGL | 999 |
| 2 | 30 | 0 | Inf. Dilution | lgK:4.5(2SF) | 4 | MGL | 999 |
| 3 | 30 | 0.1 | Unknown | lgK:4.46(3SF) | 6 | MGL | 999 |

| Reaction No. 11437 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Acac-1(2) + Acac-1 = Sm+3_Acac-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.2(2SF) | 1 | ESC | |
| 2 | 30 | 0 | Inf. Dilution | lgK:3.2(2SF) | 4 | MGL | 999 |
| 3 | 30 | 0.1 | Unknown | lgK:2.90(3SF) | 6 | MGL | 999 |

| Reaction No. 11553 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Itaconic-2 + Sm+3 = Sm+3_Itaconic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | KCl | lgK:3.10(3SF) | 0 | MGL | 11516 |
| Note (1): Makoushova G et al., Zhur. Neorg. Khim., 1989, 34, 628 | | | | | | | |
| 2 | 25 | 0.2 | NaClO4 | lgK:4.47(0.02SD) | 3 | MPT | 3677 |
| 3 | 25 | 1 | NaClO4 | lgK:2.07(3SF) | 0 | ENS | 999 |
| Note (3): Low wgt for internal consistency | | | | | | | |

| Reaction No. 11554 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1 + Itaconic-2 = Sm+3_H+1_Itaconic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:6.48(3SF) | 6 | ENS | 999 |

| Reaction No. 11555 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 2<H+1> + 2<Itaconic-2> = Sm+3_H+1(2)_Itaconic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:12.72(4SF) | 6 | ENS | 999 |

| Reaction No. 11799 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cat-2 + Sm+3 = Sm+3_Cat-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaCl | lgK:11.1(0.07SD) | 0 | MPT | 999 |
| Note (1): Low wgt for internal consistency | | | | | | | |
| 2 | 25 | 0.2 | NaClO4 | lgK:9.2(0.03SD) | 3 | MPT | 3677 |

**Reaction No. 11858**

Maltol-1 + Sm+3 = Sm+3_Maltol-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:6.51(3SF) | 6 | MGL | 999 |

**Reaction No. 11859**

Sm+3_Maltol-1 + Maltol-1 = Sm+3_Maltol-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:5.19(3SF) | 6 | MGL | 999 |

**Reaction No. 11860**

Sm+3_Maltol-1(2) + Maltol-1 = Sm+3_Maltol-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:3.73(3SF) | 6 | MGL | 999 |

**Reaction No. 11899**

Kojic-1 + Sm+3 = Sm+3_Kojic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 2 | NaClO4 | lgK:5.4(0.05SD) | 4 | MGL | 999 |
| 2 | 30 | 0.1 | Unknown | lgK:6.03(3SF) | 6 | MGL | 999 |
| 3 | 25 | 0.1 | NaClO4 | dH:-0.50(2SF) | 5 | MCL | 3231 |
| 4 | 25 | 0.1 | NaClO4 | dS:25.8(3SF) | 5 | EFE | 3231 |

**Reaction No. 11900**

Sm+3_Kojic-1 + Kojic-1 = Sm+3_Kojic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 2 | NaClO4 | lgK:5.2(0.1SD) | 4 | MGL | 999 |
| 2 | 30 | 0.1 | Unknown | lgK:5.02(3SF) | 6 | MGL | 999 |

**Reaction No. 11901**

Sm+3_Kojic-1(2) + Kojic-1 = Sm+3_Kojic-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | Unknown | lgK:3.94(3SF) | 6 | MGL | 999 |

| Reaction No. 12184 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Citric-3 + Sm+3 = Sm+3_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:9.59(3SF) | 2 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:7.99(3SF) | 0 | CRV | 31301 |
| 3 | 25 | 0.05 | NaCl | lgK:8.0(0.09SD) | 4 | MGL | 906 |
| 4 | 25 | 0.05 | NaClO4 | lgK:8.0(0.09SD) | 4 | MGL | 906 |
| 5 | 25 | 0.1 | Unknown | lgK:7.8(2SF) | 3 | CRV | 2556 |
| 6 | 25 | 0.15 | NaCl | lgK:7.38(0.03SD) | 5 | MSP | 23315 |
| 7 | 25 | 0.05 | Ethanol:Water 25:75Vol NaCl | lgK:8.8(0.05SD) | 5 | MGL | 906 |
| 8 | 25 | 0.05 | Ethanol:Water 25:75Vol NaClO4 | lgK:8.8(0.05SD) | 5 | MGL | 906 |
| 9 | 25 | 0.05 | Ethanol:Water 50:50Vol NaCl | lgK:9.7(0.06SD) | 5 | MGL | 906 |
| 10 | 25 | 0.05 | Ethanol:Water 50:50Vol NaClO4 | lgK:9.7(0.06SD) | 5 | MGL | 906 |

| Reaction No. 12245 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Adipic-2 + Sm+3 = Sm+3_Adipic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.2(0.03SD) | 4 | MGL | 1499 |
| 2 | 25 | 0.5 | Unknown | lgK:3.77(3SF) | 0 | MSL | 999 |
| 3 | 25 | 0.1 | NaClO4 | dH:15.25(0.14SD)kJ | 5 | MCL | 1499 |

| Reaction No. 12288 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Gluconic-1 + Sm+3 = Sm+3_Gluconic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:2.76(3SF) | 4 | CRV | 7271 |
| 2 | 25 | 0.2 | KCl | lgK:2.65(3SF) | 4 | MGL | 999 |
| 3 | 25 | 0.2 | NaClO4 | lgK:3.5(0.03SD) | 0 | MPT | 3677 |
| 4 | 25 | | Methanol:Water 80:20Vol | lgK:5.5(0.07SD)w | 5 | MPH | 4404 |

| Reaction No. 12289 |
|---|
| 2<Gluconic-1> + Sm+3 = Sm+3_Gluconic-1(2) |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | KCl | lgK:4.9(0.06SD) | 4 | MGL | 999 |

| Reaction No. 12379 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Picolinic-1 + Sm+3 = Sm+3_Picolinic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.97(3SF) | 6 | MGL | 3233 |
| 2 | 25 | 0.5 | NaClO4 | lgK:3.7(0.03SD) | 5 | MGL | 2650 |
| 3 | 25 | 2 | NaClO4 | lgK:3.9(0.05SD) | 4 | MGL | 999 |
| 4 | 25 | 0.5 | NaClO4 | dH:-7.32(0.59SD)kJ | 6 | MCL | 2651 |

| Reaction No. 12380 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Picolinic-1 + Picolinic-1 = Sm+3_Picolinic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.46(3SF) | 6 | MGL | 3233 |
| 2 | 25 | 2 | NaClO4 | lgK:3.0(0.08SD) | 4 | MGL | 999 |

| Reaction No. 12381 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Picolinic-1(2) + Picolinic-1 = Sm+3_Picolinic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:2.83(3SF) | 6 | MGL | 3233 |

| Reaction No. 12382 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Picolinic-1(3) + Picolinic-1 = Sm+3_Picolinic-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:2.20(3SF) | 5 | MGL | 3233 |

| Reaction No. 12434 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Picolinic-1> + Sm+3 = Sm+3_Picolinic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:7.4(0.04SD) | 4 | MGL | 999 |
| 2 | 25 | 0.5 | NaClO4 | lgK:7.0(0.04SD) | 5 | MGL | 2650 |
| 3 | 25 | 0.5 | NaClO4 | dH:-11.3(1.8SD)kJ | 6 | MCL | 2651 |

| Reaction No. 12435 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<Picolinic-1> + Sm+3 = Sm+3_Picolinic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:11.(0.3SD) | 0 | MGL | 999 |
| 2 | 25 | 0.5 | NaClO4 | lgK:9.6(0.1SD) | 5 | MGL | 2650 |
| 3 | 25 | 0.5 | NaClO4 | dH:-36.5(10.1SD)kJ | 6 | MCL | 2651 |

**Reaction No. 12451**

Nicotinic-1 + Sm+3 = Sm+3_Nicotinic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:2.07(3SF) | 6 | MGL | 999 |

**Reaction No. 12459**

3<Cupferron-1> + Sm+3 = Sm+3_Cupferron-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:14.25(4SF) | 6 | MDS | 999 |

**Reaction No. 12492**

Tiron-4 + Sm+3 = Sm+3_Tiron-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:13.92(4SF) | 0 | MGL | 999 |
| Note (1): Low wgt for internal consistency | | | | | | | |
| 2 | 25 | 0.5 | NaClO4 | lgK:12.37(0.04MD) | 5 | MGL | 3058 |

**Reaction No. 12493**

Sm+3 + H+1_Tiron-4 = Sm+3_H+1_Tiron-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:5.72(3SF) | 6 | MGL | 999 |

**Reaction No. 12565**

Sm+3 + OH-1 + NTA-3 = Sm+3_OH-1_NTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 19:21 | 0.2 | Unknown | lgK:6.17(0.01SD) | 0 | MGL | 906 |
| 2 | 25 | 0 | Inf. Dilution | lgK:19.7(3SF) | 1 | ELC | |

**Reaction No. 12580**

2<NTA-3> + Sm+3 = Sm+3_NTA-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.02 | Unknown | lgK:20.54(4SF) | 5 | MSP | 1118 |
| 2 | 25 | 0.1 | K+1 salt | lgK:20.4(0.1SD) | 6 | CRV | 552 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0.1 | KNO3 | lgK:20.53(4SF) | 5 | MGL | 11516 |
| Note (3): Moeller T et al., Inorg. Chem., 1962, 1, 55 | | | | | | | |
| 4 | 25 | 0.1 | Unknown | dH:-5.5(2SF) | 6 | CRV | 552 |

**Reaction No. 12659**

DHEGly-1 + Sm+3 = Sm+3_DHEGly-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.03 | KCl | lgK:5.7(0.04SD) | 5 | MGL | 906 |
| 2 | 20 | 0.1 | Unknown | lgK:5.72(3SF) | 6 | CRV | 552 |
| 3 | 25 | 0.1 | Unknown | lgK:5.6(2SF) | 3 | EES | 7271 |
| 4 | 30 | 0.1 | KCl | lgK:5.5(0.04SD) | 4 | MGL | 999 |
| 5 | 20 | 0.03 | Methanol:Water 20:80Wgt KCl | lgK:6.2(0.03SD) | 5 | MGL | 906 |
| 6 | 20 | 0.03 | Methanol:Water 50:50Wgt KCl | lgK:6.8(0.04SD) | 5 | MGL | 906 |
| 7 | 20 | 0.03 | Methanol:Water 80:20Wgt KCl | lgK:7.9(0.1SD) | 5 | MGL | 906 |

**Reaction No. 12660**

Sm+3_DHEGly-1 + DHEGly-1 = Sm+3_DHEGly-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | KCl | lgK:4.4(0.08SD) | 4 | MGL | 999 |

**Reaction No. 12661**

2<DHEGly-1> + Sm+3 = Sm+3_DHEGly-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:10.30(4SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | Unknown | lgK:10.2(3SF) | 3 | EES | 7271 |
| 3 | 30 | 0.1 | KCl | lgK:9.9(0.03SD) | 3 | MGL | 999 |

**Reaction No. 12763**

HIMDA-2 + Sm+3 = Sm+3_HIMDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaNO3 | lgK:8.8(2SF) | 4 | MSC | 999 |
| 2 | 20 | 1 | KNO3 | lgK:8.4(2SF) | 4 | MGL | 999 |
| 3 | 25 | 0.1 | KNO3 | lgK:9.1(0.03SD) | 4 | MGL | 999 |

| Reaction No. 12764 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HIMDA-2 + HIMDA-2 = Sm+3_HIMDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaNO3 | lgK:7.65(3SF) | 6 | MSC | 999 |
| 2 | 20 | 1 | KNO3 | lgK:7.49(3SF) | 6 | MGL | 999 |
| 3 | 25 | 0.1 | KNO3 | lgK:7.77(0.02SD) | 4 | MGL | 999 |

| Reaction No. 12765 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<HIMDA-2> + Sm+3 = Sm+3_HIMDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:18.30(4SF) | 1 | ELC | |
| 2 | 25 | 0.1 | KCl | lgK:15.74(4SF) | 0 | MPL | 999 |
| 3 | 25 | 0.1 | Unknown | lgK:16.87(4SF) | 0 | CRV | 552 |

| Reaction No. 12826 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDDA-2 + Sm+3 = Sm+3_EDDA-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:8.28(3SF) | 6 | MGL | 2028 |
| 2 | 25 | 1 | Unknown | lgK:8.26(3SF) | 6 | CRV | 552 |
| 3 | 25 | 1 | Unknown | dH:-2.9(2SF) | 6 | CRV | 552 |

| Reaction No. 12827 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDDA-2 + EDDA-2 = Sm+3_EDDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:6.07(3SF) | 6 | MGL | 2028 |

| Reaction No. 12956 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Tropolone-1 + Sm+3 = Sm+3_Tropolone-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:6.91(0.01SD) | 4 | MGL | 999 |
| 2 | 25 | 0.1 | NaClO4 | dH:-2.81(3SF) | 5 | MCL | 3231 |
| 3 | 25 | 0.1 | NaClO4 | dS:22.2(3SF) | 5 | EFE | 3231 |

| Reaction No. 12957 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Tropolone-1 + Tropolone-1 = Sm+3_Tropolone-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:5.68(0.02SD) | 6 | MGL | 999 |

**Reaction No. 12958**

Sm+3_Tropolone-1(2) + Tropolone-1 = Sm+3_Tropolone-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.60(0.02SD) | 6 | MGL | 999 |

**Reaction No. 13082**

Sm+3_HIMDA-2(2) + H2O = Sm+3_OH-1_HIMDA-2(2) + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 1 | KNO3 | lgK:10.62(4SF) | 6 | MGL | 999 |

**Reaction No. 13147**

DiEtMalon-2 + Sm+3 = Sm+3_DiEtMalon-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.328(4SF) | 6 | MGL | 4388 |

**Reaction No. 13148**

2<DiEtMalon-2> + Sm+3 = Sm+3_DiEtMalon-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:6.919(4SF) | 6 | MGL | 4388 |

**Reaction No. 13191**

35DiNitrSal-2 + Sm+3 = Sm+3_35DiNitrSal-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23:25 | 0.2 | LiCl | lgK:5.1(0.08SD) | 4 | MGL | 906 |

**Reaction No. 13294**

26PyridineDiCOO-2 + Sm+3 = Sm+3_26PyridineDiCOO-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:8.86(3SF) | 6 | MEF | 999 |

**Reaction No. 13295**

Sm+3_26PyridineDiCOO-2 + 26PyridineDiCOO-2 = Sm+3_26PyridineDiCOO-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:7.02(3SF) | 6 | MEF | 999 |

**Reaction No. 13296**

Sm+3_26PyridineDiCOO-2(2) + 26PyridineDiCOO-2 = Sm+3_26PyridineDiCOO-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:5.35(3SF) | 6 | MEF | 999 |

| Reaction No. 13370 | | | | | | | |
|---|---|---|---|---|---|---|---|
| SulfSal-3 + Sm+3 = Sm+3_SulfSal-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0 | Inf. Dilution | lgK:9.44(3SF) | 4 | MGL | 5987 |
| 2 | 20 | 0.1 | KNO3 | lgK:7.99(3SF) | 6 | MGL | 5987 |
| 3 | 20 | 0.1 | NaClO4 | lgK:7.65(3SF) | 5 | MGL | 931 |
| 4 | 20 | 1 | NaClO4 | lgK:6.8(0.04SD) | 4 | MGL | 906 |
| 5 | 25 | 1 | NaClO4 | lgK:6.339(4SF) | 6 | MGL | 5987 |
| 6 | 30 | 0.1 | KNO3 | lgK:7.75(3SF) | 6 | MGL | 5987 |
| 7 | 40 | 0.1 | KNO3 | lgK:7.37(3SF) | 6 | MGL | 5987 |
| 8 | 20 | 0.1 | KNO3 | dH:-13.1(3SF) | 6 | MTD | 5987 |

| Reaction No. 13371 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_SulfSal-3 + SulfSal-3 = Sm+3_SulfSal-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:5.93(3SF) | 5 | MGL | 931 |
| 2 | 20 | 1 | NaClO4 | lgK:5.8(0.06SD) | 4 | MGL | 906 |
| 3 | 25 | 1 | NaClO4 | lgK:5.682(4SF) | 6 | MGL | 5987 |

| Reaction No. 13483 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Phthalic-2 + Sm+3 = Sm+3_Phthalic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | NaClO4 | lgK:4.99(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.2 | Water:Ethanol 60:40Vol NaClO4 | lgK:5.0(0.05SD) | 4 | MPT | 3677 |

| Reaction No. 13484 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Phthalic-2 + Phthalic-2 = Sm+3_Phthalic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | NaClO4 | lgK:3.38(3SF) | 6 | MGL | 999 |

| Reaction No. 13507 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Mandelic-1 + Sm+3 = Sm+3_Mandelic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:2.56(0.01SD) | 6 | MGL | 2101 |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.90(0.09SD) | 5 | MGL | 6794 |
| 3 | 25 | 1 | KNO3 | lgK:2.23(0.02SD) | 6 | MGL | 2101 |
| 4 | 25 | 1 | NaClO4 | lgK:2.56(3SF) | 6 | MGL | 999 |
| 5 | 25 | 2 | NaClO4 | lgK:2.47(0.02SD) | 6 | MGL | 999 |
| 6 | 25 | 0.1 | NaClO4 | dH:-2.8(2SF)kJ | 5 | MCL | 7703 |
| 7 | 25 | 2 | NaClO4 | dH:-6.1(2SF)kJ | 5 | MCL | 7703 |

| Reaction No. 13508 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Mandelic-1 + Mandelic-1 = Sm+3_Mandelic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:1.7(0.09SD) | 6 | MGL | 2101 |
| 2 | 25 | 0.1 | NaClO4 | lgK:1.85(3SF) | 5 | MGL | 6794 |
| 3 | 25 | 1 | KNO3 | lgK:2.0(0.04SD) | 6 | MGL | 2101 |
| 4 | 25 | 1 | NaClO4 | lgK:2.01(3SF) | 6 | MGL | 999 |
| 5 | 25 | 1 | NaClO4 | lgK:1.76(3SF) | 5 | MGL | 11516 |
| Note (5): Thun H et al., J. Inorg. Nucl. Chem., 1966, 28, 1949 | | | | | | | |

| Reaction No. 13509 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Mandelic-1(2) + Mandelic-1 = Sm+3_Mandelic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.37(3SF) | 6 | MGL | 999 |

| Reaction No. 13530 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Mandelic-1(3) + Mandelic-1 = Sm+3_Mandelic-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.3(0.03SD) | 3 | MGL | 999 |

| Reaction No. 13583 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDDM-4 + Sm+3 = Sm+3_EDDM-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:12.58(4SF) | 0 | CRV | 552 |
| 2 | 25 | 0 | Inf. Dilution | lgK:13.5(3SF) | 1 | EDH | |
| 3 | 25 | 0.1 | KNO3 | lgK:10.97(4SF) | 2 | MPL | 999 |

| Reaction No. 13733 |
|---|
| Oxine-1 + Sm+3 = Sm+3_Oxine-1 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:6.84(3SF) | 6 | MDS | 999 |
| 2 | 30 | 0.3 | Dioxan:Water 50:50Wgt NaClO4 | lgK:9.05(3SF) | 4 | MGL | 906 |

| Reaction No. 13734 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<Oxine-1> + Sm+3 = Sm+3_Oxine-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:19.50(4SF) | 6 | MDS | 999 |

| Reaction No. 13816 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sulfox-2 + Sm+3 = Sm+3_Sulfox-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:6.41(3SF) | 6 | ENS | 1539 |
| 2 | 25 | 0 | Inf. Dilution | lgK:6.58(3SF) | 6 | MGL | 999 |
| 3 | 25 | 0 | Inf. Dilution | dH:-2.9(2SF) | 5 | MCL | 999 |

| Reaction No. 13817 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Sulfox-2 + Sulfox-2 = Sm+3_Sulfox-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:5.70(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0 | Inf. Dilution | dH:-3.6(2SF) | 5 | MCL | 999 |

| Reaction No. 13818 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Sulfox-2(2) + Sulfox-2 = Sm+3_Sulfox-2(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:4.76(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0 | Inf. Dilution | dH:-3.8(2SF) | 5 | MCL | 999 |

| Reaction No. 13918 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2QuinCOO-1 + Sm+3 = Sm+3_2QuinCOO-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | NaClO4 | lgK:2.64(3SF) | 6 | MGL | 1968 |

| Reaction No. 13919 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2QuinCOO-1 + 2QuinCOO-1 = Sm+3_2QuinCOO-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:2.45(3SF) | 6 | MGL | 1968 |

| Reaction No. 13953 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8QuinCOO-1 + Sm+3 = Sm+3_8QuinCOO-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | NaClO4 | lgK:2.64(3SF) | 6 | MGL | 1968 |

| Reaction No. 14050 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + EDTA-4 = Sm+3_EDTA-4(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 18:20 | 0.04 | Unknown | lgK:6.49(3SF) | 0 | MSP | 140 |
| 2 | 25 | 0.1 | KCl | lgK:3.65(3SF) | 5 | MSP | 1351 |
| 3 | 25 | 1 | KCl | lgK:3.65(3SF) | 6 | MGL | 1351 |

| Reaction No. 14113 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDDS-4 + Sm+3 = Sm+3_EDDS-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:13.77(4SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | KNO3 | lgK:13.46(0.02SD) | 6 | MPL | 999 |
| 3 | 30 | 0.1 | KNO3 | lgK:8.7(0.03SD) | 0 | MGL | 906 |

| Reaction No. 14158 | | | | | | | |
|---|---|---|---|---|---|---|---|
| HOEDTA-3 + Sm+3 = Sm+3_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 15 | 0.1 | KNO3 | lgK:15.44(4SF) | 6 | MGL | 2121 |
| 2 | 20 | 0.1 | KNO3 | lgK:15.39(4SF) | 6 | MGL | 2121 |
| 3 | 20 | 0.1 | Unknown | lgK:15.64(4SF) | 6 | MEF | 999 |
| 4 | 25 | 0.1 | KNO3 | lgK:15.28(4SF) | 6 | MGL | 2121 |
| 5 | 25 | 0.1 | KNO3 | lgK:15.17(4SF) | 5 | MGL | 8184 |
| 6 | 25 | 0.1 | Unknown | lgK:15.4(0.2SD) | 6 | CRV | 552 |
| 7 | 25 | 0.5 | NaClO4 | lgK:14.9(0.03SD) | 5 | MGL | 2650 |
| 8 | 30 | 0.1 | KNO3 | lgK:15.21(4SF) | 6 | MGL | 2121 |
| 9 | 35 | 0.1 | KNO3 | lgK:15.23(4SF) | 6 | MGL | 2121 |
| 10 | 40 | 0.1 | KNO3 | lgK:15.16(4SF) | 6 | MGL | 2121 |
| 11 | 25 | 0.1 | KNO3 | dH:-5.(0.5SD) | 6 | MCL | 2121 |
| 12 | 25 | 0.1 | Unknown | dH:-7.1(2SF) | 6 | CRV | 552 |
| 13 | 25 | 0.5 | NaClO4 | dH:-31.8(0.3SD)kJ | 6 | MCL | 2651 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 14 | 25 | 0.1 | KNO3 | dS:54.(2.SD) | 6 | EFE | 2121 |

| Reaction No. 14223 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2Thenoic-1 + Sm+3 = Sm+3_2Thenoic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.02(0.01SD) | 4 | MGL | 999 |

| Reaction No. 14224 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2Thenoic-1> + Sm+3 = Sm+3_2Thenoic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.28(0.02SD) | 4 | MGL | 999 |

| Reaction No. 14253 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_PAR-2 = Sm+3_H+1_PAR-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KCl | lgK:3.49(3SF) | 0 | MSP | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.4(2SF) | 1 | EDH | |
| 3 | 25 | 0.1 | NaNO3 | lgK:4.3(0.03SD) | 2 | MGL | 1516 |

| Reaction No. 14312 | | | | | | | |
|---|---|---|---|---|---|---|---|
| N6Me2PyridMeAsp-2 + Sm+3 = Sm+3_N6Me2PyridMeAsp-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:6.57(3SF) | 6 | MGL | 999 |

| Reaction No. 14313 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_N6Me2PyridMeAsp-2 + N6Me2PyridMeAsp-2 = Sm+3_N6Me2PyridMeAsp-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.48(3SF) | 6 | MGL | 999 |

| Reaction No. 14374 | | | | | | | |
|---|---|---|---|---|---|---|---|
| TriMDTA-4 + Sm+3 = Sm+3_TriMDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:13.14(4SF) | 6 | MIX | 1974 |
| 2 | 20 | 0.1 | KNO3 | lgK:13.21(4SF)w | 6 | MPH | 2043 |
| 3 | 20 | 0.1 | KNO3 | lgK:13.12(4SF) | 6 | MPL | 2043 |
| 4 | 20 | 0.1 | KNO3 | lgK:13.08(4SF) | 6 | MLC | 2043 |
| 5 | 25 | 0.1 | Unknown | lgK:13.17(4SF) | 6 | CRV | 552 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 20 | 0.1 | KClO4 | dH:5.52(3SF) | 5 | MIX | 1974 |
| 7 | 20 | 0.1 | KNO3 | dH:5.65(3SF) | 5 | MIX | 1974 |

**Reaction No. 14411**

MeEDTA-4 + Sm+3 = Sm+3_MeEDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:17.84(4SF) | 6 | MPL | 1978 |
| 2 | 20 | 0.1 | KNO3 | lgK:17.97(4SF) | 5 | MHG | 3272 |
| 3 | 20 | 0.1 | Unknown | lgK:18.0(0.03SD) | 6 | CRV | 552 |

**Reaction No. 14510**

dlBDTA-4 + Sm+3 = Sm+3_dlBDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.3(0.2SD) | 5 | MPL | 2005 |
| 2 | 20 | 0.1 | NaClO4 | lgK:18.22(4SF) | 6 | MSP | 1932 |

**Reaction No. 14539**

mBDTA-4 + Sm+3 = Sm+3_mBDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:16.5(0.09SD) | 6 | MGL | 2002 |
| 2 | 20 | 0.1 | NaClO4 | lgK:16.74(4SF) | 6 | MSP | 1932 |

**Reaction No. 14593**

EDDG-4 + Sm+3 = Sm+3_EDDG-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:8.30(3SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | KNO3 | lgK:7.8(0.05SD) | 4 | MGL | 999 |
| 3 | 25 | 0.1 | Unknown | lgK:8.02(3SF) | 4 | CRV | 552 |
| 4 | 30 | 0.1 | KNO3 | lgK:8.2(0.03SD) | 4 | MGL | 999 |

**Reaction No. 14695**

CDTA-4 + Sm+3 = Sm+3_CDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:19.03(4SF) | 2 | MGL | 1963 |
| 2 | 20 | 0.1 | KNO3 | lgK:18.4(0.07SD) | 3 | MGL | 2005 |
| 3 | 20 | 0.1 | Unknown | lgK:19.10(4SF) | 2 | ENS | 1539 |
| 4 | 25 | 0.1 | KNO3 | lgK:18.4(0.07SD) | 2 | MGL | 197 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5 | 25 | 0.1 | KNO3 | lgK:18.63(4SF) | 6 | MEF | 999 |
| 6 | 25 | 0.1 | KNO3 | lgK:18.6(0.04SD) | 5 | MGL | 3253 |
| 7 | 25 | 0.1 | Unknown | lgK:19.2(0.08SD) | 2 | CRV | 552 |
| 8 | 25 | 0.5 | Na+1 salt | lgK:18.61(4SF) | 0 | CRV | 552 |
| 9 | 25 | 0.5 | NaClO4 | lgK:17.9(0.04SD) | 5 | MGL | 2650 |
| 10 | 25 | 1 | K+1 salt | lgK:18.50(4SF) | 6 | CRV | 552 |
| 11 | 25 | 0.1 | KNO3 | dH:5.(0.9SD) | 3 | MCL | 3253 |
| 12 | 25 | 0.5 | NaClO4 | dH:8.58(0.33SD)kJ | 6 | MCL | 2651 |

**Reaction No. 14696**

Sm+3_CDTA-4 + H+1 = Sm+3_H+1_CDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:2.18(3SF) | 6 | MGL | 3253 |
| 2 | 25 | 0 | Inf. Dilution | lgK:2.3(2SF) | 1 | ESC | |

**Reaction No. 14754**

Sm+3_Glycolic-1 + Glycolic-1 = Sm+3_Glycolic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:2.10(3SF) | 5 | MGL | 11516 |
| Note (1): Powell J et al., Rare Earth Research II, 1964, 512 | | | | | | | |
| 2 | 20 | 2 | NaClO4 | lgK:1.97(3SF) | 5 | MQH | 11516 |
| Note (2): Sonesson A, Acta Chem. Scand., 1959, 13, 1437 | | | | | | | |
| 3 | 25 | 0.5 | NaClO4 | lgK:2.04(3SF) | 5 | MGL | 1731 |
| 4 | 25 | 2 | NaClO4 | lgK:1.94(3SF) | 6 | MGL | 2130 |

**Reaction No. 14853**

DTPA-5 + Sm+3 = Sm+3_DTPA-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:22.36(4SF) | 6 | MSP | 999 |
| 2 | 25 | 0.1 | KNO3 | lgK:22.3(0.07SD) | 3 | MEF | 3239 |
| 3 | 25 | 0.1 | Unknown | lgK:22.84(4SF) | 0 | MGL | 2143 |
| 4 | 25 | 0.5 | NaClO4 | lgK:20.7(0.07SD) | 3 | MGL | 2650 |
| 5 | 25 | 2 | NaClO4 | lgK:20.79(0.01SD) | 3 | MGL | 30513 |
| 6 | 30 | 0.1 | KNO3 | lgK:22.80(4SF) | 0 | MGL | 11516 |
| Note (6): Garg A et al., Indian J. Chem., 1976, 14A, 994 | | | | | | | |
| 7 | 25 | 0.1 | KNO3 | dH:-8.2(2SF) | 5 | MCL | 3239 |
| 8 | 25 | 0.5 | NaClO4 | dH:-44.2(0.4SD)kJ | 3 | MCL | 2651 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | 27 | 0.1 | KNO3 | dH:-8.(0.3SD) | 5 | MCL | 999 |

**Reaction No. 14864**

Sm+3 + H+1_DTPA-5 = Sm+3_H+1_DTPA-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:13.35(4SF) | 6 | MSP | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:15.8(3SF) | 2 | EAE | |

**Reaction No. 14865**

2<Sm+3> + DTPA-5 = Sm+3(2)_DTPA-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:25.47(4SF) | 6 | MSP | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:29.8(3SF) | 2 | EAE | |

**Reaction No. 14935**

EGTA-4 + Sm+3 = Sm+3_EGTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:16.88(4SF) | 6 | MEF | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:17.2(3SF) | 1 | ESC | |

**Reaction No. 14978**

TTHA-6 + Sm+3 = Sm+3_TTHA-6

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:15.7(3SF) | 0 | MGL | 906 |
| 2 | 25 | 0.1 | KNO3 | lgK:24.3(3SF) | 2 | MGL | 906 |
| 3 | 30 | 0.1 | KNO3 | lgK:19.54(4SF) | 0 | MGL | 11516 |

Note (3): Garg A et al., Indian J. Chem., 1976, 14A, 994

**Reaction No. 14979**

Sm+3_TTHA-6 + H+1 = Sm+3_H+1_TTHA-6

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.49(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.1 | KNO3 | lgK:4.5(0.2SD) | 5 | CRV | 13242 |

**Reaction No. 14980**

Sm+3_H+1_TTHA-6 + H+1 = Sm+3_H+1(2)_TTHA-6

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:2.60(3SF) | 6 | MGL | 999 |
| 2 | 25 | 0.1 | KNO3 | lgK:2.6(0.2SD) | 5 | CRV | 13242 |

**Reaction No. 14981**

2<Sm+3> + TTHA-6 = Sm+3(2)_TTHA-6

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:20.8(3SF) | 4 | MGL | 999 |

**Reaction No. 15097**

Sm+3_EDTA-4 + HOEDTA-3 = Sm+3_HOEDTA-3_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | KCl | lgK:2.49(3SF) | 6 | MGL | 1351 |

**Reaction No. 15098**

2<Sm+3_EDTA-4> + EDTA-4 = Sm+3(2)_EDTA-4(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | KCl | lgK:6.66(3SF) | 6 | MGL | 1351 |

**Reaction No. 15301**

[18]N2O4:MeCOO*2-2 + Sm+3 = Sm+3_[18]N2O4:MeCOO*2-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Me4NCl | lgK:12.12(0.04SD) | 4 | MGL | 6644 |

**Reaction No. 15321**

[18]N2O4:MeCOO-1 + Sm+3 = Sm+3_[18]N2O4:MeCOO-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Me4NCl | lgK:7.51(0.1SD) | 4 | MGL | 1335 |

**Reaction No. 15340**

[18]N2O4:EtCOO*2-2 + Sm+3 = Sm+3_[18]N2O4:EtCOO*2-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Me4NCl | lgK:7.44(0.1SD) | 4 | MGL | 1335 |

**Reaction No. 17573**

Cytidine + Sm+3 = Sm+3_Cytidine

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 35 | 0.1 | KNO3 | lgK:3.10(3SF) | 6 | MGL | 1245 |

| Reaction No. 17580 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cytidine + Sm+3 + Oxalic-2 = Sm+3_Cytidine_Oxalic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:10.4(3SF) | 1 | ESC | |
| 2 | 35 | 0.1 | KNO3 | lgK:9.98(3SF) | 6 | MGL | 1245 |

| Reaction No. 17587 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + Cytidine + H+1_Gly-1 = Sm+3_H+1_Cytidine_Gly-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 35 | 0.1 | KNO3 | lgK:8.51(3SF) | 6 | MGL | 1245 |

| Reaction No. 17594 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + Cytidine + H+1_His-1 = Sm+3_H+1_Cytidine_His-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:8.8(2SF) | 1 | ESC | |
| 2 | 35 | 0.1 | KNO3 | lgK:8.63(3SF) | 6 | MGL | 1245 |

| Reaction No. 17601 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Gly-1 = Sm+3_H+1_Gly-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 35 | 0.1 | KNO3 | lgK:3.82(3SF) | 0 | MGL | 1245 |

| Reaction No. 17607 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_His-1 = Sm+3_H+1_His-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:2.3(2SF) | 1 | ELC | |
| 2 | 35 | 0.1 | KNO3 | lgK:3.85(3SF) | 0 | MGL | 1245 |

| Reaction No. 17613 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Oxalic-2 + Sm+3 = Sm+3_Oxalic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:6.479(4SF) | 4 | CRV | 8697 |
| 2 | 25 | 0 | Inf. Dilution | lgK:6.30(3SF) | 4 | CRV | 31301 |
| 3 | 25 | 0.5 | Unknown | lgK:6.62(3SF) | 0 | MSL | 906 |
| 4 | 35 | 0.1 | KNO3 | lgK:6.61(3SF) | 3 | MGL | 1245 |

| Reaction No. 17837 |
|---|

| Sm+3_EDTA-4 + F-1 = Sm+3_F-1_EDTA-4 | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | KCl | lgK:1.68(3SF) | 6 | MGL | 6136 |
| 2 | 25 | 1 | KCl | dH:1.9(0.4SD)kJ | 5 | MCL | 6136 |

| Reaction No. 17852 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_F-1_EDTA-4 + F-1 = Sm+3_F-1(2)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | KCl | lgK:0.60(2SF) | 4 | MGL | 6136 |

| Reaction No. 18015 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_CDTA-4 = Sm+3_H+1_CDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:9.50(3SF) | 6 | ENS | 1539 |
| 2 | 25 | 0 | Inf. Dilution | lgK:9.8(2SF) | 1 | ESC | |

| Reaction No. 18076 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_EDTA-4 = Sm+3_H+1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 18:20 | 0.04 | Unknown | lgK:8.18(3SF) | 4 | MSP | 931 |
| 2 | 20 | 0.1 | Unknown | lgK:9.20(3SF) | 6 | ENS | 1539 |
| 3 | 25 | 0 | Inf. Dilution | lgK:10.6(3SF) | 2 | EAE | |

| Reaction No. 18146 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Acac-1> + Sm+3 = Sm+3_Acac-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:10.10(4SF) | 6 | ENS | 1539 |
| 2 | 25 | 0 | Inf. Dilution | lgK:10.4(3SF) | 1 | ESC | |

| Reaction No. 18147 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<Acac-1> + Sm+3 = Sm+3_Acac-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:13.00(4SF) | 6 | ENS | 1539 |
| 2 | 25 | 0 | Inf. Dilution | lgK:13.3(3SF) | 1 | ESC | |

| Reaction No. 18211 |
|---|
| 2<IDA-2> + Sm+3 = Sm+3_IDA-2(2) |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:11.88(4SF) | 6 | ENS | 1539 |

**Reaction No. 18212**

2<Sulfox-2> + Sm+3 = Sm+3_Sulfox-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:11.99(4SF) | 6 | ENS | 1539 |

**Reaction No. 18213**

3<Sulfox-2> + Sm+3 = Sm+3_Sulfox-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:16.78(4SF) | 6 | ENS | 1539 |

**Reaction No. 18253**

2<26PyridineDiCOO-2> + Sm+3 = Sm+3_26PyridineDiCOO-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:15.77(4SF) | 6 | ENS | 1539 |

**Reaction No. 18254**

3<26PyridineDiCOO-2> + Sm+3 = Sm+3_26PyridineDiCOO-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:21.06(4SF) | 6 | ENS | 1539 |

**Reaction No. 18283**

2<SulfSal-3> + Sm+3 = Sm+3_SulfSal-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:13.58(4SF) | 6 | ENS | 1539 |
| 2 | 25 | 0 | Inf. Dilution | lgK:13.9(3SF) | 1 | ESC | |

**Reaction No. 18284**

Sm+3 + H+1_SulfSal-3 = Sm+3_H+1_SulfSal-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:2.23(3SF) | 5 | MSP | 931 |
| 2 | 25 | 0 | Inf. Dilution | lgK:2.2(2SF) | 1 | ESC | |

**Reaction No. 18319**

Tartaric-2 + Sm+3 = Sm+3_Tartaric-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 0 | Inf. Dilution | lgK:5.32(3SF) | 5 | MSP | 3088 |
| 2 | 18 | 0 | Inf. Dilution | lgK:5.02(3SF) | 5 | MSP | 3088 |
| 3 | 20 | 0.1 | KNO3 | lgK:4.(0.3SD) | 3 | MPT | 999 |
| 4 | 20 | 0.1 | Unknown | lgK:3.50(3SF) | 6 | ENS | 1539 |
| 5 | 22 | 2 | NaClO4 | lgK:3.50(3SF) | 6 | MQH | 999 |
| 6 | 23:25 | 0.2 | KCl | lgK:3.5(0.05SD) | 6 | MGL | 4446 |
| 7 | 25 | 0 | Inf. Dilution | lgK:5.30(3SF) | 5 | MGL | 906 |
| 8 | 25 | 0 | Inf. Dilution | lgK:5.21(3SF) | 5 | MSP | 3088 |
| 9 | 25 | 0.05 | Unknown | lgK:4.27(0.02SD) | 5 | MGL | 906 |
| 10 | 25 | 0 | Inf. Dilution | dH:-3.(0.8SD) | 5 | MTD | 3088 |
| 11 | 25 | 0 | Ethanol:Water 25:75Wgt Inf. Dilution | lgK:6.10(3SF) | 5 | MGL | 906 |
| 12 | 25 | 0.05 | Ethanol:Water 25:75Wgt Unknown | lgK:4.8(0.1SD) | 5 | MGL | 906 |
| 13 | 25 | 0 | Ethanol:Water 40:60Wgt Inf. Dilution | lgK:6.57(3SF) | 5 | MGL | 906 |
| 14 | 25 | 0.05 | Ethanol:Water 40:60Wgt Unknown | lgK:5.1(0.2SD) | 5 | MGL | 906 |
| 15 | 25 | 0 | Ethanol:Water 50:50Wgt Inf. Dilution | lgK:7.33(3SF) | 5 | MGL | 906 |
| 16 | 25 | 0.05 | Ethanol:Water 50:50Wgt Unknown | lgK:5.6(0.1SD) | 5 | MGL | 906 |

**Reaction No. 18490**

2OHTriMDTA-4 + Sm+3 = Sm+3_2OHTriMDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:13.66(4SF) | 6 | MGL | 1492 |

**Reaction No. 18512**

NMeEDTA-3 + Sm+3 = Sm+3_NMeEDTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | K+1 salt | lgK:12.86(4SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | NaClO4 | lgK:13.00(0.02SD) | 4 | MGL | 1497 |
| 3 | 25 | 0.5 | NaClO4 | lgK:12.4(0.05SD) | 4 | MGL | 1497 |
| 4 | 25 | 0.1 | NaClO4 | dH:-8.96(0.21SD)kJ | 5 | MCL | 1497 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5 | 25 | 0.5 | NaClO4 | dH:-14.09(0.55SD)kJ | 5 | MCL | 1497 |

| Reaction No. 18558 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Succinic-2 + Sm+3 = Sm+3_Succinic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.50(0.02SD) | 0 | MGL | 1499 |
| Note (1): Low wgt for internal consistency | | | | | | | |
| 2 | 25 | 0.2 | NaClO4 | lgK:4.5(0.05SD) | 4 | MPT | 3677 |
| 3 | 25 | 0.5 | Unknown | lgK:4.96(3SF) | 4 | MSL | 999 |
| 4 | 25 | 0.1 | NaClO4 | dH:13.46(1.33SD)kJ | 4 | MCL | 1499 |

| Reaction No. 18564 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Glutaric-2 + Sm+3 = Sm+3_Glutaric-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.24(0.01SD) | 5 | MGL | 1499 |
| 2 | 25 | 0.2 | NaClO4 | lgK:4.3(0.03SD) | 0 | MPT | 3677 |
| 3 | 25 | 0.5 | Unknown | lgK:3.68(3SF) | 0 | MSL | 999 |
| 4 | 25 | 0.1 | NaClO4 | dH:14.13(0.62SD)kJ | 4 | MCL | 1499 |

| Reaction No. 18576 | | | | | | | |
|---|---|---|---|---|---|---|---|
| [Hex]14DiCOO-2 + Sm+3 = Sm+3_[Hex]14DiCOO-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:4.42(0.01SD) | 6 | MGL | 1499 |
| 2 | 25 | 0.1 | NaClO4 | dH:15.33(0.98SD)kJ | 5 | MCL | 1499 |

| Reaction No. 18849 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(2)_PO4-3 = Sm+3_H+1(2)_PO4-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:2.23(3SF) | 3 | CRV | 8781 |
| 2 | 25 | 0 | Inf. Dilution | lgK:2.35(0.20SD) | 4 | CRV | 32222 |

| Reaction No. 18953 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6BrKoj-1 + Sm+3 = Sm+3_6BrKoj-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.0053 | KCl | lgK:5.72(0.003SD) | 6 | MSP | 1526 |
| 2 | 25 | 0.1 | KCl | lgK:5.27(3SF) | 6 | MSP | 1526 |
| 3 | 25 | 0.197 | KCl | lgK:5.128(0.001SD) | 6 | MSP | 1526 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4 | 25 | 0.381 | KCl | lgK:5.008(0.002SD) | 4 | MSP | 1526 |
| 5 | 25 | 0.941 | KCl | lgK:4.856(0.001SD) | 6 | MSP | 1526 |

**Reaction No. 18967**

6IKoj-1 + Sm+3 = Sm+3_6IKoj-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:5.31(3SF) | 6 | MSP | 1526 |

**Reaction No. 19051**

2<Oxine-1> + Sm+3 = Sm+3_Oxine-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:13.3(3SF) | 4 | MDS | 999 |

**Reaction No. 19236**

Bipy + Sm+3 = Sm+3_Bipy

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaNO3 | lgK:0.9(0.1SD) | 4 | MGL | 1660 |

**Reaction No. 19544**

Sm+3 + H+1 + Malonic-2 = Sm+3_H+1_Malonic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:6.51(3SF) | 6 | MGL | 999 |

**Reaction No. 19545**

Sm+3 + H+1 + 2<Malonic-2> = Sm+3_H+1_Malonic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:9.86(3SF) | 6 | MGL | 999 |

**Reaction No. 19605**

2<Tartaric-2> + Sm+3 = Sm+3_Tartaric-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:7.3(0.2SD) | 3 | MPT | 999 |
| 2 | 25 | 2 | NaClO4 | lgK:5.54(3SF) | 5 | MEF | 11516 |

Note (2): Jercan E et al., An. Univ. Bucuresti Chim., 1969, 18, 43

**Reaction No. 19606**

Sm+3_Tartaric-2 + Tartaric-2 = Sm+3_Tartaric-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:3.80(3SF) | 3 | MEF | 11516 |
| Note (1): Dobrynina N et al., Izv. Akad. Nauk (USSR), 1969, 5, 1000 | | | | | | | |
| 2 | 22 | 2 | NaClO4 | lgK:2.04(3SF) | 6 | MQH | 999 |

| Reaction No. 19732 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + NTA-3 + EDTA-4 = Sm+3_NTA-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.9(3SF) | 1 | EDH | |
| 2 | 20 | 0.1 | KNO3 | dG:-29.71(4SF) | 5 | MCL | 999 |
| 3 | 20 | 0.1 | KNO3 | dH:-9.58(3SF) | 5 | MCL | 999 |

| Reaction No. 19930 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Malonic-2 = Sm+3_H+1_Malonic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaClO4 | lgK:1.81(0.09SD) | 3 | MIX | 5698 |
| 2 | 25 | 1 | Na+1 salt | lgK:1.53(3SF) | 3 | CRV | 7271 |
| 3 | 25 | 1 | Na+1 salt | dH:0.9(1SF) | 3 | CRV | 7271 |

| Reaction No. 20002 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_Succinic-2 + H+1_Succinic-2 = Sm+3_H+1(2)_Succinic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaClO4 | lgK:1.35(0.14SD) | 4 | MIX | 5698 |

| Reaction No. 20064 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(2)_Etbis(ImMePhos)-4 = Sm+3_H+1(2)_Etbis(ImMePhos)-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:8.43(3SF) | 6 | MGL | 999 |

| Reaction No. 20232 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(4)_EDTA-4 = Sm+3_H+1(4)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 18:20 | 0.04 | Unknown | lgK:2.58(3SF) | 4 | MSP | 931 |
| 2 | 25 | 0 | Inf. Dilution | lgK:2.58(3SF) | 2 | EAE | |

| Reaction No. 20233 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(3)_EDTA-4 = Sm+3_H+1(3)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.04 | Unknown | lgK:3.62(3SF) | 4 | MSP | 931 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.07(3SF) | 2 | EAE | |

**Reaction No. 20234**

Sm+3_EDTA-4 + H+1_EDTA-4 = Sm+3_H+1_EDTA-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.04 | Unknown | lgK:2.31(3SF) | 4 | MSP | 931 |
| 2 | 25 | 0 | Inf. Dilution | lgK:1.86(3SF) | 2 | EAE | |

**Reaction No. 21015**

Sm+3_EDTA-4 + P3O10-5 = Sm+3_EDTA-4_P3O10-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 0.1 | KNO3 | lgK:5.20(3SF) | 5 | MGL | 906 |
| 2 | 25 | 0.1 | KNO3 | lgK:5.10(3SF) | 5 | MGL | 906 |
| 3 | 35 | 0.1 | KNO3 | lgK:5.17(3SF) | 5 | MGL | 906 |
| 4 | 35 | 0.1 | KNO3 | lgK:5.17(0.04SD) | 2 | MGL | 31163 |
| 5 | 45 | 0.1 | KNO3 | lgK:5.0(0.04SD) | 5 | MGL | 906 |

**Reaction No. 21043**

5CMP-2 + Sm+3 = Sm+3_5CMP-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 35 | 0.1 | KNO3 | lgK:4.1(0.03SD) | 4 | MPT | 1760 |

**Reaction No. 21178**

Tyr-2 + Sm+3 = Sm+3_Tyr-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.71(3SF) | 6 | EIU | 904 |
| 2 | 35 | 0.1 | KNO3 | lgK:4.91(3SF) | 6 | EIU | 904 |
| 3 | 45 | 0.1 | KNO3 | lgK:5.10(3SF) | 6 | EIU | 904 |

**Reaction No. 21179**

2<Tyr-2> + Sm+3 = Sm+3_Tyr-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:8.84(3SF) | 6 | EIU | 904 |
| 2 | 35 | 0.1 | KNO3 | lgK:9.29(3SF) | 6 | EIU | 904 |
| 3 | 45 | 0.1 | KNO3 | lgK:9.72(3SF) | 6 | EIU | 904 |

| Reaction No. 21416 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_2SHPropan-2 = Sm+3_H+1_2SHPropan-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:2.06(3SF) | 6 | MGL | 3234 |
| 2 | 31 | 2 | NaClO4 | lgK:1.8(0.05SD) | 4 | MGL | 999 |
| 3 | 25 | 2 | NaClO4 | dH:0.98(0.2SD) | 5 | MCL | 3234 |

| Reaction No. 21427 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_2SHPropan-2 + H+1_2SHPropan-2 = Sm+3_H+1(2)_2SHPropan-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 31 | 2 | NaClO4 | lgK:0.9(0.1SD) | 4 | MGL | 999 |

| Reaction No. 21437 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_3SHPropan-2 + H+1_3SHPropan-2 = Sm+3_H+1(2)_3SHPropan-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 31 | 2 | NaClO4 | lgK:1.3(0.1SD) | 3 | MGL | 999 |

| Reaction No. 21704 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 12PrDiol = Sm+3_H+1(-1)_12PrDiol + H+1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 22 | 0.1 | NaClO4 | lgK:-7.55(3SF) | 6 | MGL | 2064 |

| Reaction No. 21719 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + Glycerol = Sm+3_H+1(-1)_Glycerol + H+1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 22 | 0.1 | NaClO4 | lgK:-7.25(3SF) | 6 | MGL | 2064 |
| 2 | 25 | 0.1 | NaCl | lgK:-7.25(3SF) | 6 | MGL | 999 |

| Reaction No. 22530 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + IDA-2 = Sm+3_IDA-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:4.11(3SF)w | 6 | MPH | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.1(2SF) | 1 | ESC | |
| 3 | 20 | 0.1 | KNO3 | dH:-4.26(3SF) | 5 | MCL | 999 |

| Reaction No. 22531 |
|---|
| Sm+3 + IDA-2 + EDTA-4 = Sm+3_IDA-2_EDTA-4 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:23.3(3SF) | 1 | EDH | |
| 2 | 20 | 0.1 | KNO3 | dG:-28.51(4SF) | 5 | MCL | 999 |
| 3 | 20 | 0.1 | KNO3 | dH:-7.61(3SF) | 5 | MCL | 999 |

| Reaction No. 22698 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Acac-1 = Sm+3_Acac-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:3.5(0.04SD) | 4 | MGL | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.6(2SF) | 1 | ESC | |
| 3 | 25 | 0.1 | NH4Cl | lgK:2.4(0.2SD) | 0 | MGL | 999 |

| Reaction No. 22750 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DMPA-1 + Sm+3 = Sm+3_DMPA-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.48(0.01SD) | 6 | MGL | 999 |

| Reaction No. 22751 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_DMPA-1 + DMPA-1 = Sm+3_DMPA-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.7(0.03SD) | 4 | MGL | 999 |

| Reaction No. 22752 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_DMPA-1(2) + DMPA-1 = Sm+3_DMPA-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.3(0.09SD) | 3 | MGL | 999 |

| Reaction No. 22860 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Citric-3 + Citric-3 = Sm+3_H+1_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 24:26 | 0.15 | LiClO4 | lgK:11.14(4SF) | 3 | MDS | 999 |

| Reaction No. 23216 | | | | | | | |
|---|---|---|---|---|---|---|---|
| CPTA-4 + Sm+3 = Sm+3_CPTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.2 | ClO4-1 salt | lgK:10.15(4SF) | 6 | MGL | 1929 |
| 2 | 40 | 0.2 | ClO4-1 salt | lgK:10.28(4SF) | 6 | MGL | 1929 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 50 | 0.2 | ClO4-1 salt | lgK:10.40(4SF) | 6 | MGL | 1929 |

| Reaction No. 23354 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Quinic-1 + Sm+3 = Sm+3_Quinic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:2.66(3SF) | 6 | MQH | 999 |

| Reaction No. 23355 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Quinic-1 + Quinic-1 = Sm+3_Quinic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.99(3SF) | 6 | MQH | 999 |

| Reaction No. 23356 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Quinic-1(2) + Quinic-1 = Sm+3_Quinic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.43(3SF) | 6 | MQH | 999 |

| Reaction No. 23357 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Quinic-1(3) + Quinic-1 = Sm+3_Quinic-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:0.72(2SF) | 6 | MQH | 999 |

| Reaction No. 23505 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2AmButan-1 + Sm+3 = Sm+3_2AmButan-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:5.21(0.02SD) | 4 | MGL | 1896 |
| 2 | 25 | 0.1 | KNO3 | lgK:5.26(0.02SD) | 4 | MGL | 1896 |
| 3 | 30 | 0.1 | KNO3 | lgK:5.3(0.03SD) | 4 | MGL | 1896 |
| 4 | 35 | 0.1 | KNO3 | lgK:5.38(0.02SD) | 6 | MGL | 1896 |

| Reaction No. 23605 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Sulfox-2 = Sm+3_Sulfox-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:4.35(3SF)w | 6 | MPH | 999 |
| 2 | 25 | 0 | Inf. Dilution | lgK:4.4(2SF) | 1 | ESC | |
| 3 | 20 | 0.1 | KNO3 | dH:-5.70(3SF) | 5 | MCL | 999 |

| Reaction No. 23606 |
|---|

| Sm+3 + Sulfox-2 + EDTA-4 = Sm+3_Sulfox-2_EDTA-4 | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.5(3SF) | 1 | EDH | |
| 2 | 20 | 0.1 | KNO3 | dG:-28.83(4SF) | 5 | MPH | 999 |
| 3 | 20 | 0.1 | KNO3 | dH:-9.05(3SF) | 5 | MCL | 999 |

| Reaction No. 23719 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Tiron-4 = Sm+3_EDTA-4_Tiron-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:6.7(0.07SD) | 4 | MGL | 906 |

| Reaction No. 23720 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + 5ATP-4 = Sm+3_5ATP-4_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 2 | 0.1 | KNO3 | lgK:4.8(0.1SD) | 4 | MGL | 906 |
| 2 | 25 | 0.1 | KNO3 | lgK:4.9(0.1SD) | 4 | MGL | 906 |
| 3 | 35 | 0.1 | KNO3 | lgK:4.7(0.1SD) | 4 | MGL | 906 |
| 4 | 35 | 0.1 | KNO3 | lgK:4.7(0.1SD) | 2 | MGL | 31163 |
| 5 | 45 | 0.1 | KNO3 | lgK:4.6(0.1SD) | 4 | MGL | 906 |

| Reaction No. 23804 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + IDA-2 = Sm+3_IDA-2_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.57(0.02SD) | 6 | MGL | 999 |

| Reaction No. 23805 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + HIMDA-2 = Sm+3_HIMDA-2_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.5(0.03SD) | 6 | MGL | 999 |

| Reaction No. 23806 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + EDDA-2 = Sm+3_EDDA-2_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.0(0.06SD) | 4 | MGL | 999 |

| Reaction No. 24048 |
|---|
| EEDTA-4 + Sm+3 = Sm+3_EEDTA-4 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.19(4SF) | 6 | MEF | 999 |
| 2 | 25 | 0.1 | KNO3 | lgK:18.40(4SF)w | 6 | MPH | 1946 |

**Reaction No. 24247**

PhenEDTA-4 + Sm+3 = Sm+3_PhenEDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:17.1(0.1SD) | 4 | MPL | 2010 |

**Reaction No. 24677**

Sm+3_SO4-2 + SO4-2 = Sm+3_SO4-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:1.67(3SF) | 5 | MCL | 5398 |
| 2 | 25 | 0 | Inf. Dilution | lgK:1.67(0.10SD) | 4 | MCL | 8402 |
| 3 | 25 | 2 | NaClO4 | lgK:0.6(1SF) | 3 | ESR | |
| 4 | 25 | 0 | Inf. Dilution | dH:2.1(2SF) | 5 | MCL | 5398 |
| 5 | 25 | 0 | Inf. Dilution | dH:2.1(0.3SD) | 4 | MCL | 8402 |

**Reaction No. 24690**

SO4-2 + Sm+3 = Sm+3_SO4-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:3.66(3SF) | 4 | MCN | 140 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.54(3SF) | 4 | MSP | 931 |
| 3 | 25 | 0 | Inf. Dilution | lgK:3.52(3SF) | 5 | MCL | 5398 |
| 4 | 25 | 0 | Inf. Dilution | lgK:3.68(3SF) | 4 | MCN | 5398 |
| 5 | 25 | 0 | Inf. Dilution | lgK:3.66(3SF) | 4 | MSC | 5398 |
| 6 | 25 | 0 | Inf. Dilution | lgK:3.52(0.09SD) | 4 | MCL | 8402 |
| 7 | 25 | 0 | Inf. Dilution | lgK:3.28(3SF) | 3 | CRV | 8781 |
| 8 | 25 | 0 | Inf. Dilution | lgK:3.63(0.07SD) | 5 | EOD | 22958 |
| 9 | 25 | 0 | Inf. Dilution | lgK:3.50(0.20SD) | 4 | CRV | 32222 |
| 10 | 25 | 0 | Inf. Dilution | lgK:3.67(3SF) | 4 | CRV | 32224 |
| 11 | 25 | 0.5 | Unknown | lgK:1.9(2SF) | 3 | ENS | 6496 |
| 12 | 25 | 2 | NaClO4 | lgK:1.30(3SF) | 4 | MAG | 643 |
| 13 | 25 | 0 | Inf. Dilution | dH:4.9(2SF) | 4 | CRV | 552 |
| 14 | 25 | 0 | Inf. Dilution | dH:3.70(3SF) | 0 | MCL | 5398 |
| 15 | 25 | 0 | Inf. Dilution | dH:3.42(3SF) | 0 | MCL | 5398 |
| 16 | 25 | 0 | Inf. Dilution | dH:4.34(3SF) | 4 | MCL | 5398 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 17 | 25 | 0 | Inf. Dilution | dH:4.4(2SF) | 4 | ENS | 6496 |
| 18 | 25 | 0 | Inf. Dilution | dH:3.70(0.06SD) | 3 | MCL | 8402 |
| 19 | 25 | 0 | Inf. Dilution | dH:16.575(1.916SD)kJ | 4 | CRV | 32222 |
| 20 | 25 | 2 | NaClO4 | dH:4.27(3SF) | 0 | MCL | 931 |
| 21 | 25 | 0 | D2O<br>Inf. Dilution | lgK:3.68(3SF) | 4 | MCN | 5398 |
| 22 | 25 | 0 | D2O<br>Inf. Dilution | lgK:3.68(3SF) | 4 | MSC | 5398 |
| 23 | 25 | 0 | D2O<br>Inf. Dilution | dH:3.76(3SF) | 4 | MCL | 5398 |

**Reaction No. 24760**

Sm+3 + H+1_CO3-2 = Sm+3_H+1_CO3-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:1.75(3SF) | 3 | CRV | 8781 |

**Reaction No. 25054**

Dien + Sm+3 = Sm+3_Dien

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | DMSO<br>Et4NClO4 | lgK:2.78(0.01SD) | 6 | MGL | 2221 |
| 2 | 25 | 0.1 | DMSO<br>Et4NClO4 | dG:-15.87(0.06SD)kJ | 6 | MGL | 2221 |
| 3 | 25 | 0.1 | DMSO<br>Et4NClO4 | dH:-31.1(0.07SD)kJ | 6 | MCL | 2221 |
| 4 | 25 | 0.1 | DMSO<br>Et4NClO4 | dS:-51.(2SF)J | 6 | EFE | 2221 |

**Reaction No. 25055**

2<Dien> + Sm+3 = Sm+3_Dien(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | DMSO<br>Et4NClO4 | lgK:5.40(0.02SD) | 6 | MGL | 2221 |
| 2 | 25 | 0.1 | DMSO<br>Et4NClO4 | dG:-30.8(0.01SD)kJ | 6 | MGL | 2221 |
| 3 | 25 | 0.1 | DMSO<br>Et4NClO4 | dH:-79.6(0.06SD)kJ | 6 | MCL | 2221 |
| 4 | 25 | 0.1 | DMSO<br>Et4NClO4 | dS:-164.(3SF)J | 6 | EFE | 2221 |

**Reaction No. 25159**

[2.2.2]crypt + Sm+3 = Sm+3_[2.2.2]crypt

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.25 | Me4NCl | lgK:5.9(0.06SD) | 6 | MGL | 1514 |
| 2 | 25 | 0.1 | DMF<br>NaClO4 | lgK:2.7(0.2SD) | 6 | MGL | 2277 |
| 3 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | lgK:16.0(0.2SD) | 4 | MMC | 1130 |
| 4 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | lgK:17.3(0.05SD) | 5 | MAG | 1706 |
| 5 | 25 | 0.9 | PrCarbonate<br>Et4NClO4 | lgK:16.4(0.1SD) | 4 | MMC | 1130 |
| 6 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | dH:-96.3(2.6SD)kJ | 4 | MCL | 1130 |
| 7 | 25 | 0.9 | PrCarbonate<br>Et4NClO4 | dH:-96.2(5.7SD)kJ | 4 | MCL | 1130 |

Reaction No. 25162

[2.2.1]crypt + Sm+3 = Sm+3_[2.2.1]crypt

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.25 | Me4NCl | lgK:6.76(0.02SD) | 6 | MGL | 1514 |
| 2 | 25 | 0.1 | DMF<br>NaClO4 | lgK:2.9(0.2SD) | 6 | MGL | 2277 |
| 3 | 25 | 0.1 | Methanol<br>Et4NClO4 | lgK:9.70(3SF) | 5 | MIS | 3696 |
| 4 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | lgK:19.0(3SF) | 5 | MIS | 3696 |

Reaction No. 25166

[2.1.1]crypt + Sm+3 = Sm+3_[2.1.1]crypt

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.25 | Me4NCl | lgK:6.8(0.2SD) | 6 | MGL | 1514 |
| 2 | 25 | 0.1 | DMF<br>NaClO4 | lgK:4.06(0.01SD) | 6 | MGL | 2277 |
| 3 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | lgK:15.3(0.05SD) | 5 | MAG | 1706 |

Reaction No. 25170

[18]N2O4 + Sm+3 = Sm+3_[18]N2O4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | DMF<br>NaClO4 | lgK:2.(1SF) | 3 | MGL | 2277 |
| 2 | 25 | 0.1 | PrCarbonate<br>Et4NClO4 | lgK:16.5(0.05SD) | 5 | MAG | 1706 |

**Reaction No. 25383**

Trien + Sm+3 = Sm+3_Trien

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | DMSO Et4NClO4 | lgK:4.28(0.01SD) | 6 | MGL | 2221 |
| 2 | 25 | 0.1 | DMSO Et4NClO4 | dG:-24.43(0.06SD)kJ | 6 | MGL | 2221 |
| 3 | 25 | 0.1 | DMSO Et4NClO4 | dH:-50.7(0.1SD)kJ | 6 | MCL | 2221 |
| 4 | 25 | 0.1 | DMSO Et4NClO4 | dS:-88.(2SF)J | 6 | EFE | 2221 |

**Reaction No. 25384**

2<Trien> + Sm+3 = Sm+3_Trien(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | DMSO Et4NClO4 | lgK:5.44(0.09SD) | 6 | MGL | 2221 |
| 2 | 25 | 0.1 | DMSO Et4NClO4 | dG:-31.0(0.5SD)kJ | 6 | MGL | 2221 |
| 3 | 25 | 0.1 | DMSO Et4NClO4 | dH:-89.(1.SD)kJ | 6 | MCL | 2221 |
| 4 | 25 | 0.1 | DMSO Et4NClO4 | dS:-194.(3SF)J | 6 | EFE | 2221 |

**Reaction No. 26021**

H+1 + 12BisCOOMeThioEthan-2 + Sm+3 = Sm+3_H+1_12BisCOOMeThioEthan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:5.363(4SF) | 6 | MGL | 1579 |

**Reaction No. 26022**

2<12BisCOOMeThioEthan-2> + Sm+3 = Sm+3_12BisCOOMeThioEthan-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:3.49(3SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:21.2(2.6MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26023**

12BisCOOMeThioEthan-2 + Sm+3 = Sm+3_12BisCOOMeThioEthan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:2.320(4SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:12.3(1.7MD) | 6 | MCL | 1579 |

**Reaction No. 26055**

H+1 + 2<12BisCOOMeOxEthan-2> + Sm+3 = Sm+3_H+1_12BisCOOMeOxEthan-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:10.35(4SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:-0.8(0.9MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26056**

3<12BisCOOMeOxEthan-2> + Sm+3 = Sm+3_12BisCOOMeOxEthan-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:8.95(3SF) | 6 | MGL | 1579 |

**Reaction No. 26057**

2<12BisCOOMeOxEthan-2> + Sm+3 = Sm+3_12BisCOOMeOxEthan-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:8.(0.3SD) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:7.94(0.15MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26058**

12BisCOOMeOxEthan-2 + Sm+3 = Sm+3_12BisCOOMeOxEthan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:5.07(3SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:1.58(0.07MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26081**

12BisCOOMeAmEthan-2 + Sm+3 = Sm+3_12BisCOOMeAmEthan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:8.26(3SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:-12.20(0.36MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26082**

2<12BisCOOMeAmEthan-2> + Sm+3 = Sm+3_12BisCOOMeAmEthan-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:14.44(4SF) | 6 | MGL | 1579 |
| 2 | 25 | 1 | NaClO4 | dH:25.17(0.47MD)kJ | 6 | MCL | 1579 |

**Reaction No. 26083**

2<H+1> + 12BisCOOMeAmEthan-2 + Sm+3 = Sm+3_H+1(2)_12BisCOOMeAmEthan-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:17.59(4SF) | 6 | MGL | 1579 |

| Reaction No. 26128 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DEATA-4 + Sm+3 = Sm+3_DEATA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:17.79(4SF)w | 6 | MPH | 1946 |

| Reaction No. 26151 | | | | | | | |
|---|---|---|---|---|---|---|---|
| BEATA-4 + Sm+3 = Sm+3_BEATA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:15.66(4SF)w | 6 | MPH | 1946 |

| Reaction No. 26164 | | | | | | | |
|---|---|---|---|---|---|---|---|
| TEDTA-4 + Sm+3 = Sm+3_TEDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:14.76(4SF)w | 6 | MPH | 1946 |

| Reaction No. 26395 | | | | | | | |
|---|---|---|---|---|---|---|---|
| [14]N4:Acet*4-4 + Sm+3 = Sm+3_[14]N4:Acet*4-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 80 | 1 | NaCl | lgK:15.0(0.03SD) | 5 | MGL | 1498 |

| Reaction No. 26396 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_[14]N4:Acet*4-4 + H+1 = Sm+3_H+1_[14]N4:Acet*4-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 80 | 1 | NaCl | lgK:3.9(0.1SD) | 5 | MGL | 1498 |

| Reaction No. 26422 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Ascorbic-2 = Sm+3_H+1_Ascorbic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:1.68(3SF) | 5 | MPT | 1481 |
| 2 | 25 | 2 | NaClO4 | dH:4.1(0.01SD)kJ | 4 | MCL | 1481 |

| Reaction No. 26563 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Glycolic-1(2) + Glycolic-1 = Sm+3_Glycolic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:1.46(3SF) | 6 | MGL | 2130 |

**Reaction No. 26670**

PAR-2 + Sm+3 = Sm+3_PAR-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:22 | 0.1 | NaClO4 | lgK:10.1(3SF) | 5 | MSP | 999 |
| 2 | 25 | 0.1 | NaNO3 | lgK:10.3(0.07SD) | 5 | MGL | 1516 |

**Reaction No. 26767**

Cu+2_HOEDTA-3 + H+1(3)_Tren + Sm+3 = Cu+2_Tren + Sm+3_HOEDTA-3 + 3<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 15 | 0.1 | KNO3 | lgK:-11.76(4SF) | 6 | MTD | 2121 |
| 2 | 20 | 0.1 | KNO3 | lgK:-11.48(4SF) | 6 | MTD | 2121 |
| 3 | 25 | 0.1 | KNO3 | lgK:-11.27(4SF) | 6 | MTD | 2121 |
| 4 | 30 | 0.1 | KNO3 | lgK:-11.03(4SF) | 6 | MTD | 2121 |
| 5 | 35 | 0.1 | KNO3 | lgK:-10.80(4SF) | 6 | MTD | 2121 |
| 6 | 40 | 0.1 | KNO3 | lgK:-10.57(4SF) | 6 | MTD | 2121 |
| 7 | 25 | 0.1 | KNO3 | dH:19.(0.5SD) | 6 | MCL | 2121 |

**Reaction No. 27117**

Sm+3 + H+1_TAR-2 = Sm+3_H+1_TAR-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaNO3 | lgK:4.71(3SF) | 5 | MSP | 1166 |

**Reaction No. 27118**

TAR-2 + Sm+3 = Sm+3_TAR-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaNO3 | lgK:8.10(3SF) | 5 | MSP | 1166 |

**Reaction No. 27317**

1BuEDTA-4 + Sm+3 = Sm+3_1BuEDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.2(0.1SD) | 6 | MPL | 1994 |

**Reaction No. 27376**

[Pent]11OHCOO-1 + Sm+3 = Sm+3_[Pent]11OHCOO-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.79(3SF) | 6 | MGL | 2150 |

| Reaction No. 27377 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_[Pent]11OHCOO-1 + [Pent]11OHCOO-1 = Sm+3_[Pent]11OHCOO-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.15(3SF) | 6 | MGL | 2150 |

| Reaction No. 27378 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_[Pent]11OHCOO-1(2) + [Pent]11OHCOO-1 = Sm+3_[Pent]11OHCOO-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.51(3SF) | 6 | MGL | 2150 |

| Reaction No. 27379 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_[Pent]11OHCOO-1(3) + [Pent]11OHCOO-1 = Sm+3_[Pent]11OHCOO-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.08(3SF) | 6 | MGL | 2150 |

| Reaction No. 27959 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2MeBuDioic-2 + Sm+3 = Sm+3_2MeBuDioic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.41(0.01SD) | 6 | MGL | 906 |

| Reaction No. 27960 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2MeBuDioic-2 + 2MeBuDioic-2 = Sm+3_2MeBuDioic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:1.6(0.06SD) | 6 | MGL | 906 |

| Reaction No. 28732 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2AmBenz-1 + Sm+3 = Sm+3_2AmBenz-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 12 | 0.1 | Unknown | lgK:3.38(3SF) | 4 | CRV | 2556 |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.44(0.02SD) | 4 | MGL | 6794 |
| 3 | 25 | 0.2 | Ethanol:Water 40:60Vol NaClO4 | lgK:3.4(0.05SD) | 4 | MPT | 3677 |
| 4 | 25 | 0.1 | Methanol NaCl | lgK:6.94(3SF) | 4 | MGL | 906 |

| Reaction No. 28807 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1OH4Sulf2Napthoic-3 + Sm+3 = Sm+3_1OH4Sulf2Napthoic-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:8.28(0.01SD) | 6 | MGL | 6990 |
| 2 | 25 | 0.1 | Unknown | lgK:8.50(3SF) | 5 | ENS | 233 |

**Reaction No. 28808**

2<1OH4Sulf2Napthoic-3> + Sm+3 = Sm+3_1OH4Sulf2Napthoic-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:14.31(0.02SD) | 6 | MGL | 6990 |
| 2 | 25 | 0.1 | Unknown | lgK:14.75(4SF) | 5 | ENS | 233 |

**Reaction No. 28809**

Sm+3 + H+1_1OH4Sulf2Napthoic-3 = Sm+3_H+1_1OH4Sulf2Napthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.21(3SF) | 6 | MGL | 2261 |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.43(3SF) | 6 | MGL | 5987 |
| 3 | 25 | 0.1 | NaClO4 | dH:0.55(2SF) | 6 | MCL | 5987 |

**Reaction No. 28875**

NAcAconylidOrth-2 + Sm+3 = Sm+3_NAcAconylidOrth-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:18.40(4SF) | 5 | ENS | 2261 |

**Reaction No. 28915**

2BenzNTA-3 + Sm+3 = Sm+3_2BenzNTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | K+1 salt | lgK:11.6(3SF) | 6 | CRV | 2556 |

**Reaction No. 28916**

Sm+3_2BenzNTA-3 + 2BenzNTA-3 = Sm+3_2BenzNTA-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | K+1 salt | lgK:8.86(3SF) | 4 | CRV | 2556 |

**Reaction No. 29013**

3OHPicol-1 + Sm+3 = Sm+3_3OHPicol-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:3.54(3SF) | 6 | MGL | 1968 |

**Reaction No. 29014**

2<3OHPicol-1> + Sm+3 = Sm+3_3OHPicol-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | NaClO4 | lgK:3.29(3SF) | 6 | MGL | 1968 |

| Reaction No. 29193 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Ala-1 + Sm+3 = Sm+3_Ala-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.7(0.1SD) | 3 | MGL | 2392 |
| 2 | 25 | 0.2 | NaClO4 | lgK:6.68(0.01SD) | 0 | MPT | 3677 |

| Reaction No. 30074 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Di(2MeAc)EDDA-4 + Sm+3 = Sm+3_Di(2MeAc)EDDA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:16.6(0.1SD) | 6 | MPL | 1998 |

| Reaction No. 30263 | | | | | | | |
|---|---|---|---|---|---|---|---|
| NONWAcid-2 + Sm+3 = Sm+3_NONWAcid-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:4.69(3SF) | 4 | CRV | 2556 |
| 2 | 25 | 0.1 | Unknown | lgK:3.58(3SF) | 4 | CRV | 2556 |

| Reaction No. 30291 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1NO2Naphthol-1 + Sm+3 = Sm+3_1NO2Naphthol-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:4.62(3SF) | 5 | MSP | 2261 |

| Reaction No. 30685 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EtEDTA-4 + Sm+3 = Sm+3_EtEDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:18.3(0.1SD) | 6 | MPL | 1995 |

| Reaction No. 30718 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 11DiMeEDTA-4 + Sm+3 = Sm+3_11DiMeEDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:17.04(4SF) | 6 | MPL | 1978 |

| Reaction No. 30743 | | | | | | | |
|---|---|---|---|---|---|---|---|
| HexEDTA-4 + Sm+3 = Sm+3_HexEDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.2(0.1SD) | 6 | MPT | 1981 |

**Reaction No. 30787**

CPDTA-4 + Sm+3 = Sm+3_CPDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:18.08(4SF) | 6 | CRV | 552 |
| 2 | 20 | 0.1 | Unknown | dH:-2.7(2SF) | 6 | CRV | 552 |

**Reaction No. 30818**

N2PyridMeAsp-2 + Sm+3 = Sm+3_N2PyridMeAsp-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:8.92(3SF) | 4 | MGL | 999 |

**Reaction No. 30819**

Sm+3_N2PyridMeAsp-2 + N2PyridMeAsp-2 = Sm+3_N2PyridMeAsp-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:7.96(3SF) | 4 | MGL | 999 |

**Reaction No. 31075**

Gly-1 + Sm+3 = Sm+3_Gly-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:3.5(2SF) | 5 | CRV | 2556 |
| 2 | 25 | 0.2 | NaClO4 | lgK:5.84(0.01SD) | 0 | MPT | 3677 |
| 3 | 35 | 0.1 | KNO3 | lgK:3.10(3SF) | 4 | MGL | 1245 |

**Reaction No. 31317**

Asn-1 + Sm+3 = Sm+3_Asn-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.24(3SF) | 4 | MGL | 905 |
| 2 | 30 | 0.1 | NaClO4 | lgK:3.94(3SF) | 3 | MGL | 2261 |
| 3 | 35 | 0.1 | KNO3 | lgK:4.38(3SF) | 4 | MGL | 905 |
| 4 | 45 | 0.1 | KNO3 | lgK:4.56(3SF) | 4 | MGL | 905 |

**Reaction No. 31318**

2<Asn-1> + Sm+3 = Sm+3_Asn-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:8.00(3SF) | 4 | MGL | 905 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 30 | 0.1 | NaClO4 | lgK:6.82(3SF) | 0 | MGL | 2261 |
| Note (2): Low wgt for internal consistency | | | | | | | |
| 3 | 35 | 0.1 | KNO3 | lgK:8.29(3SF) | 4 | MGL | 905 |
| 4 | 45 | 0.1 | KNO3 | lgK:8.64(3SF) | 4 | MGL | 905 |
| 5 | 35 | 0.1 | KNO3 | dH:13.05(4SF) | 4 | MTD | 905 |

**Reaction No. 31358**

HydrazineNNDiAcet-2 + Sm+3 = Sm+3_HydrazineNNDiAcet-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 60 | 0.1 | KCl | lgK:6.43(3SF) | 5 | MGL | 2261 |

**Reaction No. 31359**

2<HydrazineNNDiAcet-2> + Sm+3 = Sm+3_HydrazineNNDiAcet-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 60 | 0.1 | KCl | lgK:10.68(4SF) | 5 | MGL | 2261 |

**Reaction No. 31360**

3<HydrazineNNDiAcet-2> + Sm+3 = Sm+3_HydrazineNNDiAcet-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 60 | 0.1 | KCl | lgK:13.64(4SF) | 5 | MGL | 2261 |

**Reaction No. 32212**

Citric-3 + Sm+3 + IDA-2 = Sm+3_IDA-2_Citric-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:10.5(3SF) | 5 | MGL | 2261 |

**Reaction No. 32809**

3SHPhen2enone-1 + Sm+3 = Sm+3_3SHPhen2enone-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | Acetone:Water 3:1Vol NaClO4 | lgK:4.03(3SF) | 5 | MGL | 1878 |

**Reaction No. 32810**

Sm+3_3SHPhen2enone-1 + 3SHPhen2enone-1 = Sm+3_3SHPhen2enone-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | Acetone:Water 3:1Vol NaClO4 | lgK:3.55(3SF) | 5 | MGL | 1878 |

**Reaction No. 32811**

Sm+3_3SHPhen2enone-1(2) + 3SHPhen2enone-1 = Sm+3_3SHPhen2enone-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | Acetone:Water 3:1Vol NaClO4 | lgK:3.11(3SF) | 5 | MGL | 1878 |

**Reaction No. 32967**

2PrEDTA-4 + Sm+3 = Sm+3_2PrEDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.1(0.1SD) | 6 | MPT | 2012 |

**Reaction No. 32986**

2MePrEDTA-4 + Sm+3 = Sm+3_2MePrEDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:18.2(0.1SD) | 6 | MPT | 2012 |

**Reaction No. 33328**

3FBenz-1 + Sm+3 = Sm+3_3FBenz-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:1.95(3SF) | 5 | MGL | 2261 |
| 2 | 25 | 0.1 | NaClO4 | dH:1.46(3SF) | 5 | ENS | 2261 |

**Reaction No. 33686**

23DiOH6SulfNaphthoic-3 + Sm+3 = Sm+3_23DiOH6SulfNaphthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:9.82(0.02MD) | 5 | MGL | 3060 |

**Reaction No. 33687**

2<23DiOH6SulfNaphthoic-3> + Sm+3 = Sm+3_23DiOH6SulfNaphthoic-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:17.15(0.02MD) | 5 | MGL | 3060 |

**Reaction No. 33688**

H+1 + 23DiOH6SulfNaphthoic-3 + Sm+3 = Sm+3_H+1_23DiOH6SulfNaphthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:15.4(0.1MD) | 5 | MGL | 3060 |

| Reaction No. 33689 | | | | | | | |
|---|---|---|---|---|---|---|---|
| H+1 + 2<23DiOH6SulfNaphthoic-3> + Sm+3 = Sm+3_H+1_23DiOH6SulfNaphthoic-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.5 | NaClO4 | lgK:24.43(0.04MD) | 5 | MGL | 3060 |

| Reaction No. 33738 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Tiron-4> + Sm+3 = Sm+3_Tiron-4(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.5 | NaClO4 | lgK:20.71(0.04MD) | 5 | MGL | 3058 |

| Reaction No. 33739 | | | | | | | |
|---|---|---|---|---|---|---|---|
| H+1 + 2<Tiron-4> + Sm+3 = Sm+3_H+1_Tiron-4(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.5 | NaClO4 | lgK:28.75(0.02MD) | 5 | MGL | 3058 |

| Reaction No. 33940 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Pro-1 + Sm+3 = Sm+3_Pro-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:5.53(3SF) | 5 | MGL | 3366 |
| 2 | 25 | 0 | Inf. Dilution | lgK:6.15(3SF) | 2 | EAE | |

| Reaction No. 33956 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Hyp-1 + Sm+3 = Sm+3_Hyp-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:4.95(3SF) | 5 | MGL | 3366 |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.57(3SF) | 2 | EAE | |

| Reaction No. 34113 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_CDTA-4 + Norleu-1 = Sm+3_Norleu-1_CDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 22 | 0.1 | NaClO4 | lgK:9.79(3SF) | 5 | MGL | 3341 |
| 2 | 25 | 0 | Inf. Dilution | lgK:10.0(3SF) | 1 | ESC | |

| Reaction No. 34122 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Norleu-1 = Sm+3_Norleu-1 + H+1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 22 | 0.1 | NaClO4 | lgK:5.06(3SF) | 5 | MGL | 3341 |

Reaction No. 34134

Sm+3_Norleu-1 + H+1_Norleu-1 = Sm+3_Norleu-1(2) + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 0.1 | NaClO4 | lgK:4.87(3SF) | 5 | MGL | 3341 |

Reaction No. 34135

Sm+3_Norleu-1(2) + H+1_Norleu-1 = Sm+3_Norleu-1(3) + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 0.1 | NaClO4 | lgK:3.63(3SF) | 5 | MGL | 3341 |

Reaction No. 34324

Sm+3 + H+1(3)_HexMDTA-4 = Sm+3_H+1(3)_HexMDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:1.52(3SF) | 5 | MSP | 4428 |

Reaction No. 34325

2<Sm+3> + H+1_HexMDTA-4 + H+1(2)_HexMDTA-4 = Sm+3(2)_H+1(3)_HexMDTA-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:11.32(4SF) | 4 | MSP | 4428 |

Reaction No. 34326

Sm+3(2)_H+1(3)_HexMDTA-4(2) + H+1_HexMDTA-4 = Sm+3(2)_H+1(4)_HexMDTA-4(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:4.61(3SF) | 4 | MSP | 4428 |

Reaction No. 34327

Sm+3(2)_H+1(3)_HexMDTA-4(2) + 2<Acetic-1> = Sm+3(2)_H+1(3)_Acetic-1(2)_HexMDTA-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:2.00(3SF) | 4 | MSP | 4428 |

Reaction No. 34328

Sm+3(2)_H+1(3)_Acetic-1(3)_HexMDTA-4(2) + H+1_HexMDTA-4 = Sm+3(2)_H+1(4)_Acetic-1(3)_HexMDTA-4(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:4.06(3SF) | 4 | MSP | 4428 |

Reaction No. 34329

Sm+3(2)_H+1(2)_Acetic-1(3)_HexMDTA-4(2) + H+1_HexMDTA-4 = Sm+3(2)_H+1(3)_Acetic-1(3)_HexMDTA-4(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:7.15(3SF) | 4 | MSP | 4428 |

**Reaction No. 34330**

Sm+3(2)_H+1(3)_HexMDTA-4(2) + 3<Acetic-1> = Sm+3(2)_H+1(3)_Acetic-1(3)_HexMDTA-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 18:20 | 0.1 | Unknown | lgK:2.78(3SF) | 4 | MSP | 4428 |

**Reaction No. 34525**

NTA-3 + Sm+3 + Tropolone-1 = Sm+3_Tropolone-1_NTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 30 | 0.1 | Dioxan:Water 50:50Vol NaClO4 | lgK:5.28(3SF) | 5 | MGL | 3407 |

**Reaction No. 34796**

PAN-1 + Sm+3 = Sm+3_PAN-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 21 | 0.1 | Ethanol:Water 50:50Vol NaClO4 | lgK:10.02(4SF) | 5 | MPS | 3410 |
| 2 | 21 | 0.1 | Ethanol:Water 75:25Vol NaClO4 | lgK:11.27(4SF) | 5 | MPS | 3410 |

**Reaction No. 34797**

Sm+3 + H+1_PAN-1 = Sm+3_PAN-1 + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 21 | 0.1 | Ethanol:Water 50:50Vol NaClO4 | lgK:2.75(3SF) | 5 | MPS | 3410 |
| 2 | 21 | 0.1 | Ethanol:Water 75:25Vol NaClO4 | lgK:2.13(3SF) | 5 | MPS | 3410 |

**Reaction No. 34912**

Sm+3_EDTA-4 + Malic-2 = Sm+3_Malic-2_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:4.11(0.01SD) | 4 | MPT | 3677 |

**Reaction No. 34917**

Sm+3_EDTA-4 + Lactic-1 = Sm+3_Lactic-1_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:3.86(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 34922 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Glycolic-1 = Sm+3_Glycolic-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:3.72(0.02SD) | 3 | MPT | 3677 |

| Reaction No. 34928 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Gluconic-1 = Sm+3_Gluconic-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:2.9(0.04SD) | 3 | MPT | 3677 |

| Reaction No. 34933 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Malonic-2 = Sm+3_Malonic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:3.70(0.01SD) | 4 | MPT | 3677 |

| Reaction No. 34940 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Succinic-2 = Sm+3_Succinic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.0(0.03SD) | 5 | MPT | 3677 |

| Reaction No. 34947 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Glutaric-2 = Sm+3_Glutaric-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:3.51(0.01SD) | 4 | MPT | 3677 |

| Reaction No. 34953 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Maleic-2 = Sm+3_Maleic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.6(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 34962 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Citraconic-2 + Sm+3 = Sm+3_Citraconic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:5.3(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 34963 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Citraconic-2 = Sm+3_Citraconic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.5(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 34971 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Itaconic-2 = Sm+3_Itaconic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.32(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 34980 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Crotonic-1 + Sm+3 = Sm+3_Crotonic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.0(0.04SD) | 4 | MPT | 3677 |

| Reaction No. 34981 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Crotonic-1 = Sm+3_Crotonic-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:3.6(0.04SD) | 4 | MPT | 3677 |

| Reaction No. 34990 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Salicylic-2 + Sm+3 = Sm+3_Salicylic-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.06(3SF) | 5 | MGL | 2261 |
| 2 | 25 | 0.25 | Acetone:Water<br>50:50Vol<br>NaClO4 | lgK:2.57(3SF) | 4 | MCE | 1614 |
| 3 | 25 | 0.2 | Ethanol:Water<br>40:60Vol<br>NaClO4 | lgK:8.0(0.05SD) | 6 | MPT | 3677 |
| 4 | 25 | 0.1 | Methanol:Water<br>99.9:.1Wgt<br>NaCl | lgK:5.75(3SF) | 5 | MGL | 906 |

| Reaction No. 34991 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Salicylic-2 = Sm+3_Salicylic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | Ethanol:Water<br>40:60Vol<br>NaClO4 | lgK:7.7(0.1SD) | 4 | MPT | 3677 |

| Reaction No. 34997 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Phthalic-2 = Sm+3_Phthalic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | Ethanol:Water 40:60Vol NaClO4 | lgK:4.2(0.04SD) | 4 | MPT | 3677 |

| Reaction No. 35003 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Cat-2 = Sm+3_Cat-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:7.5(0.05SD) | 4 | MPT | 3677 |

| Reaction No. 35012 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Resorcinol-2 + Sm+3 = Sm+3_Resorcinol-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:6.0(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 35013 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Resorcinol-2 = Sm+3_Resorcinol-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:3.0(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 35022 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 34DHB-3 + Sm+3 = Sm+3_34DHB-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:8.63(0.01SD) | 4 | MPT | 3677 |

| Reaction No. 35023 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + 34DHB-3 = Sm+3_34DHB-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:5.07(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 35032 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 24DHB-3 + Sm+3 = Sm+3_24DHB-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:6.58(0.02SD) | 6 | MPT | 3677 |

| Reaction No. 35033 |
|---|
| Sm+3_EDTA-4 + 24DHB-3 = Sm+3_24DHB-3_EDTA-4 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:4.5(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 35038 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Gly-1 = Sm+3_Gly-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.97(0.02SD) | 3 | MPT | 3677 |

| Reaction No. 35044 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Ala-1 = Sm+3_Ala-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:5.37(0.01SD) | 3 | MPT | 3677 |

| Reaction No. 35049 | | | | | | | |
|---|---|---|---|---|---|---|---|
| bAla-1 + Sm+3 = Sm+3_bAla-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:6.40(0.01SD) | 0 | MPT | 3677 |

| Reaction No. 35050 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + bAla-1 = Sm+3_bAla-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:5.1(2SF) | 1 | EAE | |
| 2 | 25 | 0.2 | NaClO4 | lgK:5.0(0.1SD) | 3 | MPT | 3677 |

| Reaction No. 35059 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Val-1 + Sm+3 = Sm+3_Val-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:3.90(3SF) | 0 | MGL | 11516 |
| Note (1): Batyaev I et al., Zhur. Neorg. Khim., 1974, 19, 670 | | | | | | | |
| 2 | 25 | 0.2 | NaClO4 | lgK:6.68(0.02SD) | 3 | MPT | 3677 |

| Reaction No. 35060 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Val-1 = Sm+3_Val-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:6.05(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 35069 |
|---|
| Leu-1 + Sm+3 = Sm+3_Leu-1 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:5.09(3SF) | 0 | MGL | 2392 |
| 2 | 25 | 0.2 | NaClO4 | lgK:6.2(0.03SD) | 0 | MPT | 3677 |

**Reaction No. 35070**

Sm+3_EDTA-4 + Leu-1 = Sm+3_Leu-1_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:5.1(0.03SD) | 3 | MPT | 3677 |

**Reaction No. 35075**

Sm+3_EDTA-4 + Asp-2 = Sm+3_Asp-2_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:5.37(0.02SD) | 3 | MPT | 3677 |

**Reaction No. 35084**

2SHSuccin-3 + Sm+3 = Sm+3_2SHSuccin-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:4.83(0.01SD) | 4 | MPT | 3677 |

**Reaction No. 35085**

Sm+3_EDTA-4 + 2SHSuccin-3 = Sm+3_2SHSuccin-3_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:4.6(0.04SD) | 4 | MPT | 3677 |

**Reaction No. 35094**

2SHAcet-2 + Sm+3 = Sm+3_2SHAcet-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:3.7(0.1SD) | 4 | MPT | 3677 |

**Reaction No. 35095**

Sm+3_EDTA-4 + 2SHAcet-2 = Sm+3_2SHAcet-2_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:3.7(0.03SD) | 4 | MPT | 3677 |

**Reaction No. 35100**

Sm+3_EDTA-4 + 2AmBenz-1 = Sm+3_2AmBenz-1_EDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | Ethanol:Water 40:60Vol NaClO4 | lgK:3.2(0.03SD) | 4 | MPT | 3677 |

| Reaction No. 35169 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DiPrMalon-2 + Sm+3 = Sm+3_DiPrMalon-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.452(4SF) | 6 | MGL | 4388 |

| Reaction No. 35170 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<DiPrMalon-2> + Sm+3 = Sm+3_DiPrMalon-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:7.342(4SF) | 6 | MGL | 4388 |

| Reaction No. 35195 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DiBuMalon-2 + Sm+3 = Sm+3_DiBuMalon-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:4.433(4SF) | 6 | MGL | 4388 |

| Reaction No. 35336 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Phloroglucinol-3 + Sm+3 = Sm+3_Phloroglucinol-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:4.20(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 35337 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + Phloroglucinol-3 = Sm+3_Phloroglucinol-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | NaClO4 | lgK:2.72(0.02SD) | 4 | MPT | 3677 |

| Reaction No. 35939 | | | | | | | |
|---|---|---|---|---|---|---|---|
| AcHydroxam-1 + Sm+3 = Sm+3_AcHydroxam-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaNO3 | lgK:5.96(3SF) | 5 | MGL | 3676 |

| Reaction No. 35940 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<AcHydroxam-1> + Sm+3 = Sm+3_AcHydroxam-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaNO3 | lgK:10.73(4SF) | 5 | MGL | 3676 |

| Reaction No. 35941 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<AcHydroxam-1> + Sm+3 = Sm+3_AcHydroxam-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaNO3 | lgK:14.41(4SF) | 5 | MGL | 3676 |

| Reaction No. 36177 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Dimedone-1 + Sm+3 = Sm+3_Dimedone-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | Acetone:Water 3:1Vol Unknown | lgK:2.86(3SF) | 5 | MPT | 3184 |

| Reaction No. 36178 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Dimedone-1 + Dimedone-1 = Sm+3_Dimedone-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | Acetone:Water 3:1Vol Unknown | lgK:2.59(3SF) | 5 | MPT | 3184 |

| Reaction No. 36216 | | | | | | | |
|---|---|---|---|---|---|---|---|
| BenzHydroxam-1 + Sm+3 = Sm+3_BenzHydroxam-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 35 | 0.1 | Dioxan:Water 50:50Wgt KOH | lgK:9.95(3SF) | 5 | MGL | 3198 |

| Reaction No. 36217 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_BenzHydroxam-1 + BenzHydroxam-1 = Sm+3_BenzHydroxam-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 35 | 0.1 | Dioxan:Water 50:50Wgt KOH | lgK:8.46(3SF) | 5 | MGL | 3198 |

| Reaction No. 36218 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_BenzHydroxam-1(2) + BenzHydroxam-1 = Sm+3_BenzHydroxam-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 35 | 0.1 | Dioxan:Water 50:50Wgt KOH | lgK:7.46(3SF) | 5 | MGL | 3198 |

| Reaction No. 36286 |
|---|
| Croconic-2 + Sm+3 = Sm+3_Croconic-2 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.083(4SF) | 5 | MPT | 3231 |
| 2 | 25 | 0.1 | NaClO4 | dH:0.86(2SF) | 5 | MCL | 3231 |
| 3 | 25 | 0.1 | NaClO4 | dS:17.1(3SF) | 5 | EFE | 3231 |

**Reaction No. 36287**

2<Croconic-2> + Sm+3 = Sm+3_Croconic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.832(4SF) | 3 | MPT | 3231 |
| 2 | 25 | 0.1 | NaClO4 | dH:-0.81(2SF) | 3 | MCL | 3231 |

**Reaction No. 36480**

oAnisic-1 + Sm+3 = Sm+3_oAnisic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.10(0.02SD) | 5 | MGL | 3914 |
| 2 | 25 | 0.1 | NaClO4 | lgK:2.3(0.03SD) | 5 | MGL | 6794 |
| 3 | 25 | 0.1 | NaClO4 | dH:6.85(1.1SD)kJ | 5 | MCL | 3914 |

**Reaction No. 36494**

mAnisic-1 + Sm+3 = Sm+3_mAnisic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.2(0.05SD) | 5 | MGL | 3914 |
| 2 | 25 | 0.1 | NaClO4 | dH:8.25(0.6SD)kJ | 5 | MCL | 3914 |

**Reaction No. 36508**

pAnisic-1 + Sm+3 = Sm+3_pAnisic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.2(0.03SD) | 5 | MGL | 3914 |
| 2 | 25 | 0.1 | NaClO4 | dH:9.01(1.6SD)kJ | 5 | MCL | 3914 |

**Reaction No. 36641**

[18]N6 + Sm+3 = Sm+3_[18]N6

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaCl | lgK:10.1(0.1SD)w | 4 | MPH | 3895 |

**Reaction No. 36648**

3OHBenz-2 + Sm+3 = Sm+3_3OHBenz-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.2(0.03SD) | 6 | MGL | 3893 |
| 2 | 25 | 0.1 | NaClO4 | dG:-2.925(4SF) | 6 | EFE | 3893 |
| 3 | 25 | 0.1 | NaClO4 | dH:1.752(4SF) | 6 | MCL | 3893 |
| 4 | 25 | 0.1 | NaClO4 | dS:15.7(3SF) | 6 | EFE | 3893 |

**Reaction No. 36661**

4OHBenz-2 + Sm+3 = Sm+3_4OHBenz-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.42(0.02SD) | 4 | MGL | 3893 |
| 2 | 25 | 0.1 | NaClO4 | dG:-2.393(4SF) | 0 | EFE | 3893 |
| 3 | 25 | 0.1 | NaClO4 | dH:1.809(4SF) | 5 | MCL | 3893 |

**Reaction No. 36878**

Sm+3 + H+1_1OH47DiSulf2Naphthoic-4 = Sm+3_H+1_1OH47DiSulf2Naphthoic-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.3(0.05SD) | 6 | MGL | 3849 |
| 2 | 25 | 0.1 | NaClO4 | dH:0.693(3SF) | 6 | MCL | 3849 |

**Reaction No. 36879**

1OH47DiSulf2Naphthoic-4 + Sm+3 = Sm+3_1OH47DiSulf2Naphthoic-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:8.5(0.05SD) | 6 | MGL | 3849 |

**Reaction No. 36880**

2<1OH47DiSulf2Naphthoic-4> + Sm+3 = Sm+3_1OH47DiSulf2Naphthoic-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:14.2(0.2SD) | 6 | MGL | 3849 |

**Reaction No. 37097**

Sm+3 + H+1_Etbis(ImMePhos)-4 = Sm+3_H+1_Etbis(ImMePhos)-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:8.43(3SF) | 3 | MGL | 931 |

**Reaction No. 37256**

Sm+3 + H+1_Aluminon-5 = Sm+3_H+1_Aluminon-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | | | lgK:4.5(0.1SD) | 3 | MSP | 931 |

**Reaction No. 37444**

Benzoylacetone-1 + Sm+3 = Sm+3_Benzoylacetone-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Methanol:Water 80:20Wgt NaCl | lgK:8.03(0.02SD) | 5 | MGL | 931 |

**Reaction No. 37445**

Sm+3_Benzoylacetone-1 + Benzoylacetone-1 = Sm+3_Benzoylacetone-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Methanol:Water 80:20Wgt NaCl | lgK:6.28(0.02SD) | 5 | MGL | 931 |

**Reaction No. 37446**

Sm+3_Benzoylacetone-1(2) + Benzoylacetone-1 = Sm+3_Benzoylacetone-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Methanol:Water 80:20Wgt NaCl | lgK:4.30(0.02SD) | 5 | MGL | 931 |

**Reaction No. 37836**

TetMDTA-4 + Sm+3 = Sm+3_TetMDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:9.5(0.03SD) | 6 | MGL | 3938 |
| 2 | 25 | 0.5 | NaClO4 | dH:27.18(0.63SD)kJ | 5 | MCL | 3938 |

**Reaction No. 37837**

Sm+3_TetMDTA-4 + H+1 = Sm+3_H+1_TetMDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:6.9(0.04SD) | 6 | MGL | 3938 |

**Reaction No. 37838**

Sm+3_H+1_TetMDTA-4 + H+1 = Sm+3_H+1(2)_TetMDTA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:5.5(0.08SD) | 6 | MGL | 3938 |

**Reaction No. 37914**

[12]N4:Acet*4-4 + Sm+3 = Sm+3_[12]N4:Acet*4-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.1 | NaCl | lgK:23.04(4SF) | 4 | MSP | 1377 |
| 2 | 37 | 1 | NaCl | lgK:23.3(3SF) | 6 | MMC | 932 |

**Reaction No. 38034**

H+1 + Fumaric-2 + Sm+3 = Sm+3_H+1_Fumaric-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:6.2(0.04SD) | 6 | MGL | 3848 |

**Reaction No. 38164**

Squaric-2 + Sm+3 = Sm+3_Squaric-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | dH:2.00(3SF) | 5 | MCL | 3231 |
| 2 | 25 | 0.1 | NaClO4 | dS:19.6(3SF) | 5 | EFE | 3231 |

**Reaction No. 38189**

2<Salicylic-2> + Sm+3 = Sm+3_Salicylic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.82(3SF) | 5 | MGL | 2261 |
| 2 | 23 | 0.25 | Acetone:Water 50:50Vol NaClO4 | lgK:4.33(0.02SD) | 4 | MCE | 1614 |

**Reaction No. 38190**

3<Salicylic-2> + Sm+3 = Sm+3_Salicylic-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.25 | Acetone:Water 50:50Vol NaClO4 | lgK:6.(0.6SD) | 4 | MCE | 1614 |

**Reaction No. 38239**

EDDPrDA-4 + Sm+3 = Sm+3_EDDPrDA-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:12.86(0.01SD) | 5 | MGL | 3915 |
| 2 | 25 | 0.5 | NaClO4 | lgK:11.54(0.01SD) | 5 | MGL | 3915 |
| 3 | 25 | 0.1 | NaClO4 | dH:1.85(0.90SD)kJ | 4 | MCL | 3915 |
| 4 | 25 | 0.5 | NaClO4 | dH:-0.63(1.04SD)kJ | 4 | MCL | 3915 |

| Reaction No. 38240 | | | | | | | |
|---|---|---|---|---|---|---|---|
| H+1 + EDDPrDA-4 + Sm+3 = Sm+3_H+1_EDDPrDA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:16.15(0.02SD) | 5 | MGL | 3915 |
| 2 | 25 | 0.5 | NaClO4 | lgK:15.2(0.03SD) | 5 | MGL | 3915 |

| Reaction No. 38241 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_EDDPrDA-4 = Sm+3_H+1_EDDPrDA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:6.2(0.03SD) | 5 | MGL | 3915 |
| 2 | 25 | 0.5 | NaClO4 | lgK:5.2(0.04SD) | 5 | MGL | 3915 |
| 3 | 25 | 0.1 | NaClO4 | dH:32.38(1.60SD)kJ | 4 | MCL | 3915 |
| 4 | 25 | 0.5 | NaClO4 | dH:33.72(0.98SD)kJ | 4 | MCL | 3915 |

| Reaction No. 39199 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_HydrazineNNDiAcet-2 = Sm+3_H+1_HydrazineNNDiAcet-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:4.1(2SF)w | 4 | MPH | 4248 |

| Reaction No. 39200 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_HydrazineNNDiAcet-2 + H+1_HydrazineNNDiAcet-2 = Sm+3_H+1(2)_HydrazineNNDiAcet-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:2.9(2SF)w | 4 | MPH | 4248 |

| Reaction No. 39201 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(2)_HydrazineNNDiAcet-2(2) + H+1_HydrazineNNDiAcet-2 = Sm+3_H+1(3)_HydrazineNNDiAcet-2(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:0.1(1SF)w | 4 | MPH | 4248 |

| Reaction No. 39254 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Xylitol + Sm+3 = Sm+3_Xylitol | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:8.1(0.3SD) | 4 | MCL | 4231 |
| 2 | 25 | 0 | Inf. Dilution | dG:-5.2(0.1SD)kJ | 0 | EFE | 4231 |
| 3 | 25 | 0 | Inf. Dilution | dH:-13.7(0.5SD)kJ | 4 | MCL | 4231 |

| Reaction No. 39760 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(3)_MeThymolBlue-6 = Sm+3_H+1(3)_MeThymolBlue-6 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:4.2(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | ClO4-1 salt | lgK:4.2(0.03SD) | 4 | MGL | 1852 |

| Reaction No. 39761 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(2)_MeThymolBlue-6 = Sm+3_H+1(2)_MeThymolBlue-6 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:6.6(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | ClO4-1 salt | lgK:6.5(0.03SD) | 4 | MGL | 1852 |

| Reaction No. 39762 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(3)_MeThymolBlue-6 = Sm+3_H+1(2)_MeThymolBlue-6 + H+1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:-4.8(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | ClO4-1 salt | lgK:-4.9(0.03SD) | 4 | MGL | 1852 |

| Reaction No. 39763 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_MeThymolBlue-6 + OH-1 = Sm+3_OH-1_MeThymolBlue-6 + H+1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.1(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | ClO4-1 salt | lgK:3.1(0.03SD) | 4 | MGL | 1852 |

| Reaction No. 39899 | | | | | | | |
|---|---|---|---|---|---|---|---|
| [14]N2O3:MeCOO*2-2 + Sm+3 = Sm+3_[14]N2O3:MeCOO*2-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Me4NCl | lgK:11.7(0.09SD) | 5 | MGL | 1491 |

| Reaction No. 40041 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Tdoha + Sm+3 = Sm+3_Tdoha | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:9.5(0.05SD) | 5 | MAG | 1706 |

| Reaction No. 40042 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Tdoha> + Sm+3 = Sm+3_Tdoha(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:17.6(0.05SD) | 5 | MAG | 1706 |

| Reaction No. 40256 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Atrolactic-1 + Sm+3 = Sm+3_Atrolactic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.89(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:2.57(0.01SD) | 5 | MGL | 931 |
| 3 | 25 | 0.1 | NaClO4 | dH:-3.1(2SF)kJ | 5 | MCL | 7703 |

| Reaction No. 40257 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Atrolactic-1 + Atrolactic-1 = Sm+3_Atrolactic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.89(0.02SD) | 5 | MGL | 931 |

| Reaction No. 40258 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Atrolactic-1(2) + Atrolactic-1 = Sm+3_Atrolactic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.54(0.02SD) | 5 | MGL | 931 |

| Reaction No. 40259 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Atrolactic-1(3) + Atrolactic-1 = Sm+3_Atrolactic-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.31(0.02SD) | 5 | MGL | 931 |

| Reaction No. 40364 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Glyoxylic-1 + Sm+3 = Sm+3_Glyoxylic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.55(3SF) | 5 | MGL | 931 |

| Reaction No. 40365 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Glyoxylic-1 + Glyoxylic-1 = Sm+3_Glyoxylic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.04(3SF) | 5 | MGL | 931 |

| Reaction No. 40366 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Glyoxylic-1(2) + Glyoxylic-1 = Sm+3_Glyoxylic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:1.5(2SF) | 5 | MGL | 931 |

| Reaction No. 40674 | | | | | | | |
|---|---|---|---|---|---|---|---|
| MeoxAcet-1 + Sm+3 = Sm+3_MeoxAcet-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:2.13(3SF) | 5 | MGL | 931 |

| Reaction No. 40675 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_MeoxAcet-1 + MeoxAcet-1 = Sm+3_MeoxAcet-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:1.26(3SF) | 5 | MGL | 931 |

| Reaction No. 40717 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 26DiCOOPiperNAcet-3 + Sm+3 = Sm+3_26DiCOOPiperNAcet-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:10.6(0.05SD) | 6 | MGL | 931 |

| Reaction No. 40718 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_26DiCOOPiperNAcet-3 + 26DiCOOPiperNAcet-3 = Sm+3_26DiCOOPiperNAcet-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:7.8(0.05SD) | 6 | MGL | 931 |

| Reaction No. 40814 | | | | | | | |
|---|---|---|---|---|---|---|---|
| IBuMeGlycollic-1 + Sm+3 = Sm+3_IBuMeGlycollic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:2.71(0.02SD) | 5 | MQH | 931 |

| Reaction No. 40815 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_IBuMeGlycollic-1 + IBuMeGlycollic-1 = Sm+3_IBuMeGlycollic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:2.2(0.03SD) | 5 | MQH | 931 |

| Reaction No. 40816 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_IBuMeGlycollic-1(2) + IBuMeGlycollic-1 = Sm+3_IBuMeGlycollic-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | NaClO4 | lgK:1.5(0.04SD) | 5 | MQH | 931 |

| Reaction No. 40817 |
|---|

Sm+3_IBuMeGlycollic-1(3) + IBuMeGlycollic-1 = Sm+3_IBuMeGlycollic-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:1.0(0.04SD) | 5 | MQH | 931 |

**Reaction No. 40905**

234TriOHPentDioic-2 + Sm+3 = Sm+3_234TriOHPentDioic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23:25 | 0.2 | KCl | lgK:3.81(0.02SD) | 5 | MGL | 4446 |

**Reaction No. 40928**

2OHPentan-1 + Sm+3 = Sm+3_2OHPentan-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:2.40(0.02SD) | 5 | MGL | 931 |

**Reaction No. 41050**

23DiOH2OHMePropan-1 + Sm+3 = Sm+3_23DiOH2OHMePropan-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:2.86(3SF) | 5 | MGL | 931 |

**Reaction No. 41051**

2<23DiOH2OHMePropan-1> + Sm+3 = Sm+3_23DiOH2OHMePropan-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:5.07(3SF) | 5 | MGL | 931 |

**Reaction No. 41052**

3<23DiOH2OHMePropan-1> + Sm+3 = Sm+3_23DiOH2OHMePropan-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:6.51(3SF) | 5 | MGL | 931 |

**Reaction No. 41218**

1COOPrAmMalon-3 + Sm+3 = Sm+3_1COOPrAmMalon-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:8.94(3SF) | 6 | ENS | 233 |

**Reaction No. 41274**

Tetraglyme + Sm+3 = Sm+3_Tetraglyme

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:5.0(0.06SD) | 5 | MPT | 1032 |

**Reaction No. 41543**

DTPABMA-3 + Sm+3 = Sm+3_DTPABMA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:16.7(0.05SD) | 5 | MGL | 770 |
| 2 | 25 | 0.1 | NaClO4 | dH:-5.999(4SF) | 5 | MCL | 770 |

**Reaction No. 41575**

2OH3Naphthoic-2 + Sm+3 = Sm+3_2OH3Naphthoic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Dioxan:Water 75:25Vol NaClO4 | lgK:5.31(3SF) | 3 | MGL | 3087 |

**Reaction No. 41581**

2OH1Naphthoic-2 + Sm+3 = Sm+3_2OH1Naphthoic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Dioxan:Water 75:25Vol NaClO4 | lgK:5.90(3SF) | 3 | MGL | 3087 |

**Reaction No. 41589**

1OH2Naphthoic-2 + Sm+3 = Sm+3_1OH2Naphthoic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Dioxan:Water 75:25Vol NaClO4 | lgK:5.22(3SF) | 3 | MGL | 3087 |

**Reaction No. 41765**

2OH2PrPropan-1 + Sm+3 = Sm+3_2OH2PrPropan-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:2.65(0.02SD) | 5 | MQH | 931 |

**Reaction No. 41766**

Sm+3_2OH2PrPropan-1 + 2OH2PrPropan-1 = Sm+3_2OH2PrPropan-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:2.2(0.03SD) | 5 | MQH | 931 |

**Reaction No. 41939**

| Sm+3_Cupferron-1 + Cupferron-1 = Sm+3_Cupferron-1(2) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:4.83(3SF) | 5 | MDS | 140 |

| Reaction No. 41940 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Cupferron-1(2) + Cupferron-1 = Sm+3_Cupferron-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:3.70(3SF) | 5 | MDS | 140 |

| Reaction No. 41997 | | | | | | | |
|---|---|---|---|---|---|---|---|
| HOEDTA-3 + Sm+3 + Acac-1 = Sm+3_Acac-1_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:3.54(3SF) | 5 | MGL | 2261 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.6(2SF) | 1 | ESC | |

| Reaction No. 41998 | | | | | | | |
|---|---|---|---|---|---|---|---|
| HOEDTA-3 + Sm+3 + Benzoylacetone-1 = Sm+3_Benzoylacetone-1_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | NaClO4 | lgK:3.64(3SF) | 5 | MGL | 2261 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.7(2SF) | 1 | ESC | |

| Reaction No. 42123 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<[15]O5:Benzo> + Sm+3 = Sm+3_[15]O5:Benzo(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | | DiClMethane:Water 50:50Vol | lgK:8.12(3SF) | 4 | MPS | 2244 |

| Reaction No. 42235 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3Furoic-1 + Sm+3 = Sm+3_3Furoic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 2 | NaClO4 | lgK:1.675(4SF) | 5 | MGL | 3169 |
| 2 | 25 | 2 | NaClO4 | dH:1.53(0.01SD) | 5 | MCL | 3169 |

| Reaction No. 42342 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(-1)_Gluconic-1(2) + H+1 = Sm+3_Gluconic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | KOH | lgK:5.8(0.03SD) | 5 | MPS | 4448 |

| Reaction No. 42343 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(-2)_Gluconic-1(2) + 2<H+1> = Sm+3_Gluconic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.2 | KOH | lgK:13.2(0.03SD) | 5 | MPS | 4448 |

| Reaction No. 42938 | | | | | | | |
|---|---|---|---|---|---|---|---|
| [9]N3:Acet*3-3 + Sm+3 = Sm+3_[9]N3:Acet*3-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | 0.1 | NaCl | lgK:13.5(3SF) | 3 | MSP | 1377 |

| Reaction No. 43352 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Asp-2 + Asp-2 = Sm+3_Asp-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KCl | lgK:4.17(3SF) | 3 | MGL | 11516 |
| Note (1): Dreyer R et al., Z. Phys. Chem., 1968, 238, 417 | | | | | | | |
| 2 | 30 | 0.1 | NaClO4 | lgK:4.51(3SF) | 4 | MGL | 11516 |
| Note (2): Kemin Y et al., Chem. J. of Chin. Univ., 1984, , 603 | | | | | | | |
| 3 | 40 | 0.1 | NaClO4 | lgK:6.4(2SF) | 0 | MGL | 906 |
| 4 | 50 | 0.1 | NaClO4 | lgK:8.4(2SF) | 0 | MGL | 906 |

| Reaction No. 43406 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_DiMeGlyox-2 = Sm+3_H+1_DiMeGlyox-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Dioxan:Water 50:50Vol NaClO4 | lgK:8.01(3SF) | 4 | MGL | 906 |

| Reaction No. 43407 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_DiMeGlyox-2 + H+1_DiMeGlyox-2 = Sm+3_H+1(2)_DiMeGlyox-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Dioxan:Water 50:50Vol NaClO4 | lgK:6.97(3SF) | 4 | MGL | 906 |

| Reaction No. 43744 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + OH-1 + H+1(-1)_Citric-3 = Sm+3_H+1(-1)_OH-1_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | | | lgK:18.5(0.1SD) | 4 | MDS | 906 |

| Reaction No. 43745 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 2<OH-1> + H+1(-1)_Citric-3 = Sm+3_H+1(-1)_OH-1(2)_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | | | lgK:24.4(0.1SD) | 4 | MDS | 906 |

| Reaction No. 43943 | | | | | | | |
|---|---|---|---|---|---|---|---|
| ClKojic-1 + Sm+3 = Sm+3_ClKojic-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | Unknown | lgK:5.83(3SF) | 5 | MGL | 906 |

| Reaction No. 43944 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_ClKojic-1 + ClKojic-1 = Sm+3_ClKojic-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | Unknown | lgK:5.08(3SF) | 5 | MGL | 906 |

| Reaction No. 44050 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EtThioAcet-1(2) + EtThioAcet-1 = Sm+3_EtThioAcet-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.1 | Acetone:Water 75:25Wgt Unknown | lgK:5.52(3SF) | 5 | MGL | 906 |

| Reaction No. 44193 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Salicylic-2 + Salicylic-2 = Sm+3_Salicylic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Methanol:Water 99.9:.1Wgt NaCl | lgK:4.67(3SF) | 5 | MGL | 906 |

| Reaction No. 44194 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Salicylic-2(2) + Salicylic-2 = Sm+3_Salicylic-2(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Methanol:Water 99.9:.1Wgt NaCl | lgK:3.24(3SF) | 5 | MGL | 906 |

| Reaction No. 44378 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2AmBenz-1 + 2AmBenz-1 = Sm+3_2AmBenz-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Methanol NaCl | lgK:5.42(3SF) | 4 | MGL | 906 |

| Reaction No. 44379 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2AmBenz-1(2) + 2AmBenz-1 = Sm+3_2AmBenz-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Methanol NaCl | lgK:3.26(3SF) | 4 | MGL | 906 |

| Reaction No. 44380 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_2AmBenz-1(3) + 2AmBenz-1 = Sm+3_2AmBenz-1(4) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Methanol NaCl | lgK:2.55(3SF) | 4 | MGL | 906 |

| Reaction No. 44428 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_SalHydroxam-2 = Sm+3_H+1_SalHydroxam-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Acetone:Water 75:25Wgt NaClO4 | lgK:7.20(3SF) | 4 | MGL | 906 |

| Reaction No. 44429 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_SalHydroxam-2 + H+1_SalHydroxam-2 = Sm+3_H+1(2)_SalHydroxam-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Acetone:Water 75:25Wgt NaClO4 | lgK:6.77(3SF) | 4 | MGL | 906 |

| Reaction No. 44430 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(2)_SalHydroxam-2(2) + H+1_SalHydroxam-2 = Sm+3_H+1(3)_SalHydroxam-2(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Acetone:Water 75:25Wgt NaClO4 | lgK:5.01(3SF) | 4 | MGL | 906 |

| Reaction No. 44652 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Oxine-1 + Oxine-1 = Sm+3_Oxine-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 30 | 0.3 | Dioxan:Water 50:50Wgt NaClO4 | lgK:8.26(3SF) | 4 | MGL | 906 |

| Reaction No. 44880 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTA-4 + H+1_P3O10-5 = Sm+3_H+1_EDTA-4_P3O10-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.20(3SF) | 2 | EES | |
| 2 | 35 | 0.1 | KNO3 | lgK:3.41(3SF) | 2 | MGL | 31163 |

| Reaction No. 44958 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + H+1(3)_HOEDTA-3 = Sm+3_H+1(3)_HOEDTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | | | lgK:0.06(1SF) | 4 | MGL | 906 |
| 2 | 25 | 1 | NaClO4 | lgK:1.68(3SF) | 4 | MGL | 906 |

| Reaction No. 44959 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + H+1(2)_HOEDTA-3 = Sm+3_H+1(2)_HOEDTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | | | lgK:0.62(2SF) | 5 | MGL | 906 |
| 2 | 25 | 1 | NaClO4 | lgK:0.66(2SF) | 5 | MGL | 906 |

| Reaction No. 44960 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + H+1_HOEDTA-3 = Sm+3_H+1_HOEDTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | | | lgK:1.92(3SF) | 5 | MGL | 906 |
| 2 | 25 | 1 | NaClO4 | lgK:1.94(3SF) | 5 | MGL | 906 |

| Reaction No. 44961 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + HOEDTA-3 = Sm+3_HOEDTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | | | lgK:3.36(3SF) | 4 | MGL | 906 |
| 2 | 25 | 1 | NaClO4 | lgK:2.89(3SF) | 4 | MGL | 906 |

| Reaction No. 45153 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Dithizone-1 = Sm+3_H+1_Dithizone-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Ethanol:Water 50:50Wgt NaClO4 | lgK:3.50(3SF)w | 4 | MPH | 906 |

| Reaction No. 45204 |
|---|

| Dithizone-1 + Sm+3 = Sm+3_Dithizone-1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Ethanol:Water 50:50Wgt NaClO4 | lgK:1.90(3SF)w | 4 | MPH | 906 |

| Reaction No. 45441 | | | | | | | |
|---|---|---|---|---|---|---|---|
| CEDTA-5 + Sm+3 = Sm+3_CEDTA-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | 0.1 | KCl | lgK:16.16(4SF) | 4 | MPT | 906 |
| 2 | 23 | | | lgK:18.4(3SF) | 4 | MPL | 906 |

| Reaction No. 45516 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDD3Me2BuDA-4 + Sm+3 = Sm+3_EDD3Me2BuDA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:12.9(0.05SD) | 5 | MGL | 2008 |

| Reaction No. 45544 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDDPentDA-4 + Sm+3 = Sm+3_EDDPentDA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:16.1(0.1SD) | 5 | MPL | 2008 |

| Reaction No. 45573 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(2)_Arsenazo1-6 = Sm+3_H+1(2)_Arsenazo1-6 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 19:21 | 0.1 | Unknown | lgK:8.4(0.03SD) | 5 | MSP | 906 |

| Reaction No. 45650 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(4)_Arsenazo3-8 = Sm+3_H+1(4)_Arsenazo3-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 19:21 | | | lgK:16.(0.3SD) | 4 | MSP | 906 |

| Reaction No. 45688 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1(2)_XylenolOrange-6 = Sm+3_H+1(2)_XylenolOrange-6 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | | | lgK:6.52(3SF) | 4 | MSP | 906 |

| Reaction No. 45698 |
|---|
| Bilirubin + Sm+3 = Sm+3_Bilirubin |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | | CHCl3:Ethanol 1:1Vol | lgK:5.52(3SF) | 4 | MSP | 906 |

| Reaction No. 45699 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Bilirubin + Sm+3 = Sm+3(2)_Bilirubin | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 23 | | CHCl3:Ethanol 1:1Vol | lgK:4.68(3SF) | 4 | MSP | 906 |

| Reaction No. 45851 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTMP-8 + Sm+3 = Sm+3_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:20.71(0.08SD) | 5 | MSP | 23315 |
| 2 | 37 | 0.15 | NaCl | lgK:14.44(0.02SD) | 0 | MGL | 6359 |

| Reaction No. 45852 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTMP-8 + H+1 = Sm+3_H+1_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:6.98(0.03SD) | 5 | MSP | 23315 |
| 2 | 37 | 0.15 | NaCl | lgK:7.13(0.02SD) | 5 | MGL | 6359 |

| Reaction No. 45853 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_EDTMP-8 + H+1 = Sm+3_H+1(2)_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:5.91(0.03SD) | 5 | MSP | 23315 |
| 2 | 37 | 0.15 | NaCl | lgK:6.00(0.02SD) | 5 | MGL | 6359 |

| Reaction No. 45950 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_HOEDTA-3 + OH-1 = Sm+3_OH-1_HOEDTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:3.70(3SF) | 5 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.8(2SF) | 1 | ESC | |

| Reaction No. 46311 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Cl-1 + [18]O6 = Sm+3_[18]O6_Cl-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.005 | Methanol KCl | lgK:2.0(0.07SD) | 6 | MCL | 4975 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 25 | 0.005 | Methanol<br>KCl | dH:-0.877(3SF) | 6 | MCL | 4975 |

| Reaction No. 52528 | | | | | | | |
|---|---|---|---|---|---|---|---|
| GlyGly-1 + Sm+3 = Sm+3_GlyGly-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:4.4(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | NaClO4 | lgK:4.08(3SF) | 5 | MGL | 4509 |
| 3 | 40 | 0.1 | NaClO4 | lgK:3.88(3SF) | 5 | MGL | 4509 |

| Reaction No. 52529 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_GlyGly-1 + GlyGly-1 = Sm+3_GlyGly-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.1(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | NaClO4 | lgK:2.85(3SF) | 5 | MGL | 4509 |
| 3 | 40 | 0.1 | NaClO4 | lgK:2.67(3SF) | 5 | MGL | 4509 |

| Reaction No. 52530 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_GlyGly-1(2) + GlyGly-1 = Sm+3_GlyGly-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.1(2SF) | 1 | ESC | |
| 2 | 30 | 0.1 | NaClO4 | lgK:2.80(3SF) | 5 | MGL | 4509 |
| 3 | 40 | 0.1 | NaClO4 | lgK:2.63(3SF) | 5 | MGL | 4509 |

| Reaction No. 52848 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Br-1 + Sm+3 = Sm+3_Br-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 3 | Unknown | lgK:-0.5(1SF) | 0 | CRV | 552 |
| 2 | 25 | 0 | Inf. Dilution | lgK:0.23(2SF) | 4 | CRV | 32222 |
| 3 | 25 | 3 | LiClO4 | lgK:-0.16(2SF) | 5 | MSP | 5398 |
| 4 | 0:50 | 1 | Many Media | dH:-2.3(0.1SD) | 0 | MSE | 5260 |
| 5 | 25 | 0 | Inf. Dilution | dH:17.023(5SF)kJ | 2 | CRV | 32222 |
| 6 | 25 | 0.2 | DiMeAcetamide<br>Bu4NClO4 | lgK:2.8(0.07MD) | 5 | MCL | 7233 |
| 7 | 25 | 0.2 | DiMeAcetamide<br>Bu4NClO4 | dH:3.39(3SF) | 5 | MCL | 7233 |
| 8 | 25 | 3 | Methanol:Water<br>50:50Wgt<br>LiClO4 | lgK:0.27(2SF) | 5 | MSP | 5398 |

| Reaction No. 52972 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cl-1 + Sm+3 = Sm+3_Cl-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 3 | LiClO4 | lgK:-0.39(2SF) | 0 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-2.03(3SF) | 0 | MSP | 5398 |
| 3 | 25 | 0 | Inf. Dilution | lgK:0.36(2SF) | 3 | EOD | 7141 |
| 4 | 25 | 0 | Inf. Dilution | lgK:0.30(2SF) | 3 | CRV | 8781 |
| 5 | 25 | 0 | Inf. Dilution | lgK:0.40(0.10SD) | 3 | CRV | 32222 |
| 6 | 25 | 0.5 | HClO4 | lgK:-0.46(2SF) | 3 | MSL | 931 |
| 7 | 25 | 0.5 | Inert | lgK:-0.51(2SF) | 4 | MLC | 8917 |
| 8 | 25 | 1 | HClO4 | lgK:-0.02(0.05SD) | 2 | CMN | 5260 |
| 9 | 25 | 1 | Inert | lgK:-0.57(2SF) | 2 | MLC | 8917 |
| 10 | 25 | 1 | NaClO4 | lgK:0.2(0.05SD) | 0 | CMN | 5260 |
| 11 | 25 | 1 | Unknown | lgK:-0.1(1SF) | 3 | ENS | 6496 |
| 12 | 25 | 3 | LiClO4 | lgK:-0.02(1SF) | 2 | MSP | 5398 |
| 13 | 100 | 0 | Inf. Dilution | lgK:1.00(0.13SD) | 1 | MSP | 22846 |
| 14 | 150 | 0 | Inf. Dilution | lgK:1.40(0.11SD) | 0 | MSP | 22846 |
| 15 | 200 | 0 | Inf. Dilution | lgK:2.28(0.11SD) | 0 | MSP | 22846 |
| 16 | 250 | 0 | Inf. Dilution | lgK:3.28(0.13SD) | 0 | MSP | 22846 |
| 17 | 0:50 | 1 | Many Media | dH:-1.6(0.1SD) | 2 | MSE | 5260 |
| 18 | 25 | 0 | Inf. Dilution | dH:0.0(1SF) | 2 | ENS | 6496 |
| 19 | 25 | 0 | Inf. Dilution | dH:24.103(6.079SD)kJ | 2 | CRV | 32222 |
| 20 | 25 | 0.2 | DMF<br>Et4NClO4 | lgK:3.0(0.1MD) | 5 | MCL | 7216 |
| 21 | 25 | 0.2 | DMF<br>Et4NClO4 | dH:3.3(2SF) | 5 | MCL | 7216 |
| 22 | 25 | 3 | Ethanol:Water<br>50:50Vol<br>LiClO4 | lgK:0.54(2SF) | 5 | MSP | 5398 |
| 23 | 25 | 3 | Ethanol:Water<br>90:10Vol<br>LiClO4 | lgK:0.97(2SF) | 5 | MSP | 5398 |
| 24 | 25 | 3 | Methanol<br>LiClO4 | lgK:0.96(2SF) | 5 | MSP | 5398 |
| 25 | 25 | 3 | Methanol:Water<br>50:50Wgt<br>LiClO4 | lgK:0.52(2SF) | 5 | MSP | 5398 |

| Reaction No. 53460 |
|---|

Sm+3 + 2<P3O10-5> = Sm+3_P3O10-5(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:9.846(4SF) | 1 | EOR | |
| 2 | 25 | 0.1 | KNO3 | lgK:8.7(0.1SD) | 5 | MPT | 4602 |
| 3 | 35 | 0.1 | KNO3 | lgK:8.9(0.1SD) | 5 | MPT | 4602 |
| 4 | 45 | 0.1 | KNO3 | lgK:8.4(0.1SD) | 5 | MPT | 4602 |
| 5 | 25 | 0.1 | KNO3 | dH:-6.8(2SF) | 5 | MTD | 5398 |

**Reaction No. 53623**

NO3-1 + Sm+3 = Sm+3_NO3-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 19:21 | 4.1:4.2 | NaClO4 | lgK:-0.02(1SF) | 0 | MSP | 5398 |
| Note (1): Low wgt for internal consistency | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:1.1(2SF) | 3 | EAC | |
| 3 | 25 | 0 | Inf. Dilution | lgK:0.78(2SF) | 0 | EDH | 8781 |
| 4 | 25 | 0 | Inf. Dilution | lgK:0.90(2SF) | 4 | CRV | 32222 |
| 5 | 25 | 1 | Inert | lgK:0.41(2SF) | 4 | MLC | 8917 |
| 6 | 25 | 1 | NaClO4 | lgK:0.31(2SF) | 5 | MDS | 931 |
| 7 | 0:50 | 1 | Many Media | dH:-4.0(0.05SD) | 6 | MSE | 5260 |

**Reaction No. 53916**

2<Sm+3> + P2O7-4 = Sm+3(2)_P2O7-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:20.16(4SF) | 1 | EOR | |
| 2 | 25 | 0 | Inf. Dilution | lgK:20.16(4SF) | 4 | MKN | 931 |

**Reaction No. 53963**

Sm+3 + H+1_P3O10-5 = Sm+3_H+1_P3O10-5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.1 | NaClO4 | lgK:4.89(3SF) | 4 | MGL | 140 |
| 2 | 30 | 0.3 | NaClO4 | lgK:7.42(3SF) | 0 | MGL | 931 |

**Reaction No. 54041**

2<SO4-2> + Sm+3 = Sm+3_SO4-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.2(2SF) | 4 | CRV | 2556 |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.20(0.10SD) | 4 | CRV | 32222 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0 | Inf. Dilution | lgK:5.20(3SF) | 4 | CRV | 32224 |
| 4 | 25 | 2 | NaClO4 | lgK:1.91(3SF) | 4 | MAG | 643 |
| 5 | 25 | 0 | Inf. Dilution | dH:7.0(2SF) | 4 | CRV | 2556 |
| 6 | 25 | 0 | Inf. Dilution | dH:24.910(1.641SD)kJ | 4 | CRV | 32222 |

**Reaction No. 55205**

Sm+3 + MoO4-2 = Sm+3_MoO4-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.101(4SF) | 1 | EOR | |
| 2 | 25 | 0.0012 | Unknown | lgK:4.9(0.05SD) | 5 | MCN | 931 |

**Reaction No. 55661**

Sm+3 + e-1 = Sm+2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | E:-1.55(3SF) | 4 | CRV | 6330 |
| 2 | 25 | 0 | Inf. Dilution | E:-1.55(3SF) | 4 | CRV | 7761 |
| 3 | 25 | 0 | Inf. Dilution | E:-1.55(3SF) | 4 | CRV | 22519 |
| 4 | 25 | 0 | Unknown | E:-1.6(0.2SD) | 0 | ENS | 5256 |
| Note (4): Low wgt for internal consistency | | | | | | | |
| 5 | 25 | 1 | HClO4 | E:-1.40(3SF) | 4 | ENS | 7276 |
| 6 | 25 | 0 | Inf. Dilution | dH:189.8(4SF)kJ | 3 | CRV | 7761 |

**Reaction No. 55698**

Sm+3 + I-1 = Sm+3_I-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 0:50 | 1 | Many Media | dH:-2.2(0.1SD) | 5 | MSE | 5260 |

**Reaction No. 57093**

Sm+3 + 4<CO3-2> = Sm+3_CO3-2(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 2.5 | Unknown | lgK:14.20(0.01SD) | 7 | ENS | 5181 |

**Reaction No. 57112**

Sm+3(2)_CO3-2(3)_(s) = 2<Sm+3> + 3<CO3-2>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-32.5(0.1SD) | 1 | ENS | 5181 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-34.5(2.00SD) | 2 | CRV | 32222 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0 | Inf. Dilution | lgK:-32.50(4SF) | 0 | CRV | 32224 |

**Reaction No. 57313**

Sm+3 + H2O = Sm+3_OH-1 + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 1 | NaClO4 | lgK:-7.5(0.1SD) | 0 | MSL | 5215 |
| Note (1): Low wgt for internal consistency | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:-7.9(2SF) | 4 | CRV | 6478 |
| 3 | 25 | 0 | Inf. Dilution | lgK:-7.84(3SF) | 4 | MPS | 8699 |
| 4 | 25 | 0 | Inf. Dilution | lgK:-7.96(0.06SD) | 4 | EFE | 8712 |
| 5 | 25 | 0 | Inf. Dilution | lgK:-7.90(0.10SD) | 4 | CRV | 32222 |
| 6 | 25 | 0 | Inf. Dilution | lgK:-7.90(3SF) | 4 | CRV | 32224 |
| 7 | 25 | 0.7 | NaClO4 | lgK:-8.15(0.08SD) | 6 | MPS | 8699 |
| 8 | 25 | 0.7 | Unknown | lgK:-8.44(3SF) | 0 | ENS | 6496 |
| 9 | 25 | 0 | Inf. Dilution | dH:13.8(3SF) | 0 | CRV | 6478 |
| 10 | 25 | 0 | Inf. Dilution | dH:81.304(6.438SD)kJ | 2 | CRV | 32222 |
| 11 | 25 | 0.7 | NaClO4 | dH:10.2(1.3SD) | 0 | MTD | 8699 |
| 12 | 25 | 0.7 | Unknown | dH:13.8(3SF) | 0 | ENS | 6496 |

**Reaction No. 57314**

Sm+3 + 2<H2O> = Sm+3_OH-1(2) + 2<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 1 | NaClO4 | lgK:-15.0(0.1SD) | 3 | MSL | 5215 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-16.51(0.13SD) | 4 | EFE | 8712 |
| 3 | 25 | 0 | Inf. Dilution | lgK:-16.50(0.20SD) | 4 | CRV | 32222 |
| 4 | 25 | 0.7 | Unknown | lgK:-17.43(4SF) | 0 | ENS | 6496 |
| 5 | 25 | 0 | Inf. Dilution | dH:150.260(6.514SD)kJ | 4 | CRV | 32222 |
| 6 | 25 | 0.7 | Unknown | dH:27.6(3SF) | 0 | ENS | 6496 |

**Reaction No. 57315**

Sm+3 + 3<H2O> = Sm+3_OH-1(3) + 3<H+1>

Note: Doubtful species

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 1 | NaClO4 | lgK:-22.7(0.1SD) | 0 | MSL | 5215 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-25.91(0.59SD) | 1 | EFE | 8712 |
| 3 | 25 | 0 | Inf. Dilution | lgK:-25.90(1.00SD) | 4 | CRV | 32222 |
| 4 | 25 | 0 | Inf. Dilution | dH:226.680(8.585SD)kJ | 4 | CRV | 32222 |

**Reaction No. 57316**

3<Sm+3> + 4<H2O> = Sm+3(3)_OH-1(4) + 4<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 1 | NaClO4 | lgK:-19.5(0.1SD) | 5 | MSL | 5215 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-19.8(3SF) | 2 | EAE | |

**Reaction No. 57317**

Sm+3 + 3<H2O> = Sm+3_OH-1(3)_(s) + 3<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 1 | NaClO4 | lgK:-17.5(0.1SD) | 5 | MSL | 5215 |
| Note (1): Sign changed | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:-16.5(0.3SD) | 4 | CRV | 6478 |
| 3 | 25 | 0 | Inf. Dilution | lgK:-16.50(1.00SD) | 4 | CRV | 32222 |
| 4 | 25 | 0 | Inf. Dilution | lgK:-16.50(4SF) | 4 | CRV | 32224 |
| 5 | 25 | 0 | Inf. Dilution | dH:32.25(0.7SD) | 4 | CRV | 6478 |

**Reaction No. 57362**

Sm+3 + H+1(2)_P3O10-5(2) = Sm+3_H+1(2)_P3O10-5(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:6.5(0.1SD) | 5 | MPT | 4602 |
| 2 | 35 | 0.1 | KNO3 | lgK:6.9(0.1SD) | 5 | MPT | 4602 |
| 3 | 45 | 0.1 | KNO3 | lgK:6.4(0.1SD) | 5 | MPT | 4602 |

**Reaction No. 57401**

Sm+3 + SCN-1 = Sm+3_SCN-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.5 | Unknown | lgK:0.09(1SF) | 6 | CRV | 552 |
| 2 | 25 | 1 | NaClO4 | lgK:-0.9(0.07SD) | 5 | MSP | 5261 |
| 3 | 25 | 2 | NaClO4 | lgK:-0.8(0.1SD) | 5 | MSP | 5261 |
| 4 | 25 | 2.5 | NaClO4 | lgK:-0.9(0.1SD) | 5 | MSP | 5261 |
| 5 | 25 | 3 | NaClO4 | lgK:-1.1(0.1SD) | 5 | MSP | 5261 |
| 6 | 25 | 4 | NaClO4 | lgK:-0.6(0.08SD) | 5 | MSP | 5261 |

**Reaction No. 57402**

Sm+3 + 2<SCN-1> = Sm+3_SCN-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:-0.9(0.1SD) | 5 | MSP | 5261 |
| 2 | 25 | 2 | NaClO4 | lgK:-1.3(0.2SD) | 5 | MSP | 5261 |
| 3 | 25 | 2.5 | NaClO4 | lgK:-1.(0.3SD) | 5 | MSP | 5261 |
| 4 | 25 | 3 | NaClO4 | lgK:-1.1(0.2SD) | 5 | MSP | 5261 |
| 5 | 25 | 4 | NaClO4 | lgK:-1.2(0.2SD) | 5 | MSP | 5261 |

**Reaction No. 57780**

2<F-1> + Sm+3 = Sm+3_F-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:6.57(3SF) | 4 | EDH | 8917 |
| 2 | 25 | 0.025 | HNO3 | lgK:6.10(0.09SD) | 4 | MCE | 8221 |
| 3 | 25 | 0.5 | NH4NO3 | lgK:7.0(0.2SD) | 0 | MIS | 5387 |
| Note (3): Algebraic error [JS, Apr 00] - all B120s suspect | | | | | | | |
| 4 | 25 | 0.68 | NaClO4 | lgK:5.23(3SF) | 5 | EFE | 8917 |
| 5 | 25 | 1 | NaCl | lgK:4.56(0.02SD) | 4 | MPT | 5385 |

**Reaction No. 57948**

Sm+3_F-1(3)_(s) = Sm+3 + 3<F-1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-17.7(3SF) | 1 | EDH | |
| 2 | 25 | 0 | Inf. Dilution | lgK:-19.0(3SF) | 0 | EES | |
| 3 | 25 | 0.1 | Unknown | lgK:-17.9(3SF) | 0 | CRV | 2556 |
| 4 | 25 | 0.5 | NH4NO3 | dG:21.8(3SF) | 5 | MPT | 5604 |
| 5 | 25 | 0.5 | NH4NO3 | dH:12.(2SF) | 5 | MPT | 5604 |

**Reaction No. 58471**

3<S2O3-2> + 2<Sm+3> = Sm+3(2)_S2O3-2(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 32 | | | lgK:11.26(4SF) | 0 | MCN | 140 |

**Reaction No. 59154**

Sm+3 + EtGlycol = Sm+3_H+1(-1)_EtGlycol + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 22 | 0.1 | NaClO4 | lgK:-7.50(3SF) | 5 | MGL | 2064 |

**Reaction No. 59307**

Cys-2 + Sm+3 = Sm+3_Cys-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0 | Inf. Dilution | lgK:7.300(4SF) | 5 | MGL | 905 |
| 2 | 20 | 0.1 | KCl | lgK:4.8(2SF) | 0 | MGL | 905 |
| 3 | 25 | 0 | Inf. Dilution | lgK:7.25(3SF) | 2 | EAE | |
| 4 | 35 | 0 | Inf. Dilution | lgK:7.150(4SF) | 5 | MGL | 905 |
| 5 | 45 | 0 | Inf. Dilution | lgK:7.075(4SF) | 5 | MGL | 905 |
| 6 | 25 | 0 | Inf. Dilution | dH:-4.13(3SF) | 4 | MCL | 905 |

**Reaction No. 59308**

2<Cys-2> + Sm+3 = Sm+3_Cys-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0 | Inf. Dilution | lgK:14.00(4SF) | 5 | MGL | 905 |
| 2 | 25 | 0 | Inf. Dilution | lgK:13.9(3SF) | 2 | EAE | |
| 3 | 35 | 0 | Inf. Dilution | lgK:13.650(5SF) | 5 | MGL | 905 |
| 4 | 45 | 0 | Inf. Dilution | lgK:13.475(5SF) | 5 | MGL | 905 |
| 5 | 25 | 0 | Inf. Dilution | dH:-9.63(3SF) | 4 | MCL | 905 |

**Reaction No. 59773**

35DiSulfSal-4 + Sm+3 = Sm+3_35DiSulfSal-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:8.2(0.03SD) | 6 | MGL | 3059 |

**Reaction No. 59774**

2<35DiSulfSal-4> + Sm+3 = Sm+3_35DiSulfSal-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:13.5(0.04SD) | 6 | MGL | 3059 |

**Reaction No. 59906**

5ATP-4 + Sm+3 = Sm+3_5ATP-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.05 | Unknown | lgK:6.66(3SF) | 4 | MKN | 1437 |
| 2 | 25 | 0.1 | KCl | lgK:6.71(3SF) | 4 | MGL | 1437 |

**Reaction No. 59924**

Sm+3 + H+1_5ATP-4 = Sm+3_H+1_5ATP-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:4.51(3SF) | 4 | MGL | 1437 |

**Reaction No. 59925**

Sm+3 + H+1(2)_5ATP-4 = Sm+3_H+1(2)_5ATP-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:3.95(3SF) | 4 | MGL | 1437 |

**Reaction No. 59926**

Sm+3 + 2<5ATP-4> = Sm+3_5ATP-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:10.53(4SF) | 4 | MGL | 1437 |

**Reaction No. 59927**

Sm+3_5ATP-4 + OH-1 = Sm+3_OH-1_5ATP-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:3.9(2SF) | 4 | MGL | 1437 |

**Reaction No. 59928**

Sm+3 + H+1(-1)_5ATP-4 + OH-1 = Sm+3_H+1(-1)_OH-1_5ATP-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KCl | lgK:12.28(4SF) | 4 | MGL | 1437 |

**Reaction No. 60088**

EtDiAm + Sm+3 = Sm+3_EtDiAm

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.03 | MeCN<br>ClO4-1 salt | dH:-19.3(3SF) | 4 | MCL | 148 |

**Reaction No. 60089**

Sm+3_EtDiAm + EtDiAm = Sm+3_EtDiAm(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.03 | MeCN<br>ClO4-1 salt | dH:-17.9(3SF) | 4 | MCL | 148 |

**Reaction No. 60090**

Sm+3_EtDiAm(2) + EtDiAm = Sm+3_EtDiAm(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.03 | MeCN<br>ClO4-1 salt | dH:-13.4(3SF) | 4 | MCL | 148 |

**Reaction No. 60091**

Sm+3_EtDiAm(3) + EtDiAm = Sm+3_EtDiAm(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 23 | 0.03 | MeCN ClO4-1 salt | dH:-9.8(2SF) | 4 | MCL | 148 |

**Reaction No. 60399**

Sm+3 + OH-1 = Sm+3_OH-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.81(3SF) | 3 | EDH | 8781 |
| 2 | 25 | 0.5 | Unknown | lgK:5.39(3SF) | 5 | CRV | 2556 |
| 3 | 25 | 3 | Unknown | lgK:5.4(2SF) | 5 | CRV | 2556 |
| 4 | 0:60 | 1 | Unknown | dH:-5.(1SF) | 5 | CRV | 2556 |

**Reaction No. 60400**

2<Sm+3> + 2<OH-1> = Sm+3(2)_OH-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 3 | Unknown | lgK:13.90(4SF) | 5 | CRV | 2556 |
| 2 | 0:60 | 1 | Unknown | dH:-6.(1SF) | 5 | CRV | 2556 |

**Reaction No. 60401**

Sm+3_OH-1(3)_(s) = Sm+3 + 3<OH-1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-25.(2SF) | 0 | ENS | 773 |
| 2 | 25 | 0 | Inf. Dilution | lgK:-25.4(3SF) | 4 | CRV | 2556 |
| 3 | 25 | 1 | Unknown | lgK:-23.8(3SF) | 4 | EES | |
| 4 | 25 | 1 | Unknown | lgK:-23.4(3SF) | 0 | ENS | 773 |
| 5 | 100 | 0 | Inf. Dilution | lgK:-23.5(3SF) | 3 | EES | |

Note (5): Split the difference with Sm+3 + 3<H2O> = Sm+3_OH-1(3)_(s) + 3<H+1>

**Reaction No. 60819**

Sm+3 + H+1_Salicylic-2 = Sm+3_H+1_Salicylic-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.1(0.03SD) | 6 | MGL | 6794 |

**Reaction No. 60859**

Sm+3_H+1_Salicylic-2 + H+1_Salicylic-2 = Sm+3_H+1(2)_Salicylic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:1.76(3SF) | 6 | MGL | 5987 |

**Reaction No. 60892**

Sm+3 + H+1(2)_24DHB-3 = Sm+3_H+1(2)_24DHB-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.04(3SF) | 6 | MGL | 5987 |

**Reaction No. 60932**

Sm+3 + H+1(2)_25DHB-3 = Sm+3_H+1(2)_25DHB-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.07(3SF) | 6 | MGL | 5987 |
| 2 | 25 | 0.1 | NaClO4 | dH:0.38(2SF) | 6 | MCL | 5987 |

**Reaction No. 61103**

Sm+3_35DiSulfSal-4 + 35DiSulfSal-4 = Sm+3_35DiSulfSal-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.5 | NaClO4 | lgK:5.34(3SF) | 6 | MGL | 5987 |

**Reaction No. 61149**

3Br5SulfSal-3 + Sm+3 = Sm+3_3Br5SulfSal-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:7.39(0.04MD) | 6 | MGL | 2270 |

**Reaction No. 61150**

2<3Br5SulfSal-3> + Sm+3 = Sm+3_3Br5SulfSal-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:12.4(0.1MD) | 6 | MGL | 2270 |

**Reaction No. 61151**

Sm+3_3Br5SulfSal-3 + 3Br5SulfSal-3 = Sm+3_3Br5SulfSal-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:5.00(3SF) | 6 | MGL | 5987 |

**Reaction No. 61152**

H+1 + 3Br5SulfSal-3 + Sm+3 = Sm+3_H+1_3Br5SulfSal-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:12.48(0.05MD) | 6 | MGL | 2270 |
| 2 | 25 | 0.1 | NaClO4 | dH:-2.3(0.2MD)kJ | 6 | MCL | 2270 |

**Reaction No. 61153**

3Br5SulfSal-3 + Sm+3 = H+1 + Sm+3_H+1(-1)_3Br5SulfSal-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:-0.4(0.2MD) | 6 | MGL | 2270 |

**Reaction No. 61154**

3Br5SulfSal-3 + Sm+3 = 2<H+1> + Sm+3_H+1(-2)_3Br5SulfSal-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:-8.2(0.1MD) | 6 | MGL | 2270 |

**Reaction No. 61281**

Sm+3_1OH4Sulf2Napthoic-3 + 1OH4Sulf2Napthoic-3 = Sm+3_1OH4Sulf2Napthoic-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:6.025(4SF) | 6 | MGL | 5987 |

**Reaction No. 61282**

H+1 + 1OH4Sulf2Napthoic-3 + Sm+3 = Sm+3_H+1_1OH4Sulf2Napthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:13.9(0.02SD) | 6 | MGL | 6990 |
| 2 | 25 | 0.1 | NaClO4 | dH:0.55(2SF) | 6 | MCL | 5987 |

**Reaction No. 61324**

Sm+3_1OH47DiSulf2Naphthoic-4 + 1OH47DiSulf2Naphthoic-4 = Sm+3_1OH47DiSulf2Naphthoic-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:5.7(2SF) | 6 | MGL | 5987 |

**Reaction No. 61362**

3OH7Sulf2Naphthoic-3 + Sm+3 = Sm+3_3OH7Sulf2Naphthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:7.44(3SF) | 6 | MGL | 5987 |

**Reaction No. 61363**

Sm+3 + H+1_3OH7Sulf2Naphthoic-3 = Sm+3_H+1_3OH7Sulf2Naphthoic-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.31(3SF) | 6 | MGL | 5987 |

**Reaction No. 61528**

Arg-1 + Sm+3 = Sm+3_Arg-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:1.76(3SF) | 2 | EAE | |
| 2 | 27 | 0.2 | Unknown | lgK:1.(0.3SD) | 5 | MMR | 6000 |

**Reaction No. 62187**

2<EDDA-2> + Sm+3 = Sm+3_EDDA-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:14.35(4SF) | 6 | CRV | 552 |
| 2 | 25 | 1 | Unknown | lgK:14.44(4SF) | 6 | CRV | 552 |
| 3 | 25 | 1 | Unknown | dH:-6.0(2SF) | 6 | CRV | 552 |

**Reaction No. 62188**

Sm+3_EDDA-2 + H+1 = Sm+3_H+1_EDDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | Unknown | lgK:3.59(3SF) | 6 | CRV | 552 |

**Reaction No. 62189**

Sm+3_EDDA-2 + 2<H+1> = Sm+3_H+1(2)_EDDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | Unknown | lgK:9.34(3SF) | 6 | CRV | 552 |

**Reaction No. 62219**

2<EDDM-4> + Sm+3 = Sm+3_EDDM-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:16.79(4SF) | 6 | CRV | 552 |

**Reaction No. 62274**

NNBis2OHEtAla-1 + Sm+3 = Sm+3_NNBis2OHEtAla-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:5.36(3SF) | 6 | CRV | 552 |

**Reaction No. 62275**

2<NNBis2OHEtAla-1> + Sm+3 = Sm+3_NNBis2OHEtAla-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:9.75(3SF) | 4 | CRV | 552 |

**Reaction No. 62336**

HBED-4 + Sm+3 = Sm+3_HBED-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:19.24(4SF) | 6 | CRV | 552 |

**Reaction No. 62337**

Sm+3_HBED-4 + H+1 = Sm+3_H+1_HBED-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:5.5(2SF) | 6 | CRV | 552 |

**Reaction No. 62338**

Sm+3_H+1_HBED-4 + H+1 = Sm+3_H+1(2)_HBED-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:5.0(2SF) | 6 | CRV | 552 |

**Reaction No. 62535**

CarbMeAsp-3 + Sm+3 = Sm+3_CarbMeAsp-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:9.24(3SF) | 6 | CRV | 552 |

**Reaction No. 62576**

MIDA-2 + Sm+3 = Sm+3_MIDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:6.85(3SF) | 6 | CRV | 552 |

**Reaction No. 62577**

2<MIDA-2> + Sm+3 = Sm+3_MIDA-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:12.26(4SF) | 6 | CRV | 552 |

**Reaction No. 62648**

NBenzIDA-2 + Sm+3 = Sm+3_NBenzIDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:6.06(3SF) | 6 | CRV | 552 |

**Reaction No. 62649**

2<NBenzIDA-2> + Sm+3 = Sm+3_NBenzIDA-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:10.70(4SF) | 6 | CRV | 552 |

| Reaction No. 62741 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2MeNTA-3 + Sm+3 = Sm+3_2MeNTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:12.18(4SF) | 6 | CRV | 552 |

| Reaction No. 62742 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2MeNTA-3> + Sm+3 = Sm+3_2MeNTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:21.43(4SF) | 4 | CRV | 552 |

| Reaction No. 62765 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2EtNTA-3 + Sm+3 = Sm+3_2EtNTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:11.13(4SF) | 6 | CRV | 552 |

| Reaction No. 62766 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2EtNTA-3> + Sm+3 = Sm+3_2EtNTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:19.57(4SF) | 4 | CRV | 552 |

| Reaction No. 62791 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2PrNTA-3 + Sm+3 = Sm+3_2PrNTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:10.87(4SF) | 6 | CRV | 552 |

| Reaction No. 62792 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<2PrNTA-3> + Sm+3 = Sm+3_2PrNTA-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:19.07(4SF) | 4 | CRV | 552 |

| Reaction No. 62815 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 22PrNTA-3 + Sm+3 = Sm+3_22PrNTA-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:9.42(3SF) | 6 | CRV | 552 |

| Reaction No. 62816 |
|---|

2<22PrNTA-3> + Sm+3 = Sm+3_22PrNTA-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:16.51(4SF) | 4 | CRV | 552 |

Reaction No. 62842

2HexNTA-3 + Sm+3 = Sm+3_2HexNTA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | K+1 salt | lgK:10.93(4SF) | 6 | CRV | 552 |

Reaction No. 62900

N2OHBenzIDA-3 + Sm+3 = Sm+3_N2OHBenzIDA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:13.34(4SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | Unknown | lgK:12.94(4SF) | 6 | CRV | 552 |

Reaction No. 62901

2<N2OHBenzIDA-3> + Sm+3 = Sm+3_N2OHBenzIDA-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:23.12(4SF) | 6 | CRV | 552 |
| 2 | 25 | 0.1 | Unknown | lgK:22.88(4SF) | 6 | CRV | 552 |

Reaction No. 62902

Sm+3 + H+1_N2OHBenzIDA-3 = Sm+3_H+1_N2OHBenzIDA-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:5.96(3SF) | 4 | CRV | 552 |
| 2 | 25 | 0.1 | Unknown | lgK:5.77(3SF) | 4 | CRV | 552 |

Reaction No. 62903

Sm+3 + 2<H+1_N2OHBenzIDA-3> = Sm+3_H+1(2)_N2OHBenzIDA-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:12.32(4SF) | 6 | CRV | 552 |

Reaction No. 62984

N23DiOHPrIDA-2 + Sm+3 = Sm+3_N23DiOHPrIDA-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | Unknown | lgK:8.50(3SF) | 6 | CRV | 552 |

Reaction No. 62985

| 2<N23DiOHPrIDA-2> + Sm+3 = Sm+3_N23DiOHPrIDA-2(2) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | Unknown | lgK:16.02(4SF) | 4 | CRV | 552 |

| Reaction No. 63013 | | | | | | | |
|---|---|---|---|---|---|---|---|
| N2MeoxEtIDA-2 + Sm+3 = Sm+3_N2MeoxEtIDA-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:7.98(3SF) | 6 | CRV | 552 |

| Reaction No. 63014 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<N2MeoxEtIDA-2> + Sm+3 = Sm+3_N2MeoxEtIDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:15.57(4SF) | 6 | CRV | 552 |

| Reaction No. 65606 | | | | | | | |
|---|---|---|---|---|---|---|---|
| N2MeSEtIDA-2 + Sm+3 = Sm+3_N2MeSEtIDA-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:6.31(3SF) | 6 | CRV | 552 |

| Reaction No. 65607 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<N2MeSEtIDA-2> + Sm+3 = Sm+3_N2MeSEtIDA-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:11.30(4SF) | 6 | CRV | 552 |

| Reaction No. 65654 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Fe+3_CN-1(6) + Sm+3 = Fe+3_Sm+3_CN-1(6) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.8(0.04SD) | 4 | CRV | 552 |
| 2 | 25 | 0 | Inf. Dilution | dG:-5.08(0.04SD) | 4 | EFE | 5760 |
| 3 | 25 | 0 | Inf. Dilution | dH:0.9(1SF) | 4 | CRV | 552 |
| 4 | 25 | 0 | Inf. Dilution | dH:0.91(0.02SD) | 4 | MTD | 5760 |

| Reaction No. 66146 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1PrEDTA-4 + Sm+3 = Sm+3_1PrEDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 20 | 0.1 | KNO3 | lgK:18.3(0.1SD) | 6 | MPL | 1994 |

| Reaction No. 66253 |
|---|

| Sm+3 + 3<e-1> = Sm(s) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | dG:159.1(4SF) | 4 | CRV | 6488 |
| 2 | 25 | 0 | Inf. Dilution | dG:666.6(4SF)kJ | 5 | CRV | 7421 |
| 3 | 25 | 0 | Inf. Dilution | dG:159.100(6SF) | 4 | CRV | 9029 |
| 4 | 25 | 0 | Inf. Dilution | dG:159.10(5SF) | 0 | CRV | 10174 |
| 5 | 25 | 0 | Inf. Dilution | dG:665.670(1.400SD)kJ | 4 | CRV | 32222 |
| 6 | 50 | 0 | Inf. Dilution | dG:157.79(5SF) | 3 | CRV | 10174 |
| 7 | 75 | 0 | Inf. Dilution | dG:156.42(5SF) | 3 | CRV | 10174 |
| 8 | 100 | 0 | Inf. Dilution | dG:154.98(5SF) | 2 | CRV | 10174 |
| 9 | 125 | 0 | Inf. Dilution | dG:153.49(5SF) | 2 | CRV | 10174 |
| 10 | 150 | 0 | Inf. Dilution | dG:151.93(5SF) | 1 | CRV | 10174 |
| 11 | 175 | 0 | Inf. Dilution | dG:150.30(5SF) | 1 | CRV | 10174 |
| 12 | 200 | 0 | Inf. Dilution | dG:148.60(5SF) | 0 | CRV | 10174 |
| 13 | 225 | 0 | Inf. Dilution | dG:146.82(5SF) | 0 | CRV | 10174 |
| 14 | 250 | 0 | Inf. Dilution | dG:144.93(5SF) | 0 | CRV | 10174 |
| 15 | 300 | 0 | Inf. Dilution | dG:140.82(5SF) | 0 | CRV | 10174 |
| 16 | 25 | 0 | Inf. Dilution | E:-2.41(3SF) | 0 | MDG | 140 |
| 17 | 25 | 0 | Inf. Dilution | E:-2.304(4SF) | 5 | CRV | 6330 |
| 18 | 25 | 0 | Inf. Dilution | E:-2.30(3SF) | 4 | ENS | 7276 |
| 19 | 25 | 0 | Inf. Dilution | E:-2.304(4SF) | 5 | CRV | 7761 |
| 20 | 25 | 0 | Inf. Dilution | E:-2.304(4SF) | 4 | CRV | 22519 |
| 21 | 25 | 0 | Inf. Dilution | dH:165.2(4SF) | 4 | CRV | 6488 |
| 22 | 25 | 0 | Inf. Dilution | dH:691.0(4SF)kJ | 5 | CRV | 7761 |
| 23 | 25 | 0 | Inf. Dilution | dH:165.200(6SF) | 4 | CRV | 9029 |
| 24 | 25 | 0 | Inf. Dilution | dH:691.200(2.138SD)kJ | 4 | CRV | 32222 |

| Reaction No. 66418 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Sm(s)> + 2<Zr(s)> + 3.5<O2(g)> = Sm2Zr2O7(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:677.6(4SF) | 4 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:673.2(4SF) | 0 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:496.0(4SF) | 2 | ENS | 6422 |
| 4 | 227 | 0 | Inf. Dilution | lgK:389.8(4SF) | 1 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:319.0(4SF) | 0 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-924.3(4SF) | 4 | EFE | 6422 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 7 | 25 | 0 | Inf. Dilution | dH:-973.0(4SF) | 2 | MTD | 6422 |
| 8 | 127 | 0 | Inf. Dilution | dH:-972.4(4SF) | 1 | MTD | 6422 |
| 9 | 227 | 0 | Inf. Dilution | dH:-971.7(4SF) | 0 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-971.0(4SF) | 0 | MTD | 6422 |

**Reaction No. 66419**

Sm(s) + 0.5<O2(g)> + 0.5<F2(g)> = SmOF(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:191.9(4SF) | 4 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:190.7(4SF) | 4 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:140.7(4SF) | 4 | ENS | 6422 |
| 4 | 227 | 0 | Inf. Dilution | lgK:110.7(4SF) | 4 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:90.76(4SF) | 4 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-261.8(4SF) | 4 | EFE | 6422 |
| 7 | 25 | 0 | Inf. Dilution | dH:-274.6(4SF) | 4 | MTD | 6422 |
| 8 | 127 | 0 | Inf. Dilution | dH:-274.3(4SF) | 4 | MTD | 6422 |
| 9 | 227 | 0 | Inf. Dilution | dH:-274.1(4SF) | 4 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-274.0(4SF) | 4 | MTD | 6422 |

**Reaction No. 66420**

1.5<O2(g)> + 2<Sm(s)> = Sm2O3(monoclinic,s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:304.1(4SF) | 3 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:302.1(4SF) | 0 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:222.7(4SF) | 2 | ENS | 6422 |
| 4 | 227 | 0 | Inf. Dilution | lgK:175.1(4SF) | 1 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:143.4(4SF) | 0 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-414.8(4SF) | 3 | EFE | 6422 |
| 7 | 25 | 0 | Inf. Dilution | dG:-414.79(5SF) | 3 | CRV | 31291 |
| 8 | 25 | 0 | Inf. Dilution | dH:-435.9(4SF) | 2 | MTD | 6422 |
| 9 | 127 | 0 | Inf. Dilution | dH:-435.5(4SF) | 1 | MTD | 6422 |
| 10 | 227 | 0 | Inf. Dilution | dH:-435.3(4SF) | 0 | MTD | 6422 |
| 11 | 327 | 0 | Inf. Dilution | dH:-435.1(4SF) | 0 | MTD | 6422 |

**Reaction No. 66421**

1.5<O2(g)> + 2<Sm(s)> = Sm2O3(cubic,s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:304.4(4SF) | 4 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:302.4(4SF) | 0 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:222.9(4SF) | 3 | ENS | 6422 |
| 4 | 227 | 0 | Inf. Dilution | lgK:175.2(4SF) | 2 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:143.5(4SF) | 0 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-415.2(4SF) | 4 | EFE | 6422 |
| 7 | 25 | 0 | Inf. Dilution | dG:-1737.(4SF)kJ | 5 | CRV | 30332 |
| Note (7): Value from LW3JTH option 2 | | | | | | | |
| 8 | 25 | 0 | Inf. Dilution | dG:-415.24(5SF) | 3 | CRV | 31291 |
| 9 | 25 | 0 | Inf. Dilution | dH:-436.8(4SF) | 4 | MTD | 6422 |
| 10 | 25 | 0 | Inf. Dilution | dH:-1827.2(5SF)kJ | 5 | CRV | 30332 |
| 11 | 127 | 0 | Inf. Dilution | dH:-436.4(4SF) | 1 | MTD | 6422 |
| 12 | 227 | 0 | Inf. Dilution | dH:-436.1(4SF) | 0 | MTD | 6422 |
| 13 | 327 | 0 | Inf. Dilution | dH:-435.9(4SF) | 0 | MTD | 6422 |

**Reaction No. 66422**

1.5<Cl2(g)> + Sm(s) = Sm+3_Cl-1(3)_(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:166.5(4SF) | 4 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:165.4(4SF) | 0 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:120.7(4SF) | 4 | ENS | 6422 |
| 4 | 227 | 0 | Inf. Dilution | lgK:94.02(4SF) | 3 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:76.23(4SF) | 0 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-227.1(4SF) | 4 | EFE | 6422 |
| 7 | 25 | 0 | Inf. Dilution | dH:-1025.9(5SF)kJ | 3 | CRV | 6330 |
| 8 | 25 | 0 | Inf. Dilution | dH:-245.2(4SF) | 4 | MTD | 6422 |
| 9 | 127 | 0 | Inf. Dilution | dH:-244.7(4SF) | 4 | MTD | 6422 |
| 10 | 227 | 0 | Inf. Dilution | dH:-244.3(4SF) | 3 | MTD | 6422 |
| 11 | 327 | 0 | Inf. Dilution | dH:-243.9(4SF) | 0 | MTD | 6422 |

**Reaction No. 66423**

Sm(s) + Cl2(g) = Sm+2_Cl-1(2)_(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:134.3(4SF) | 4 | ENS | 6422 |
| 2 | 27 | 0 | Inf. Dilution | lgK:133.4(4SF) | 0 | ENS | 6422 |
| 3 | 127 | 0 | Inf. Dilution | lgK:97.91(4SF) | 2 | ENS | 6422 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4 | 227 | 0 | Inf. Dilution | lgK:76.68(4SF) | 1 | ENS | 6422 |
| 5 | 327 | 0 | Inf. Dilution | lgK:62.54(4SF) | 0 | ENS | 6422 |
| 6 | 25 | 0 | Inf. Dilution | dG:-183.2(4SF) | 4 | EFE | 6422 |
| 7 | 25 | 0 | Inf. Dilution | dH:-194.9(4SF) | 2 | MTD | 6422 |
| 8 | 127 | 0 | Inf. Dilution | dH:-194.5(4SF) | 1 | MTD | 6422 |
| 9 | 227 | 0 | Inf. Dilution | dH:-194.2(4SF) | 0 | MTD | 6422 |
| 10 | 327 | 0 | Inf. Dilution | dH:-193.9(4SF) | 0 | MTD | 6422 |

**Reaction No. 66869**

Sm+3_IO3-1(3)_(s) = Sm+3 + 3<IO3-1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-11.(2SF) | 1 | ENS | 773 |
| 2 | 25 | 0 | Inf. Dilution | dG:15.41(4SF) | 1 | MSL | 21965 |

**Reaction No. 67469**

CO3-2 + Sm+3 = Sm+3_CO3-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.33(3SF) | 0 | ENS | 6496 |
| 2 | 25 | 0 | Inf. Dilution | lgK:7.71(3SF) | 4 | MSE | 7346 |
| 3 | 25 | 0 | Inf. Dilution | lgK:7.93(0.05SD) | 4 | EFE | 8712 |
| 4 | 25 | 0 | Inf. Dilution | lgK:7.30(3SF) | 3 | CRV | 8781 |
| 5 | 25 | 0 | Inf. Dilution | lgK:7.80(0.50SD) | 4 | CRV | 32222 |
| 6 | 25 | 0.7 | NaClO4/m | lgK:5.73(0.01SD) | 5 | MSE | 7346 |
| 7 | 25 | 0 | Inf. Dilution | dH:0.0(1SF) | 0 | ENS | 6496 |
| 8 | 25 | 0 | Inf. Dilution | dH:163.390(6.859SD)kJ | 4 | CRV | 32222 |

**Reaction No. 67470**

2<CO3-2> + Sm+3 = Sm+3_CO3-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:8.99(3SF) | 0 | ENS | 6496 |
| 2 | 25 | 0 | Inf. Dilution | lgK:13.09(4SF) | 4 | MSE | 7346 |
| 3 | 25 | 0 | Inf. Dilution | lgK:13.37(0.11SD) | 4 | EFE | 8712 |
| 4 | 25 | 0 | Inf. Dilution | lgK:12.11(4SF) | 3 | CRV | 8781 |
| 5 | 25 | 0 | Inf. Dilution | lgK:12.80(0.60SD) | 4 | CRV | 32222 |
| 6 | 25 | 0.7 | NaClO4/m | lgK:10.01(0.01SD) | 5 | MSE | 7346 |
| 7 | 25 | 0 | Inf. Dilution | dH:0.0(1SF) | 3 | ENS | 6496 |

| Reaction No. 67541 | | | | | | | |
|---|---|---|---|---|---|---|---|
| DETAP-4 + Sm+3 = Sm+3_DETAP-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:15.29(4SF) | 5 | MGL | 6529 |

| Reaction No. 67542 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_DETAP-4 = Sm+3_H+1_DETAP-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:9.59(3SF) | 5 | MGL | 6529 |

| Reaction No. 67654 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 23DiOH2MePropan-1 + Sm+3 = Sm+3_23DiOH2MePropan-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.05(3SF) | 5 | MGL | 6516 |

| Reaction No. 67655 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_23DiOH2MePropan-1 + 23DiOH2MePropan-1 = Sm+3_23DiOH2MePropan-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:2.34(3SF) | 5 | MGL | 6516 |

| Reaction No. 67656 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_23DiOH2MePropan-1(2) + 23DiOH2MePropan-1 = Sm+3_23DiOH2MePropan-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:1.64(3SF) | 5 | MGL | 6516 |

| Reaction No. 67699 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 23DiOH2MeButan-1 + Sm+3 = Sm+3_23DiOH2MeButan-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:3.14(3SF) | 5 | MGL | 6516 |

| Reaction No. 67700 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_23DiOH2MeButan-1 + 23DiOH2MeButan-1 = Sm+3_23DiOH2MeButan-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:2.32(3SF) | 5 | MGL | 6516 |

| Reaction No. 67701 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_23DiOH2MeButan-1(2) + 23DiOH2MeButan-1 = Sm+3_23DiOH2MeButan-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:1.45(3SF) | 5 | MGL | 6516 |

**Reaction No. 68058**

IDS-4 + Sm+3 = Sm+3_IDS-4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Unknown | lgK:10.41(4SF) | 5 | ENS | 7168 |

**Reaction No. 68422**

Sm+3_Cl-1 + Cl-1 = Sm+3_Cl-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | DMF Et4NClO4 | lgK:1.9(0.1MD) | 5 | MCL | 7216 |
| 2 | 25 | 0.2 | DMF Et4NClO4 | dH:4.8(2SF) | 5 | MCL | 7216 |

**Reaction No. 68423**

Sm+3_Cl-1(2) + Cl-1 = Sm+3_Cl-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | DMF Et4NClO4 | lgK:1.5(0.1MD) | 5 | MCL | 7216 |
| 2 | 25 | 0.2 | DMF Et4NClO4 | dH:3.6(2SF) | 5 | MCL | 7216 |

**Reaction No. 68424**

Sm+3_Cl-1(3) + Cl-1 = Sm+3_Cl-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | DMF Et4NClO4 | lgK:1.(0.09MD) | 5 | MCL | 7216 |

**Reaction No. 68463**

Sm+3_Br-1 + Br-1 = Sm+3_Br-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | DiMeAcetamide Bu4NClO4 | lgK:2.0(0.09MD) | 5 | MCL | 7233 |
| 2 | 25 | 0.2 | DiMeAcetamide Bu4NClO4 | dH:1.45(3SF) | 5 | MCL | 7233 |

**Reaction No. 68464**

Sm+3_Br-1(2) + Br-1 = Sm+3_Br-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | DiMeAcetamide Bu4NClO4 | lgK:1.1(0.08MD) | 5 | MCL | 7233 |
| 2 | 25 | 0.2 | DiMeAcetamide Bu4NClO4 | dH:2.63(3SF) | 5 | MCL | 7233 |

**Reaction No. 68544**

[Hexanone]2CH3CO-1 + Sm+3 = Sm+3_[Hexanone]2CH3CO-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Acetone:Water 3:1Vol Me4NOH | lgK:8.92(3SF) | 5 | MGL | 6837 |

**Reaction No. 68545**

Sm+3_[Hexanone]2CH3CO-1 + [Hexanone]2CH3CO-1 = Sm+3_[Hexanone]2CH3CO-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Acetone:Water 3:1Vol Me4NOH | lgK:8.07(3SF) | 5 | MGL | 6837 |

**Reaction No. 68546**

Sm+3_[Hexanone]2CH3CO-1(2) + [Hexanone]2CH3CO-1 = Sm+3_[Hexanone]2CH3CO-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Acetone:Water 3:1Vol Me4NOH | lgK:8.16(3SF) | 5 | MGL | 6837 |

**Reaction No. 68667**

2<Mandelic-1> + Sm+3 = Sm+3_Mandelic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.75(0.08SD) | 5 | MGL | 6794 |
| 2 | 25 | 1 | KNO3 | lgK:4.23(3SF) | 5 | MGL | 2101 |
| 3 | 25 | 1 | NaClO4 | lgK:4.57(3SF) | 5 | MGL | 10090 |
| 4 | 25 | 1 | NaClO4 | lgK:4.32(3SF) | 5 | MGL | 11516 |
| Note (4): Thun H et al., J. Inorg. Nucl. Chem., 1966, 28, 1949 | | | | | | | |
| 5 | 25 | 0.1 | NaClO4 | dH:-10.7(3SF)kJ | 5 | MCL | 7703 |

**Reaction No. 68680**

Sm+3 + 2<H+1_Salicylic-2> = Sm+3_H+1(2)_Salicylic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.8(0.2SD) | 6 | MGL | 6794 |

**Reaction No. 68692**

2<oAnisic-1> + Sm+3 = Sm+3_oAnisic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.2(0.1SD) | 5 | MGL | 6794 |

**Reaction No. 68707**

Tropic-1 + Sm+3 = Sm+3_Tropic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.2(0.06SD) | 5 | MGL | 6794 |

**Reaction No. 68708**

2<Tropic-1> + Sm+3 = Sm+3_Tropic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.1(0.2SD) | 5 | MGL | 6794 |

**Reaction No. 68730**

2<2AmBenz-1> + Sm+3 = Sm+3_2AmBenz-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.8(0.05SD) | 4 | MGL | 6794 |

**Reaction No. 68910**

2<Malic-2> + Sm+3 = Sm+3_Malic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | KNO3 | lgK:7.82(0.02SD) | 4 | MGL | 6670 |
| 2 | 25 | 0.1 | NaClO4 | lgK:8.16(3SF) | 5 | MGL | 11516 |
| Note (2): Roulet R et al., Helv. Chim. Acta, 1970, 53, 1876 | | | | | | | |
| 3 | 25 | 2 | NaClO4 | lgK:6.48(3SF) | 5 | MEF | 11516 |
| Note (3): Jercan E et al., An. Univ. Bucuresti Chim., 1969, 18, 43 | | | | | | | |

**Reaction No. 68960**

MeMalon-2 + Sm+3 = Sm+3_MeMalon-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | KCl | lgK:4.13(3SF) | 5 | MGL | 6662 |

**Reaction No. 68961**

Sm+3_MeMalon-2 + MeMalon-2 = Sm+3_MeMalon-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | KCl | lgK:2.18(3SF) | 5 | MGL | 6662 |

**Reaction No. 69654**

[12]O4 + Sm+3 = Sm+3_[12]O4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:5.2(0.1SD) | 5 | MIS | 2922 |

**Reaction No. 69655**

2<[12]O4> + Sm+3 = Sm+3_[12]O4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:6.8(0.1SD) | 5 | MIS | 2922 |

**Reaction No. 69674**

[15]O5 + Sm+3 = Sm+3_[15]O5

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:6.11(3SF) | 5 | MIS | 3696 |

**Reaction No. 69684**

[15]N2O3 + Sm+3 = Sm+3_[15]N2O3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | Methanol Et4NClO4 | lgK:7.00(3SF) | 5 | MIS | 3696 |
| 2 | 25 | 0.1 | PrCarbonate Et4NClO4 | lgK:14.9(3SF) | 5 | MIS | 3696 |

**Reaction No. 70775**

4<2OH2MePropan-1> + Sm+3 = Sm+3_2OH2MePropan-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.2 | NaClO4 | lgK:7.38(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:7.24(3SF) | 5 | MQH | 7703 |

**Reaction No. 70814**

2OH2MeButan-1 + Sm+3 = Sm+3_2OH2MeButan-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:2.80(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:2.60(3SF) | 5 | MQH | 7703 |

**Reaction No. 70815**

2<2OH2MeButan-1> + Sm+3 = Sm+3_2OH2MeButan-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:4.95(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:4.74(3SF) | 5 | MQH | 7703 |

**Reaction No. 70816**

3<2OH2MeButan-1> + Sm+3 = Sm+3_2OH2MeButan-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:6.46(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:6.00(3SF) | 5 | MQH | 7703 |

**Reaction No. 70829**

4<2OH2MeButan-1> + Sm+3 = Sm+3_2OH2MeButan-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:7.14(3SF) | 4 | MQH | 7703 |

**Reaction No. 70892**

2OH2Et3MeButan-1 + Sm+3 = Sm+3_2OH2Et3MeButan-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.44(3SF) | 5 | CRV | 7271 |

**Reaction No. 70893**

2<2OH2Et3MeButan-1> + Sm+3 = Sm+3_2OH2Et3MeButan-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.17(3SF) | 5 | CRV | 7271 |

**Reaction No. 70907**

2<2OH2PrPropan-1> + Sm+3 = Sm+3_2OH2PrPropan-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:4.82(3SF) | 5 | MQH | 7703 |

**Reaction No. 70966**

2<Atrolactic-1> + Sm+3 = Sm+3_Atrolactic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.99(3SF) | 5 | MGL | 7703 |
| 2 | 25 | 1 | NaClO4 | lgK:4.46(3SF) | 5 | MQH | 7703 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 25 | 0.1 | NaClO4 | dH:-9.2(2SF)kJ | 5 | MCL | 7703 |

Reaction No. 70974

3<Atrolactic-1> + Sm+3 = Sm+3_Atrolactic-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:6.00(3SF) | 5 | MQH | 7703 |

Reaction No. 70975

4<Atrolactic-1> + Sm+3 = Sm+3_Atrolactic-1(4)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 1 | NaClO4 | lgK:7.31(3SF) | 5 | MQH | 7703 |

Reaction No. 70999

Sm+2 + 2<e-1> = Sm(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | dG:123.0(4SF) | 3 | CRV | 6488 |
| 2 | 25 | 0 | Inf. Dilution | dG:514.(3SF)kJ | 5 | CRV | 7421 |
| 3 | 25 | 0 | Inf. Dilution | dG:123.00(5SF) | 2 | CRV | 10174 |
| 4 | 25 | 0 | Inf. Dilution | dG:123.00(5SF) | 2 | EOD | 31585 |
| 5 | 50 | 0 | Inf. Dilution | dG:122.85(5SF) | 4 | CRV | 10174 |
| 6 | 75 | 0 | Inf. Dilution | dG:122.71(5SF) | 3 | CRV | 10174 |
| 7 | 100 | 0 | Inf. Dilution | dG:122.57(5SF) | 2 | CRV | 10174 |
| 8 | 125 | 0 | Inf. Dilution | dG:122.44(5SF) | 2 | CRV | 10174 |
| 9 | 150 | 0 | Inf. Dilution | dG:122.30(5SF) | 2 | CRV | 10174 |
| 10 | 175 | 0 | Inf. Dilution | dG:122.16(5SF) | 4 | CRV | 10174 |
| 11 | 200 | 0 | Inf. Dilution | dG:122.00(5SF) | 4 | CRV | 10174 |
| 12 | 225 | 0 | Inf. Dilution | dG:121.82(5SF) | 4 | CRV | 10174 |
| 13 | 250 | 0 | Inf. Dilution | dG:121.60(5SF) | 1 | CRV | 10174 |
| 14 | 300 | 0 | Inf. Dilution | dG:121.03(5SF) | 0 | CRV | 10174 |
| 15 | 25 | 0 | Inf. Dilution | E:-2.68(3SF) | 1 | CRV | 6330 |
| 16 | 25 | 0 | Inf. Dilution | E:-2.68(3SF) | 1 | CRV | 7761 |
| 17 | 25 | 0 | Inf. Dilution | E:-2.68(3SF) | 1 | CRV | 22519 |
| 18 | 25 | 0 | Inf. Dilution | dH:120.5(4SF) | 3 | CRV | 6488 |
| 19 | 25 | 0 | Inf. Dilution | dH:501.1(4SF)kJ | 0 | CRV | 7761 |
| 20 | 25 | 0 | Inf. Dilution | dH:120.50(5SF) | 2 | EOD | 31585 |

Reaction No. 72060

Sm+3_OH-1 + H+1 + 3<e-1> = Sm(s) + H2O

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-108.5(4SF) | 1 | EOR | |
| 2 | 25 | 0 | Inf. Dilution | E:-2.15(3SF) | 1 | CRV | 7761 |

**Reaction No. 72061**

Sm+3_OH-1(3)_(am.,s) + 3<e-1> = Sm(s) + 3<OH-1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | E:-2.75(3SF) | 3 | CRV | 7761 |
| 2 | 25 | 0 | Inf. Dilution | dH:710.(3SF)kJ | 3 | EES | |
| Note (2): By analogy with cristalline solid | | | | | | | |

**Reaction No. 72062**

Sm+2_OH-1(2)_H2O_(s) + 2<e-1> = Sm(s) + 2<OH-1> + H2O

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-94.2(3SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | E:-2.77(3SF) | 0 | CRV | 7761 |
| Note (2): Low wgt for inter-reaction consistency | | | | | | | |
| 3 | 25 | 0 | Inf. Dilution | dH:482.7(4SF)kJ | 3 | CRV | 7761 |

**Reaction No. 72063**

Sm+3_OH-1(3)_(s) + 3<e-1> = Sm(s) + 3<OH-1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-142.2(4SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | E:-2.78(3SF) | 0 | CRV | 7761 |
| Note (2): Low wgt for inter-reaction consistency | | | | | | | |
| 3 | 25 | 0 | Inf. Dilution | dH:720.1(4SF)kJ | 3 | CRV | 7761 |

**Reaction No. 72064**

Sm+3_OH-1(3)_(s) + H2O + e-1 = Sm+2_OH-1(2)_H2O_(s) + OH-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-48.3(3SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | E:-2.8(2SF) | 0 | CRV | 7761 |
| 3 | 25 | 0 | Inf. Dilution | dH:235.6(4SF)kJ | 3 | CRV | 7761 |

**Reaction No. 72190**

Sm+3 + H+1_NTA-3 = Sm+3_NTA-3 + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:3.11(3SF) | 4 | EDH | 8698 |

**Reaction No. 72196**

Sm+3 + H+1_PO4-3 = Sm+3_H+1_PO4-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.35(3SF) | 3 | CRV | 8781 |

**Reaction No. 72212**

Sm+3 + 2<H+1_PO4-3> = Sm+3_H+1(2)_PO4-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:8.96(3SF) | 3 | CRV | 8781 |

**Reaction No. 72231**

0.5<Sm+3(2)_Oxalic-2(3)_(s)> = Sm+3 + 1.5<Oxalic-2>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-16.1(3SF) | 3 | EES | |
| Note (1): Bhat & Rao [8868] by 1lgK unit to match Gammons & Wood [8807] for Yb+3 | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:-15.15(4SF) | 0 | MSL | 8868 |

**Reaction No. 72252**

Sm+3_PO4-3_(s) = Sm+3 + PO4-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-25.31(4SF) | 4 | EDH | 8934 |
| Note (1): Corrected to inf. dil. by Byrne & Kim, GCA, 93, 57, 519 [8915] | | | | | | | |

**Reaction No. 72294**

PO4-3 + Sm+3 = Sm+3_PO4-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:12.11(0.06SD) | 4 | EFE | 8712 |

**Reaction No. 72295**

2<PO4-3> + Sm+3 = Sm+3_PO4-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:20.42(0.15SD) | 4 | EFE | 8712 |

**Reaction No. 72351**

2<Oxalic-2> + Sm+3 = Sm+3_Oxalic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.7 | NaClO4 | lgK:11.85(4SF) | 2 | MSL | 3035 |
| 2 | 25 | 0 | Inf. Dilution | lgK:10.555(5SF) | 4 | CRV | 8697 |
| 3 | 25 | 0 | Inf. Dilution | lgK:10.13(4SF) | 4 | CRV | 31301 |

**Reaction No. 72508**

Sm+3_H+1_Malonic-2 + H+1_Malonic-2 = Sm+3_H+1(2)_Malonic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.15 | NaClO4 | lgK:1.15(0.12SD) | 4 | MIX | 5698 |

**Reaction No. 74830**

2<Cl-1> + Sm+3 = Sm+3_Cl-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-4.39(3SF) | 2 | EAE | |
| 2 | 200 | 0 | Inf. Dilution | lgK:3.18(0.24SD) | 5 | MSP | 22846 |
| 3 | 250 | 0 | Inf. Dilution | lgK:4.46(0.11SD) | 5 | MSP | 22846 |

**Reaction No. 75018**

CrO4-2 + Sm+3 = Sm+3_CrO4-2

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 0.01 | 0 | Inf. Dilution | lgK:4.2320(5SF) | 3 | EOD | 22958 |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.8930(5SF) | 3 | EOD | 22958 |
| 3 | 60 | 0 | Inf. Dilution | lgK:3.5710(5SF) | 3 | EOD | 22958 |
| 4 | 100 | 0 | Inf. Dilution | lgK:3.3459(5SF) | 3 | EOD | 22958 |
| 5 | 150 | 0 | Inf. Dilution | lgK:3.2404(5SF) | 3 | EOD | 22958 |
| 6 | 200 | 0 | Inf. Dilution | lgK:3.3295(5SF) | 3 | EOD | 22958 |

**Reaction No. 75417**

Sm+3_Citric-3 + H+1 = Sm+3_H+1_Citric-3

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.15 | NaCl | lgK:3.38(0.02SD) | 5 | MSP | 23315 |

**Reaction No. 75418**

Sm+3_Citric-3 + Citric-3 = Sm+3_Citric-3(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.15 | NaCl | lgK:5.08(0.04SD) | 5 | MSP | 23315 |

| Reaction No. 75419 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Citric-3(2) + H+1 = Sm+3_H+1_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:4.21(0.05SD) | 5 | MSP | 23315 |

| Reaction No. 75420 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(-1)_Citric-3 + H+1 = Sm+3_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:5.82(3SF) | 5 | MSP | 23315 |

| Reaction No. 75421 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(-2)_Citric-3 + H+1 = Sm+3_H+1(-1)_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:7.02(0.07SD) | 5 | MSP | 23315 |

| Reaction No. 75422 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(-1)_Citric-3(2) + H+1 = Sm+3_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:6.85(0.07SD) | 5 | MSP | 23315 |

| Reaction No. 75430 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1(2)_EDTMP-8 + H+1 = Sm+3_H+1(3)_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:4.80(0.03SD) | 5 | MSP | 23315 |

| Reaction No. 75438 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_EDTMP-8 + Ca+2 = Sm+3_Ca+2_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:3.47(0.05SD) | 5 | MSP | 23315 |

| Reaction No. 75439 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Ca+2_EDTMP-8 + H+1 = Sm+3_Ca+2_H+1_EDTMP-8 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.15 | NaCl | lgK:6.22(0.08SD) | 5 | MSP | 23315 |

| Reaction No. 75755 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Lys-1 + Sm+3 = Sm+3_Lys-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:7.71(3SF) | 3 | MGL | 11245 |
| 2 | 25 | 0 | Inf. Dilution | lgK:8.33(3SF) | 2 | EAE | |
| 3 | 30 | 0.1 | NaClO4 | lgK:7.51(3SF) | 3 | MGL | 22560 |

Note (3): DSaxena & Dhawan, Indian J. Chem., 1983, 22A, 89

**Reaction No. 75756**

Sm+3_Lys-1 + Lys-1 = Sm+3_Lys-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:6.95(3SF) | 3 | MGL | 11245 |
| 2 | 25 | 0 | Inf. Dilution | lgK:7.36(3SF) | 2 | EAE | |
| 3 | 30 | 0.1 | NaClO4 | lgK:6.85(3SF) | 3 | MGL | 22560 |

Note (3): DSaxena & Dhawan, Indian J. Chem., 1983, 22A, 89

**Reaction No. 75757**

Sm+3 + 2<Lys-1> = Sm+3_Lys-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:14.66(4SF) | 3 | MGL | 11245 |
| 2 | 25 | 0 | Inf. Dilution | lgK:15.7(3SF) | 2 | EAE | |

**Reaction No. 75767**

Sm+3_His-1 + His-1 = Sm+3_His-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:4.8(2SF) | 1 | ERG | |
| 2 | 25 | 3 | NaClO4 | lgK:4.25(3SF) | 4 | MGL | 11497 |
| 3 | 37 | 3 | NaClO4 | lgK:4.41(3SF) | 4 | MGL | 11497 |

**Reaction No. 75785**

Sm+3_Asn-1 + Asn-1 = Sm+3_Asn-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:4.3(2SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | lgK:3.33(3SF) | 0 | EAE | |
| 3 | 30 | 0.1 | NaClO4 | lgK:2.88(3SF) | 2 | MGL | 11516 |

Note (3): Kemin Y et al., Chem. J. of Chin. Univ., 1984, , 603

**Reaction No. 76009**

Sm+3 + Pyruvic-1 = Sm+3_Pyruvic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 2 | NaClO4 | lgK:1.53(3SF) | 5 | MMR | 11516 |
| Note (1): Choppin G et al., Inorg. Chem., 1980, 19, 1889 | | | | | | | |

**Reaction No. 76109**

Sm+3 + 2<Hyp-1> = Sm+3_Hyp-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:4.95(3SF) | 5 | MGL | 3366 |

**Reaction No. 76224**

Sm+3 + 2<EDTA-4> = Sm+3_EDTA-4(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 19 | 0.04 | Unknown | lgK:22.83(4SF) | 1 | MSP | 11516 |
| Note (1): Galaktionov Y et al., Zhur. Neorg. Khim., 1963, 8, 460 | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:22.4(3SF) | 1 | ELC | |

**Reaction No. 76296**

Sm+3 + 2<Propanoic-1> = Sm+3_Propanoic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 20 | 0.1 | NaClO4 | lgK:3.70(3SF) | 5 | MGL | 11516 |
| Note (1): Powell J et al., Inorg. Chem., 1964, 3, 518 | | | | | | | |
| 2 | 25 | 2 | NaClO4 | lgK:3.24(3SF) | 5 | MGL | 11516 |
| Note (2): Choppin G et al., Inorg. Chem., 1965, 4, 1254 | | | | | | | |

**Reaction No. 76313**

Sm+3_Malonic-2 + Malonic-2 = Sm+3_Malonic-2(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | KNO3 | lgK:2.65(3SF) | 5 | MGL | 4388 |
| 2 | 25 | 1 | NaClO4 | lgK:2.40(3SF) | 5 | MGL | 10134 |

**Reaction No. 76407**

Sm+3 + Benzoic-1 = Sm+3_Benzoic-1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.21(3SF) | 5 | MCL | 11516 |
| Note (1): Choppin G et al., Inorg. Chem., 1982, 21, 3722 | | | | | | | |

**Reaction No. 76408**

Sm+3_Benzoic-1 + Benzoic-1 = Sm+3_Benzoic-1(2)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0.1 | NaClO4 | lgK:1.71(3SF) | 5 | MCL | 11516 |
| Note (1): Choppin G et al., Inorg. Chem., 1982, 21, 3722 | | | | | | | |

| Reaction No. 76442 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + PhenAcet-1 = Sm+3_PhenAcet-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | NaClO4 | lgK:2.13(3SF) | 5 | MGL | 11516 |
| Note (1): Hasegawa Y et al., Bull. Chem. Soc. Jpn., 1990, 63, 269 | | | | | | | |

| Reaction No. 76873 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2Thiobarb-2 + Sm+3 = Sm+3_2Thiobarb-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | Unknown | lgK:3.163(4SF) | 1 | EOD | 22560 |
| Note (1): Tabassum et al., Indian J. Chem., 1987, 26A, 489 & 523 | | | | | | | |

| Reaction No. 76892 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_Malonic-2(2) + H+1 = Sm+3_H+1_Malonic-2(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 1 | Na+1 salt | lgK:3.8(2SF) | 3 | CRV | 7271 |
| 2 | 25 | 1 | Na+1 salt | dH:-0.7(1SF) | 3 | CRV | 7271 |

| Reaction No. 77402 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 2<OH-1> = Sm+3_OH-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:11.21(4SF) | 1 | ELC | |

| Reaction No. 77405 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 3<OH-1> = Sm+3_OH-1(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:16.1(3SF) | 1 | ELC | |

| Reaction No. 77429 | | | | | | | |
|---|---|---|---|---|---|---|---|
| ClO4-1 + Sm+3 = Sm+3_ClO4-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:1.001(4SF) | 2 | EAC | |

| Reaction No. 77761 |
|---|

| Sm+3 + 2<P3O10-5> + 2<H+1> = Sm+3_H+1(2)_P3O10-5(2) | | | | | | | |
|---|---|---|---|---|---|---|---|
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:27.5(3SF) | 1 | ELC | |

| Reaction No. 77955 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Asp-2 + Sm+3 = Sm+3_Asp-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:24.4(4SF) | 1 | ELC | |

| Reaction No. 77959 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Gly-1 + EDTA-4 + Sm+3 = Sm+3_Gly-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:24.25(4SF) | 1 | ELC | |

| Reaction No. 77980 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + 2SHAcet-2 + Sm+3 = Sm+3_2SHAcet-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:22.73(4SF) | 1 | ELC | |

| Reaction No. 78044 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + P3O10-5 + H+1 = Sm+3_H+1_EDTA-4_P3O10-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:31.5(3SF) | 1 | ELC | |

| Reaction No. 78223 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Ala-1 + Sm+3 = Sm+3_Ala-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:24.65(4SF) | 1 | ELC | |

| Reaction No. 78224 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Cat-2 + Sm+3 = Sm+3_Cat-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:26.53(4SF) | 1 | ELC | |

| Reaction No. 78256 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + bAla-1 + Sm+3 = Sm+3_bAla-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:24.28(4SF) | 1 | ELC | |

| Reaction No. 78257 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Citraconic-2 + Sm+3 = Sm+3_Citraconic-2_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.53(4SF) | 1 | ELC | |

| Reaction No. 78267 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + F-1 = Sm+3_F-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:20.92(4SF) | 1 | ELC | |

| Reaction No. 78282 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + 2<F-1> = Sm+3_F-1(2)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:20.92(4SF) | 1 | ELC | |

| Reaction No. 78305 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + 24DHB-3 + Sm+3 = Sm+3_24DHB-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.28(4SF) | 1 | ELC | |

| Reaction No. 78373 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + H+1 = Sm+3_H+1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:21.39(4SF) | 1 | ELC | |

| Reaction No. 78384 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + 5ATP-4 + Sm+3 = Sm+3_5ATP-4_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.45(4SF) | 1 | ELC | |

| Reaction No. 78392 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3<EDTA-4> + 2<Sm+3> = Sm+3(2)_EDTA-4(3) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:43.04(4SF) | 1 | ELC | |

| Reaction No. 78396 |
|---|
| 2<EDTA-4> + Sm+3 + H+1 = Sm+3_H+1_EDTA-4(2) |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:32.44(4SF) | 1 | ELC | |

| Reaction No. 78416 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Citric-3> + Sm+3 + H+1 = Sm+3_H+1_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:19.49(4SF) | 1 | ELC | |

| Reaction No. 78424 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + 3<H+1> = Sm+3_H+1(3)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:24.85(4SF) | 1 | ELC | |

| Reaction No. 78434 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Crotonic-1 + Sm+3 = Sm+3_Crotonic-1_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:22.88(4SF) | 1 | ELC | |

| Reaction No. 78440 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + Sm+3 + 4<H+1> = Sm+3_H+1(4)_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:25.54(4SF) | 1 | ELC | |

| Reaction No. 78451 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + 34DHB-3 + Sm+3 = Sm+3_34DHB-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.85(4SF) | 1 | ELC | |

| Reaction No. 78557 | | | | | | | |
|---|---|---|---|---|---|---|---|
| EDTA-4 + 2SHSuccin-3 + Sm+3 = Sm+3_2SHSuccin-3_EDTA-4 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.38(4SF) | 1 | ELC | |

| Reaction No. 78923 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 3<Cl-1> = Sm+3_Cl-1(3)_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:-18.9(4SF) | 1 | ELC | |

| Reaction No. 78924 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+2 + 2<Cl-1> = Sm+2_Cl-1(2)_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:-1.886(4SF) | 1 | ELC | |

| Reaction No. 79182 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 3<OH-1> = Sm+3_OH-1(3)_(am.,s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:23.06(4SF) | 1 | ELC | |

| Reaction No. 79183 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+2 + 2<OH-1> + H2O = Sm+2_OH-1(2)_H2O_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.793(4SF) | 1 | ELC | |

| Reaction No. 79207 | | | | | | | |
|---|---|---|---|---|---|---|---|
| H+1 + CO3-2 + Sm+3 = Sm+3_H+1_CO3-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:12.1(4SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | lgK:12.43(0.50SD) | 4 | CRV | 32222 |

| Reaction No. 79212 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + PO4-3 + 2<H+1> = Sm+3_H+1(2)_PO4-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:21.75(4SF) | 1 | ELC | |

| Reaction No. 79223 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + PO4-3 + H+1 = Sm+3_H+1_PO4-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:17.68(4SF) | 1 | ELC | |

| Reaction No. 79234 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + 2<PO4-3> + 2<H+1> = Sm+3_H+1(2)_PO4-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:33.61(4SF) | 1 | ELC | |

| Reaction No. 79401 |
|---|
| Citric-3 + Sm+3 + H+1 = Sm+3_H+1_Citric-3 |

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:12.88(4SF) | 1 | ELC | |

| Reaction No. 79402 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Citric-3> + Sm+3 = Sm+3_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:14.58(4SF) | 1 | ELC | |

| Reaction No. 79403 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Citric-3 + Sm+3 - H+1 = Sm+3_H+1(-1)_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:3.446(4SF) | 1 | ELC | |

| Reaction No. 79404 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Citric-3 + Sm+3 - 2<H+1> = Sm+3_H+1(-2)_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:-4.042(4SF) | 1 | ELC | |

| Reaction No. 79405 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<Citric-3> + Sm+3 - H+1 = Sm+3_H+1(-1)_Citric-3(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:6.795(4SF) | 1 | ELC | |

| Reaction No. 79597 | | | | | | | |
|---|---|---|---|---|---|---|---|
| BrO3-1 + Sm+3 = Sm+3_BrO3-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:-1.(1SF) | 1 | EES | |

| Reaction No. 80222 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_DTPA-5 + H+1 = Sm+3_H+1_DTPA-5 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0.1 | KNO3 | lgK:2.20(3SF) | 3 | EOD | 30513 |
| 2 | 25 | 2 | NaClO4 | lgK:1.59(0.02SD) | 3 | MGL | 30513 |

| Reaction No. 80229 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Lys-1 = Sm+3_H+1_Lys-1 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 15 | 1 | KNO3 | lgK:5.26(3SF) | 3 | MGL | 30533 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Note (1): Rxn from Pettit database [05PeP_22560] | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:5.4(2SF) | 1 | ESC | 30533 |
| 3 | 30 | 1 | KNO3 | lgK:4.84(3SF) | 2 | MGL | 30533 |

| Reaction No. 80230 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3_H+1_Lys-1 + H+1_Lys-1 = Sm+3_H+1(2)_Lys-1(2) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 15 | 1 | KNO3 | lgK:2.57(3SF) | 3 | MGL | 30533 |
| Note (1): Rxn from Pettit database [05PeP_22560] | | | | | | | |
| 2 | 25 | 0 | Inf. Dilution | lgK:2.6(2SF) | 1 | ESC | 30533 |
| 3 | 30 | 1 | KNO3 | lgK:2.55(3SF) | 2 | MGL | 30533 |

| Reaction No. 80542 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8<H2(g)> + 10<O2(g)> + 3<S(s)> + 2<Sm(s)> = Sm+3(2)_SO4-2(3)_H2O(8)_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | dG:-1322.0(5SF) | 3 | ERP | 31293 |

| Reaction No. 80543 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5<H2(g)> + 8.5<O2(g)> + 3<S(s)> + 2<Sm(s)> = Sm+3(2)_SO4-2(3)_H2O(5)_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | dG:-1149.1(5SF) | 3 | ERP | 31293 |

| Reaction No. 80544 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<H2(g)> + 2<Na(s)> + 9<O2(g)> + 4<S(s)> + 2<Sm(s)> = Sm+3(2)_Na+1(2)_SO4-2(4)_H2O(2)_(s) | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | dG:-1203.30(6SF) | 3 | ERP | 31293 |

| Reaction No. 80568 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2<O2(g)> + S(s) + Sm(s) - e-1 = Sm+3_SO4-2 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:250.8(4SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | dG:-340.29(5SF) | 0 | ERP | 31293 |

| Reaction No. 80632 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Sm+3 + H+1_Citric-3 = Sm+3_H+1_Citric-3 | | | | | | | |
| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
| 1 | 25 | 0 | Inf. Dilution | lgK:6.5(2SF) | 1 | ELC | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 25 | 0 | Inf. Dilution | lgK:5.31(3SF) | 0 | CRV | 31301 |

Note (2): Suspect B110

**Reaction No. 80716**

Sm+2 + H+1 + 0.25<O2> = Sm+3 + 0.5<H2O>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:47.96(4SF) | 4 | CRV | 31585 |

**Reaction No. 80734**

Sm(s) - 4<e-1> = Sm+4

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | dG:-39.900(5SF) | 2 | EOD | 31585 |
| 2 | 25 | 0 | Inf. Dilution | dH:-56.400(5SF) | 2 | EOD | 31585 |

**Reaction No. 81062**

Sm+3 + 4<H2O> = Sm+3_OH-1(4) + 4<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-36.90(1.00SD) | 4 | CRV | 32222 |
| 2 | 25 | 0 | Inf. Dilution | dH:276.900(8.585SD)kJ | 4 | CRV | 32222 |

**Reaction No. 81064**

Sm+3 + 3<H2O> = Sm+3_OH-1(3)_(am.,s) + 3<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-18.60(1.00SD) | 4 | CRV | 32222 |

**Reaction No. 81065**

2<Sm+3> + 3<H2O> = Sm2O3(cubic,s) + 6<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-53.7(3SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | lgK:-43.11(0.61SD) | 0 | CRV | 32222 |
| 3 | 25 | 0 | Inf. Dilution | dH:355.040(5.051SD)kJ | 1 | CRV | 32222 |

**Reaction No. 81071**

Sm+3 + CO3-2 + H2O = Sm+3_OH-1_CO3-2_(s) + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:7.70(3SF) | 4 | CRV | 32222 |

**Reaction No. 81073**

Sm+3 + H+1(2)_PO4-3 = Sm+3_H+1_PO4-3 + H+1

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-1.61(0.50SD) | 4 | CRV | 32222 |

Reaction No. 81074

Sm+3 + 2<H+1(2)_PO4-3> = Sm+3_H+1(2)_PO4-3(2) + 2<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-5.02(0.50SD) | 4 | CRV | 32222 |

Reaction No. 81076

Sm+3 + 2<H+1(2)_PO4-3> = Sm+3_PO4-3(2) + 4<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-18.72(4SF) | 4 | CRV | 32222 |

Reaction No. 81079

Sm+3 + H+1(2)_PO4-3 = Sm+3_PO4-3 + 2<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-7.46(3SF) | 4 | CRV | 32222 |

Reaction No. 81080

Sm+3 + H+1(2)_PO4-3 + H2O = Sm+3_PO4-3_H2O_(s) + 2<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:4.94(3SF) | 4 | CRV | 32222 |

Reaction No. 81083

Sm+3 + H+1(2)_PO4-3 = Sm+3_PO4-3_(s) + 2<H+1>

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:5.8(2SF) | 1 | ELC | |
| 2 | 25 | 0 | Inf. Dilution | lgK:6.67(0.03SD) | 0 | CRV | 32222 |

Reaction No. 81085

Sm+3 + 3<F-1> + 0.5<H2O> = Sm+3_F-1(3)_H2O(0.5)_(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:17.50(4SF) | 4 | CRV | 32222 |

Reaction No. 81088

Sm+3 + 3<Cl-1> + 6<H2O> = Sm+3_Cl-1(3)_H2O(6)_(s)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 25 | 0 | Inf. Dilution | lgK:-4.80(3SF) | 4 | CRV | 32222 |
| 2 | 25 | 0 | Inf. Dilution | dH:38.311(5SF)kJ | 4 | CRV | 32222 |

## Data Assessment

| Weight | Assessment |
|---|---|
| 9 | highest accuracy |
| 8 | multi-determined; excellent dependability |
| 7 | trustworthy; outstanding single determination |
| 6 | better than average reliability |
| 5 | average single determination |
| 4 | some doubt |
| 3 | poorly known |
| 2 | guess to achieve consistency (or just as bad) |
| 1 | no information but consistent |
| 0 | problematic/inconsistent |

## Techniques

| Abbrev | Method |
|---|---|
| CMN | Several experimental values (crit.) |
| CRV | Critical review |
| EAC | Activity Coefficient |
| EAE | Automated extrapolation |
| EDH | Debye-Hueckel corrections |
| EES | Personal estimate |
| EFE | Free Energy relations |
| EIU | Critical IUPAC review (estim.) |
| ELC | Linear Combination of Reactions |
| ENS | Not specified |
| EOD | Calculated from data of others |
| EOR | Other Reaction Constants |
| ERG | Relative Gibbs energy |

| | |
|---|---|
| ERP | Regression procedure |
| ESC | Standard conditions anchor |
| ESR | Similar reaction(s) |
| MAG | Silver electrode e.m.f. |
| MCE | Cation exchange |
| MCL | Calorimetry |
| MCN | Conductivity |
| MDG | Combination of Thermodynamic Data |
| MDS | Distribution between two phases |
| MEF | Electromotoric force, not specified |
| MEP | Electrophoresis |
| MGL | Glass electrode |
| MHG | Emf with amalgam electrode |
| MIS | Ion selective electrode |
| MIX | Ion exchange |
| MKN | Rate of reaction |
| MLC | Ligand competition |
| MMC | Metal competition |
| MMR | Nuclear magnetic resonance |
| MPH | pH method, not specified |
| MPL | Polarography |
| MPS | Potentiometry and Spectrophotometry |
| MPT | Potentiometry |
| MQH | Quinhydrone electrode |
| MRX | Emf with redox electrode |
| MSC | Miscellaneous |
| MSE | Solvent Extraction |
| MSL | Solubility |
| MSP | Spectroscopy |
| MTD | Temperature difference |

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